

Basic AI

Computer Vision for Department of Lands

<https://kaopanboonyuen.github.io/>

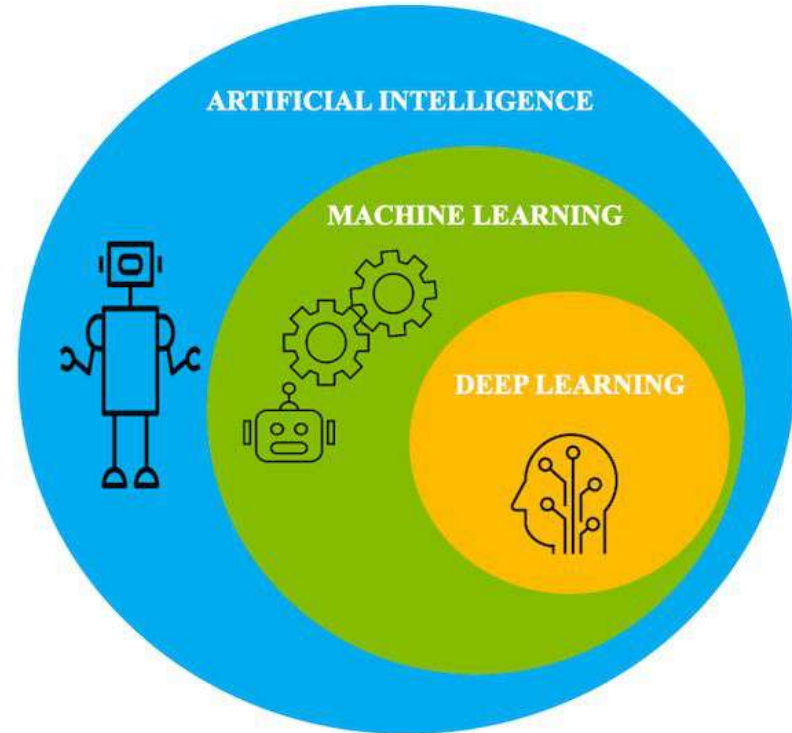


Reference:

1. https://pytorch.org/tutorials/beginner/basics/quickstart_tutorial.html
2. <http://introtodeeplearning.com/>
3. <https://www.simplilearn.com/tutorials/deep-learning-tutorial/introduction-to-deep-learning>
4. <https://www.geeksforgeeks.org/introduction-deep-learning/>
5. <https://stanford.edu/~shervine/teaching/cs-230/cheatsheet-convolutional-neural-networks>

AI vs Machine Learning vs Deep Learning

- **Artificial Intelligence (AI)** is the concept of creating smart intelligent machines.
- **Machine Learning (ML)** is a subset of artificial intelligence that helps you build AI-driven applications.
- **Deep Learning** is a subset of machine learning that uses vast volumes of data and complex algorithms to train a model.



Artificial Intelligence

Is the field of study

Machine Learning

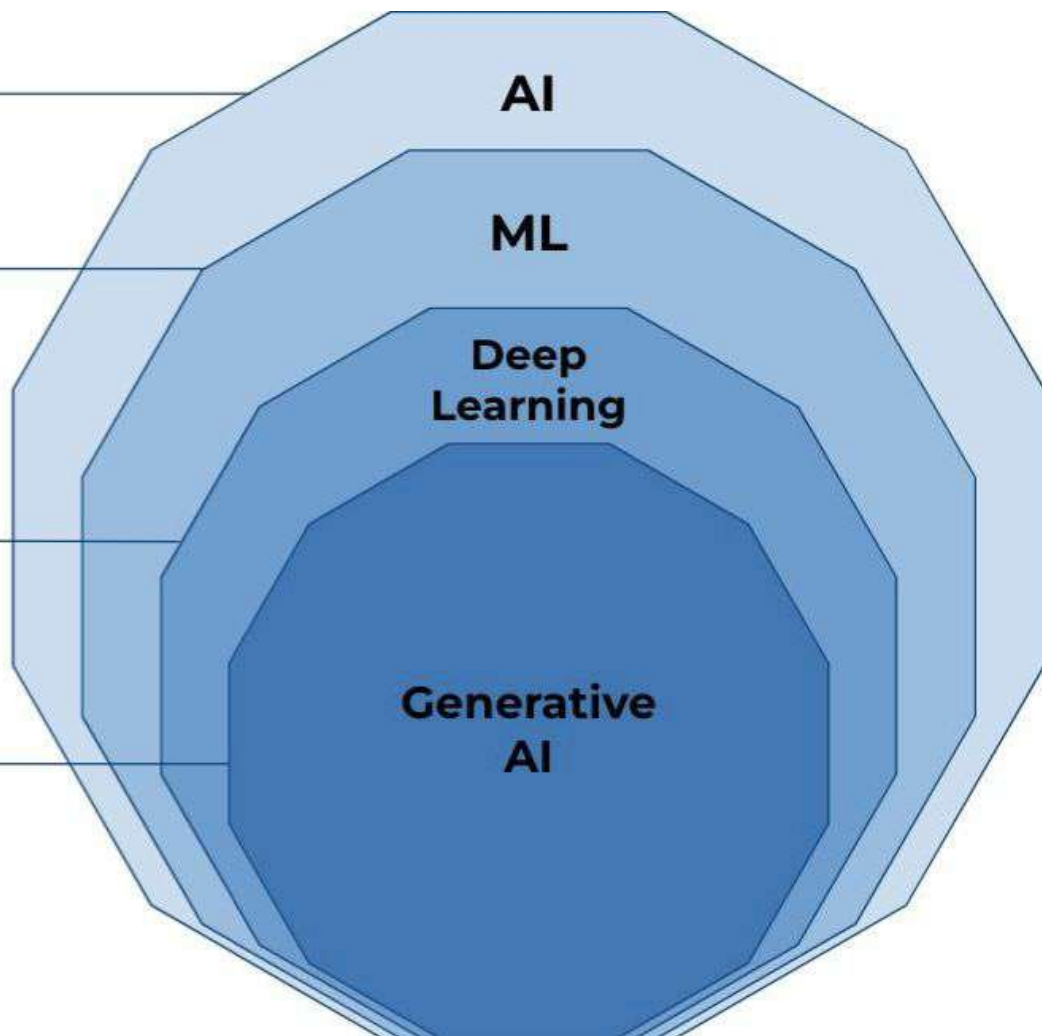
Is a branch of AI that focus on the creation of intelligent machines that learn from data. Another very well know branch inside AI is **Optimization**.

Deep Learning

Is a subset of Machine Learning methods, based on **Artificial Neural Networks**.
Examples: CNNs, RNNs

Generative AI

A type of ANNs that generate data that is similar to the data it was trained on.
Examples: GANs, LLMs



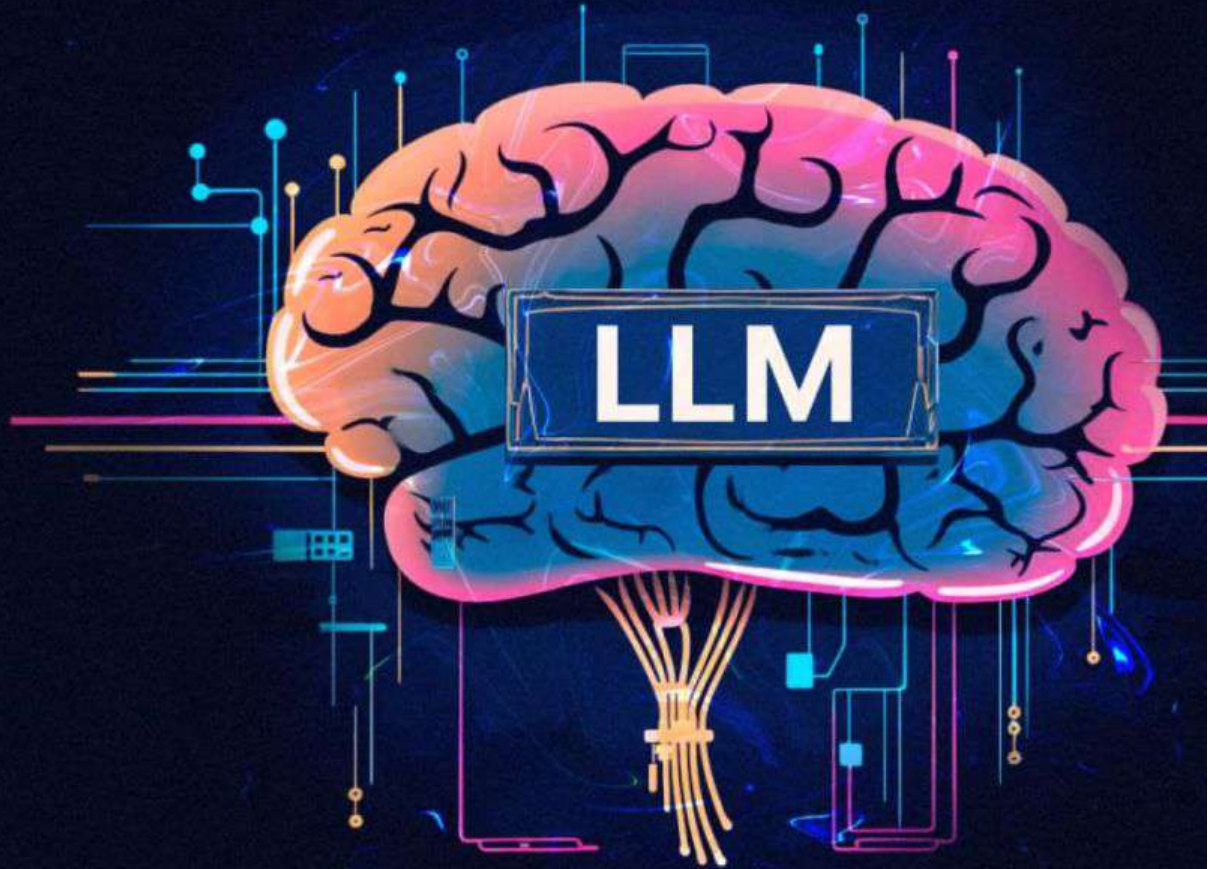


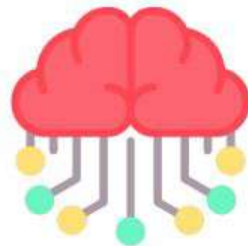
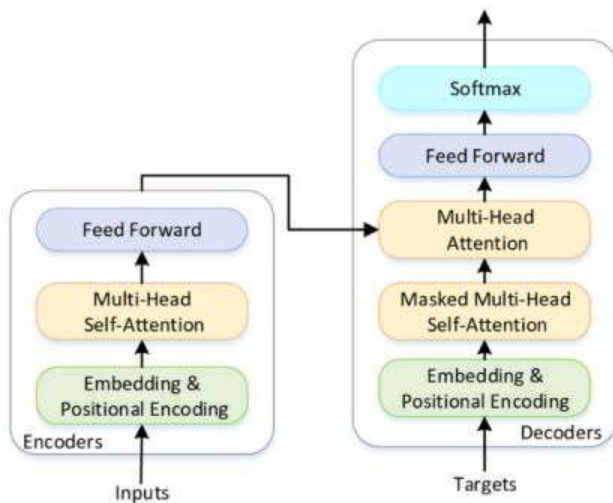
GPT

Generative Pre-trained Transformer



Large Language Models (LLMs)?





Data



Architecture



Training



LLM



Thailand title deeds

along with 4 methods to check
for fake title deeds
(easy to understand)



ด่วน
ที่ นท ๐๕๖๕.๑/ ๑๒๕๕๖

กรมที่ดิน
ศูนย์ราชการเฉลิมพระเกียรติ ๘๐ พรรษา
อาคารรัฐประศาสนภักดี ถนนแจ้งวัฒนะ
นครหลวง กรุงเทพมหานคร ๑๐๑๑๐

๒๙ มิถุนายน ๒๕๖๑

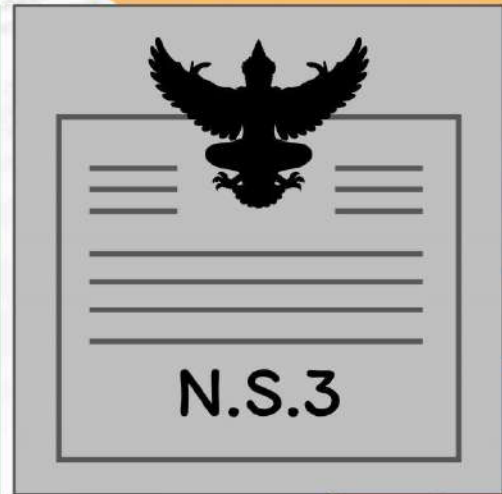
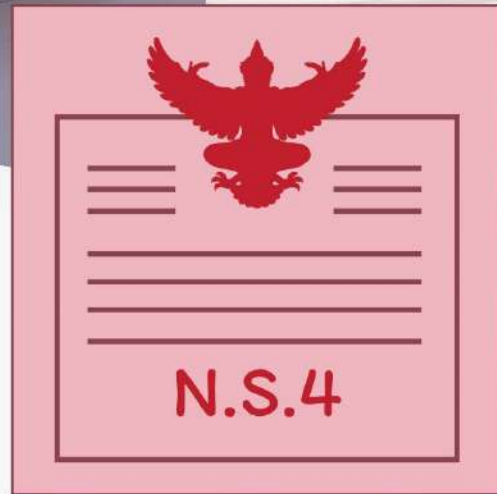
เนื่อง: ประชาชนยื่นคำร้องเกี่ยวกับกลุ่มเจ้าของที่ดินขอตรวจสอบที่ดินและทรัพย์สินที่เช่าทำ
เงิน: ผู้ว่าราชการจังหวัดสุพรรณบุรี

อ้างถึง: หนังสือกรมที่ดิน ที่ นท ๐๕๖๕.๑/ ๑๒๕๕๖ ลงวันที่ ๙ มีนาคม ๒๕๖๑
ถึงที่สำนักงาน: สำนักงานที่ดินจังหวัดสุพรรณบุรี

เตือนภัยรูปแบบใหม่!
หลอกซื้อที่ดิน-ก่อนล่อทำผิดสัญญา

PPTV HD 36

Types of land documents for each color



4 Methods to check for fake title deeds



1. Illuminate the title deed through the light.



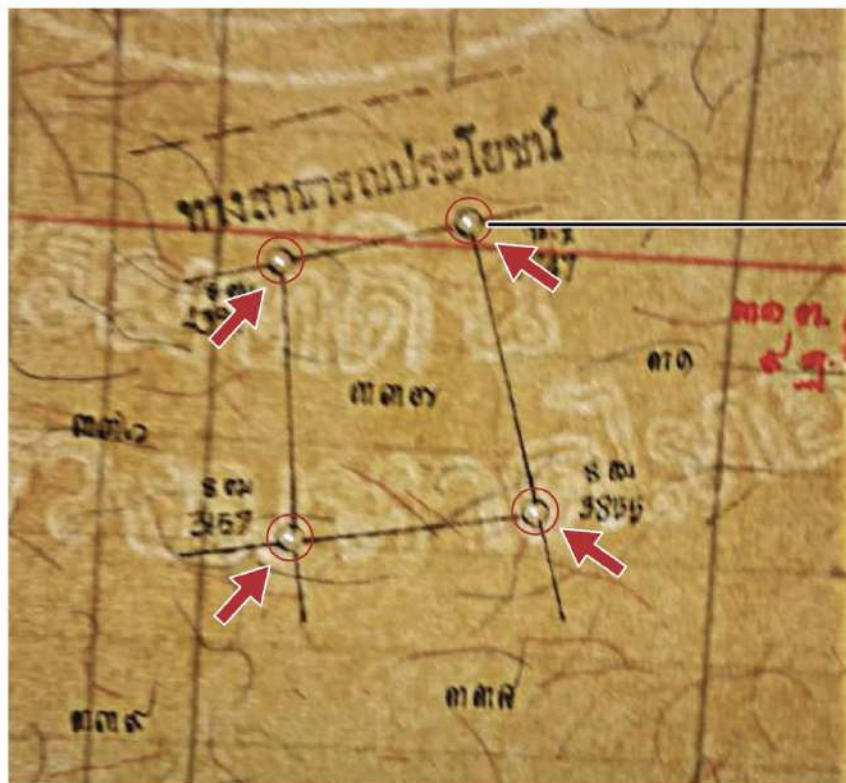
Title Deed

is a certificate of Right

issued in accordance with the land code

Garuda in 2 circles

Department of Lands, Ministry of Interior



Luminous hole

2. Check with the signature and stamp of the officer.

ชอก น วันที่ พฤษภาคม เดือน พฤษภาคม พุทธศักราช สองพันห้าร้อยสี่สิบสี่

(นายวิชาญ หอวิจิตร)
เจ้าพนักงานทดแทน

นายวิชาญ หอวิจิตร
(นางสาววันวิสาข์ หอวิจิตร)
นางสาววันวิสาข์ หอวิจิตร

22 พ.ค. 2544
368415

(นายวิชาญ หอวิจิตร)
นายวิชาญ หอวิจิตร

231

Officer's signature
and stamp

[illegible]

Cadastral map
Parcel file
Survey page
Sub-district

4928 II 0876-11

72194 DE

1897/1898 1898/1899

၆၇၅၇၁ မဟာဗျူဟ

LAND TITLE

โขนมมมม

ស្រាវជ្រាវ និង អភិវឌ្ឍន៍

ออกโดยอาศัยอำนาจตามประมวลกฎหมายวิธีพิจารณาความอาชญา

First owner

Size of land in	Rai	Ngan	Talang Wah
100	100	100	100
200	200	200	200
300	300	300	300
400	400	400	400
500	500	500	500
600	600	600	600
700	700	700	700
800	800	800	800
900	900	900	900
1000	1000	1000	1000

In red, separation from another title deed. That's rare

Description of the land

Boundary post
with number

Date _____

Month

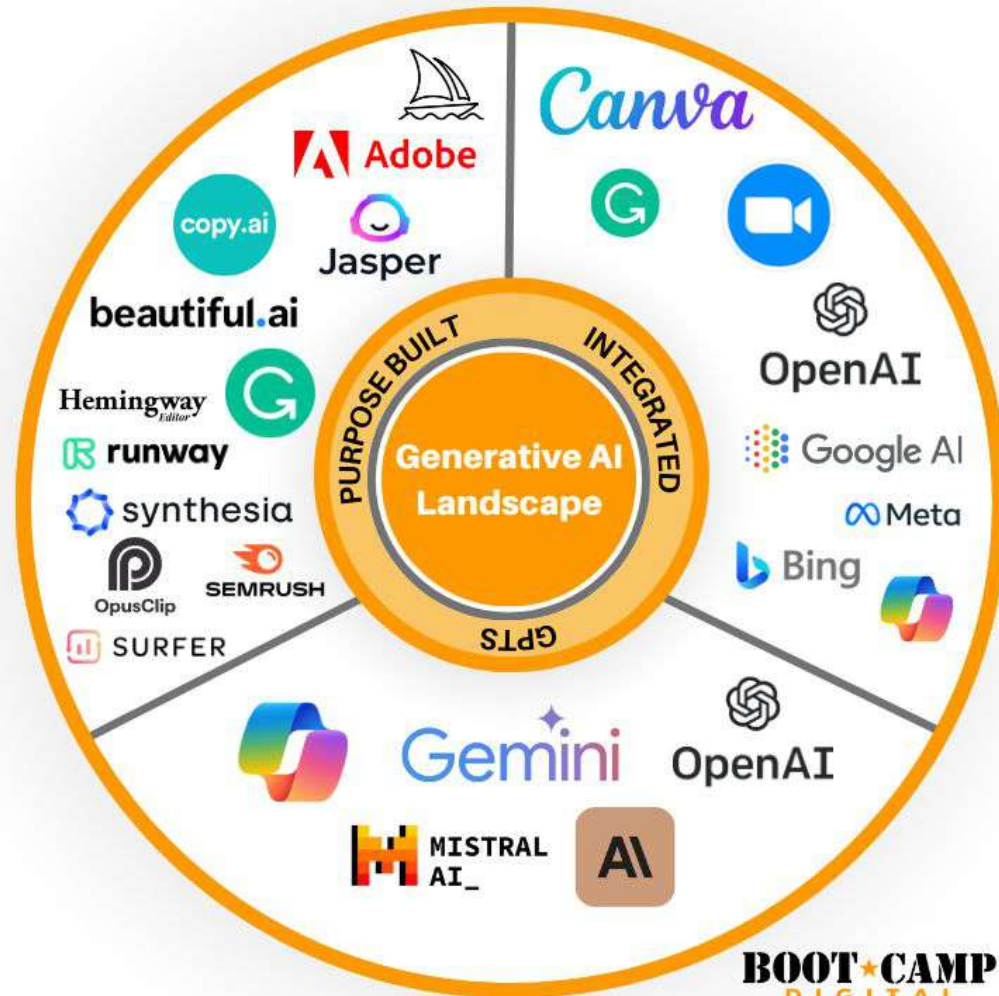
Year

This title deed is from 15 August 2566 (2023)

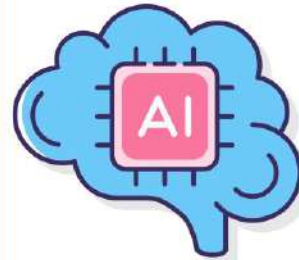
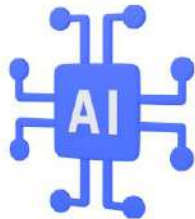
— Land Department
Seal

158501

16



Types Of Artificial Intelligence



Three types of Artificial Intelligence



Artificial Narrow
Intelligence (ANI)

Stage-1

Machine
Learning

Specialises in
one area and
solves one problem



Artificial General
Intelligence (AGI)

Stage-2

Machine
Intelligence

Refers to a computer that
is as smart as a human
across the board



Artificial Super
Intelligence (ASI)

Stage-3

Machine
Consciousness

An intellect that is much
smarter than the best human
brains in practically every field



Hi, I'm DeepSeek.

How can I help you today?

DEEPSEEK VS CHATGPT



Category

DeepSeek

ChatGPT

DeepSeek AI



OpenAI

Open-Source LLM



Proprietary LLM

Mixture-of-Experts



Transformer-Based (GPT - 4)

Multilingual, Chinese Focused



Extensive Multilingual and Strong in English

Efficient, Cost-Effective



Strong in reasoning, Creativity and text coherence

Excels in Coding and technical tasks



Best For Conversational AI, Coding assistance and creative writing

Lacks Multi-Modal capabilities



Limited Real-Time Capabilities

Free and Open-Source



Free (3.5), Paid (GPT-4 : \$20/Month)

~ \$6 Million



More than \$100 million

Complex Deployment and Integration ~ Higher Development and Deployment cost



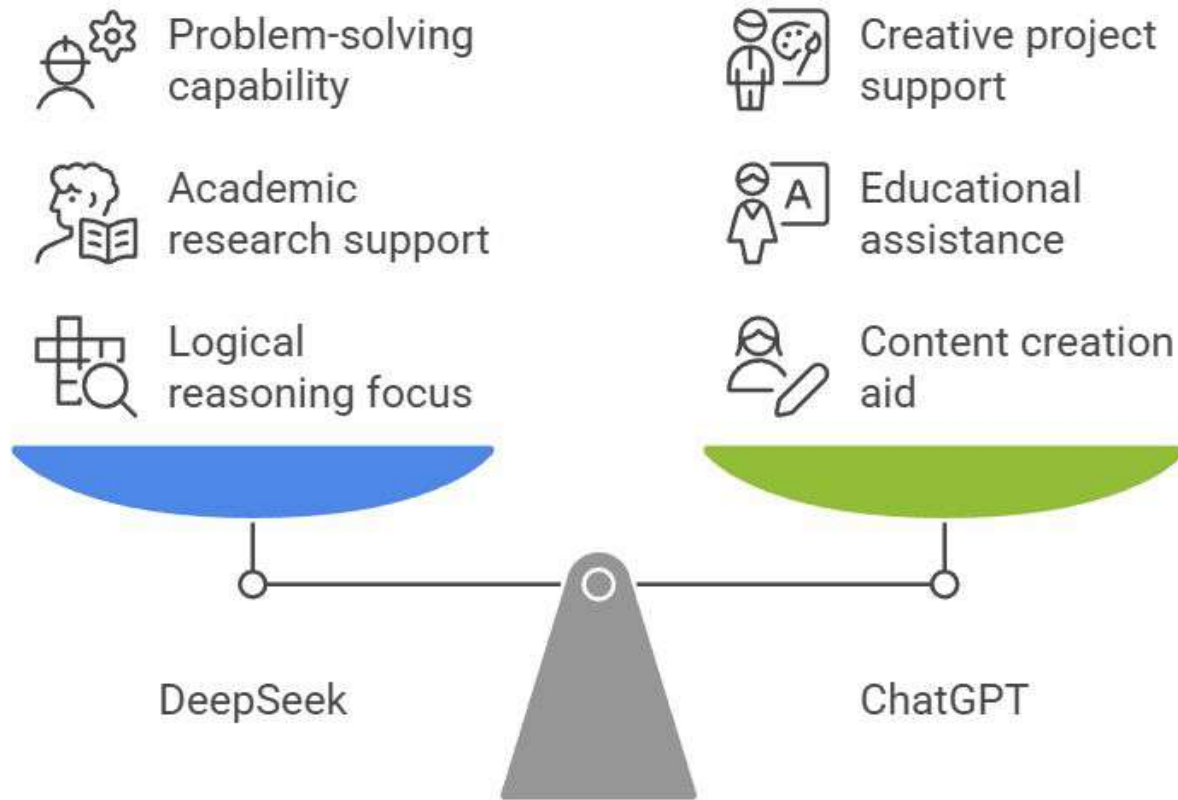
Larger datasets and advanced models like GPT - 4, Focus on reducing inaccuracies & Enhancing domain specific features



LEARN AI AUTOMATION STEP-BY-STEP TO BUILD WORKFLOWS THAT EARN MONEY AND FUTURE-PROOF YOUR CAREER.

CHECKOUT THE LINK IN BIO !





Comparing AI Platforms: DeepSeek vs. ChatGPT

Generative AI vs. Large Language Models (LLMs)



Generative AI

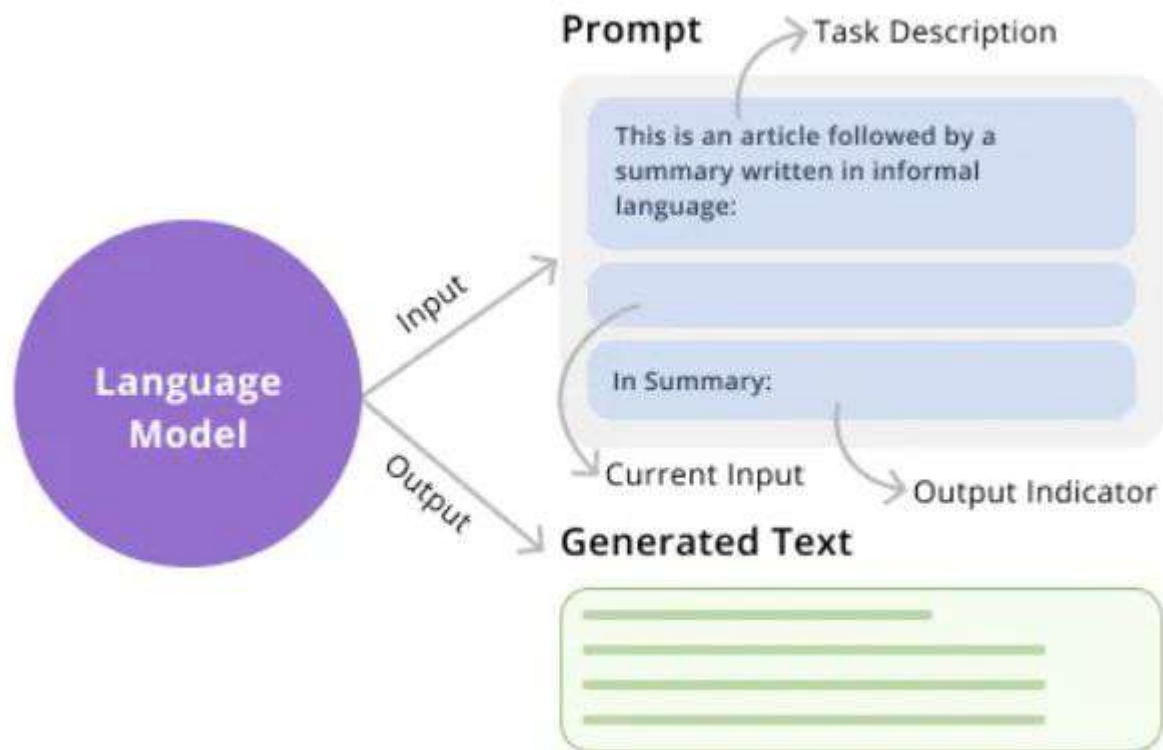
- Creates many outputs
- Myriad of generative AI tools
- Built on LLMs but also other types of machine learning models

vs.



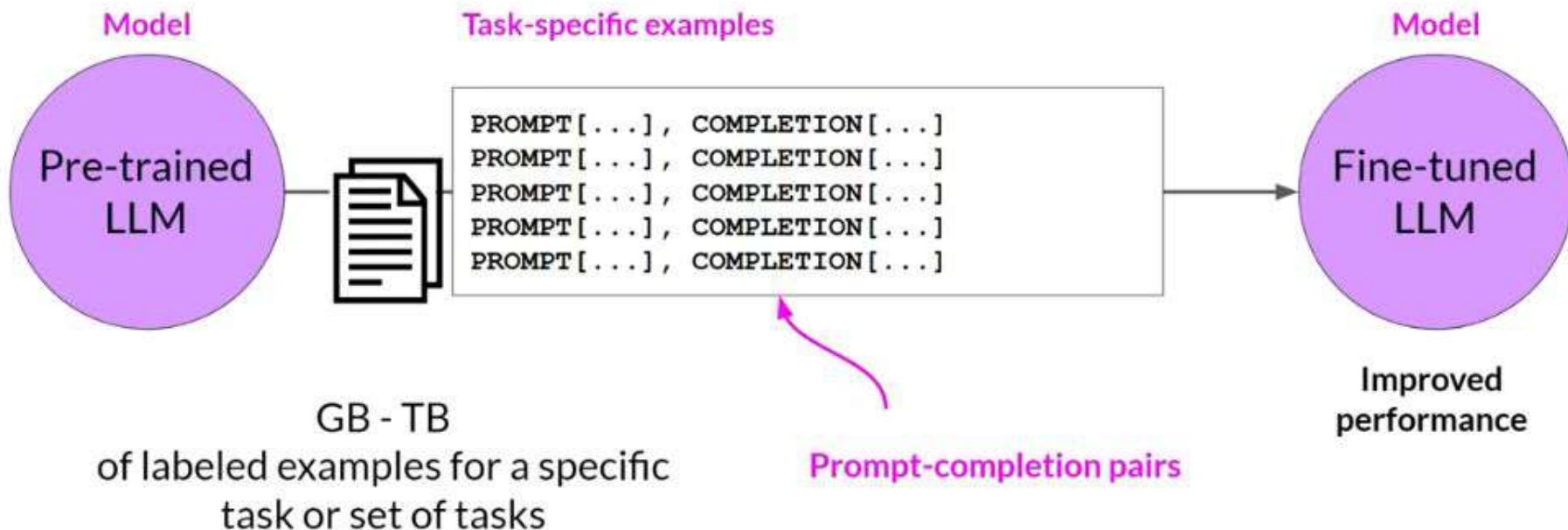
Large Language Models

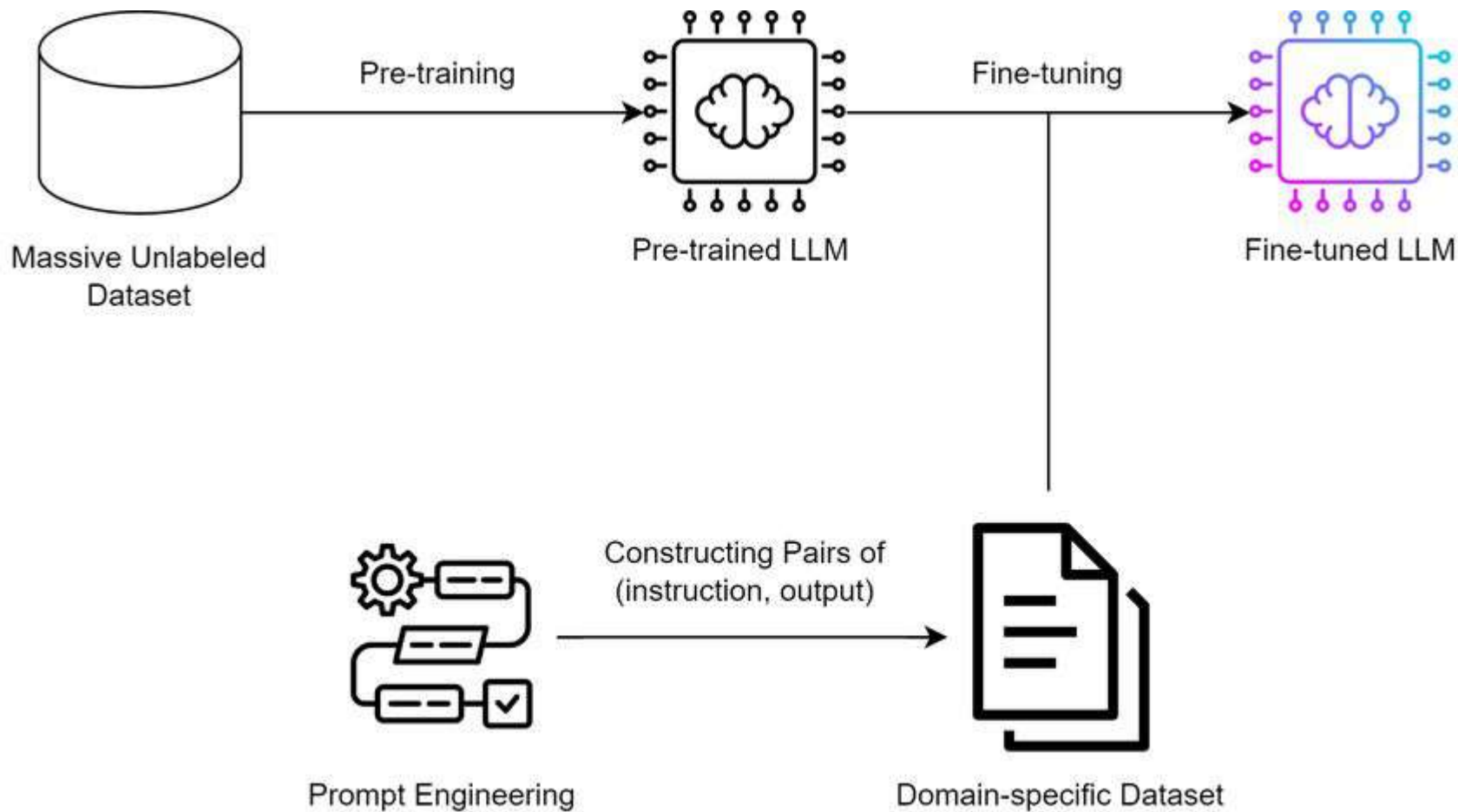
- Create text-only outputs
- Myriad of parameters to understand and produce text
- Foundation for text-based generative AI tools like ChatGPT

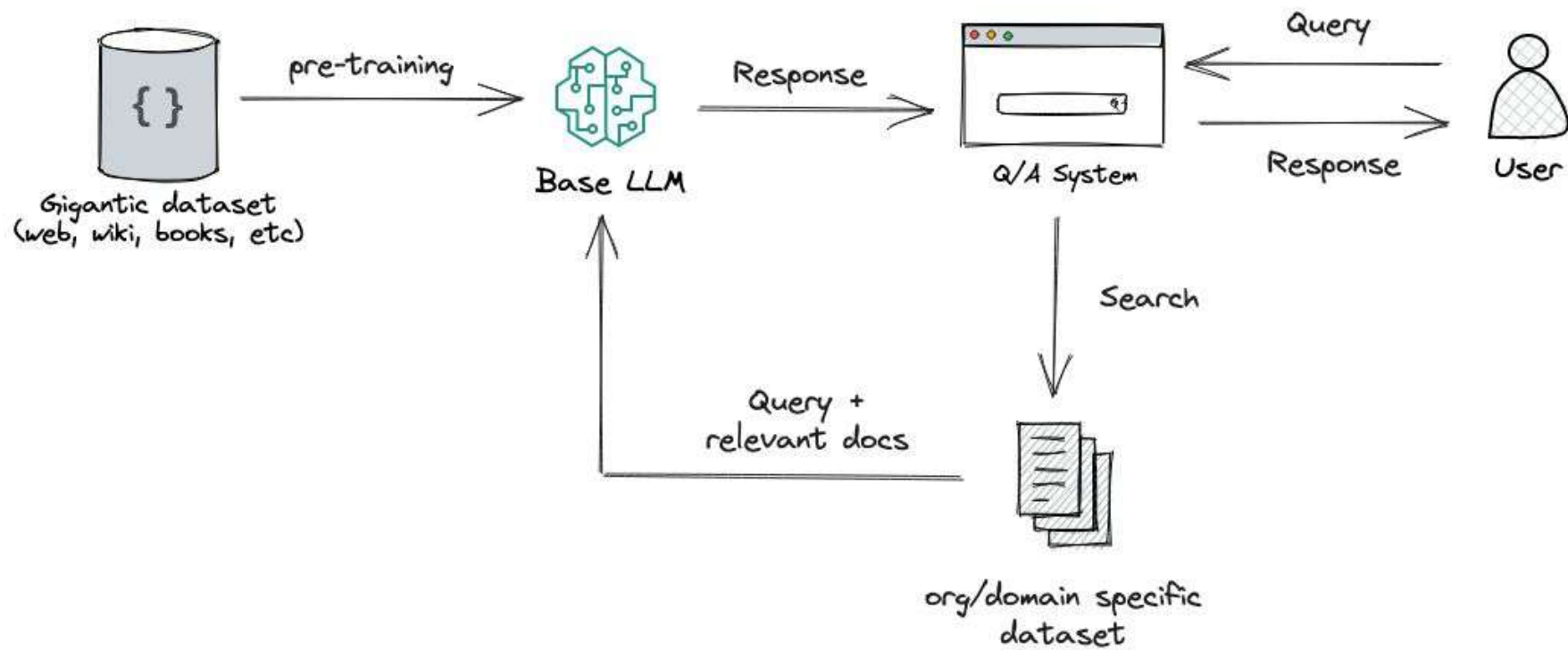


LLM fine-tuning at a high level

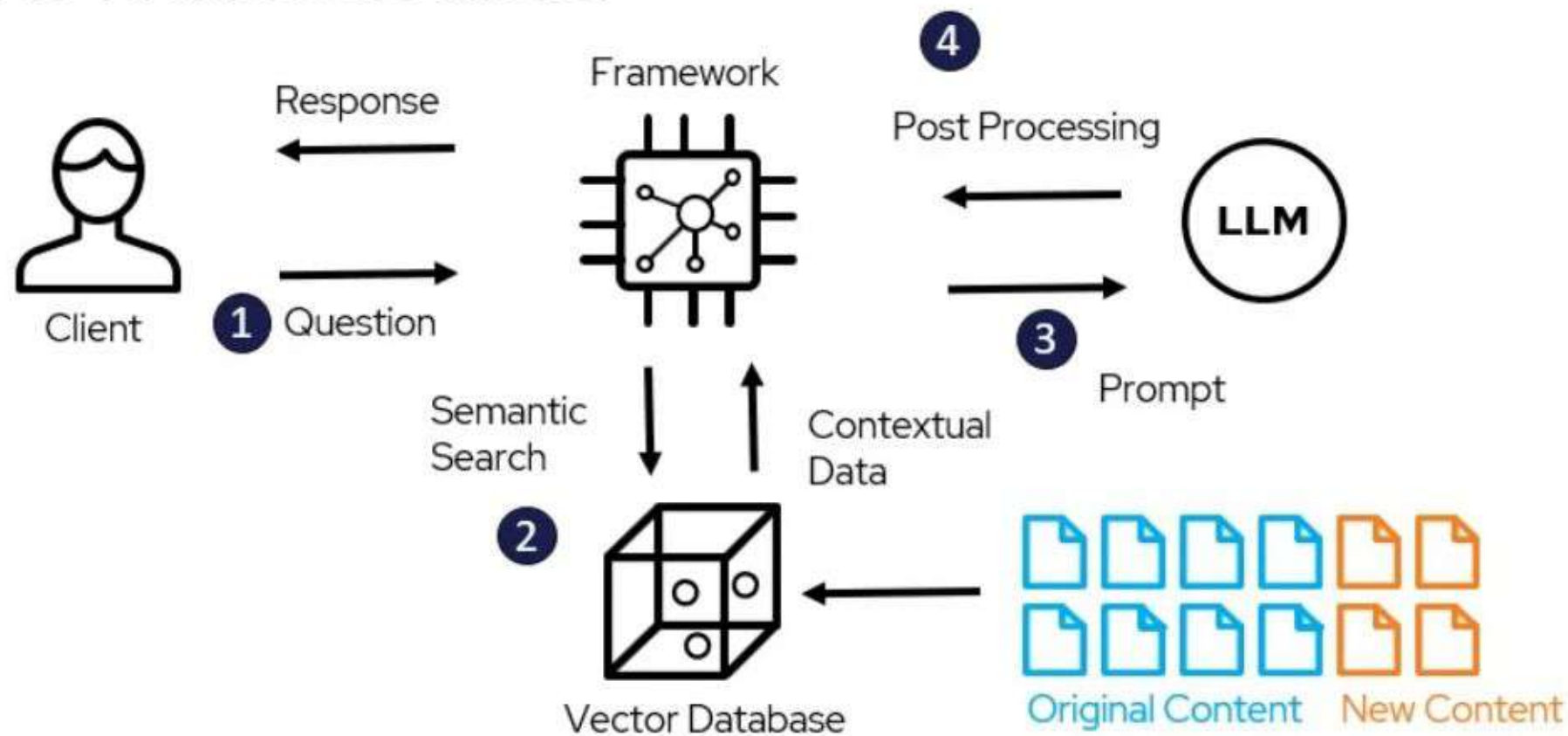
LLM fine-tuning

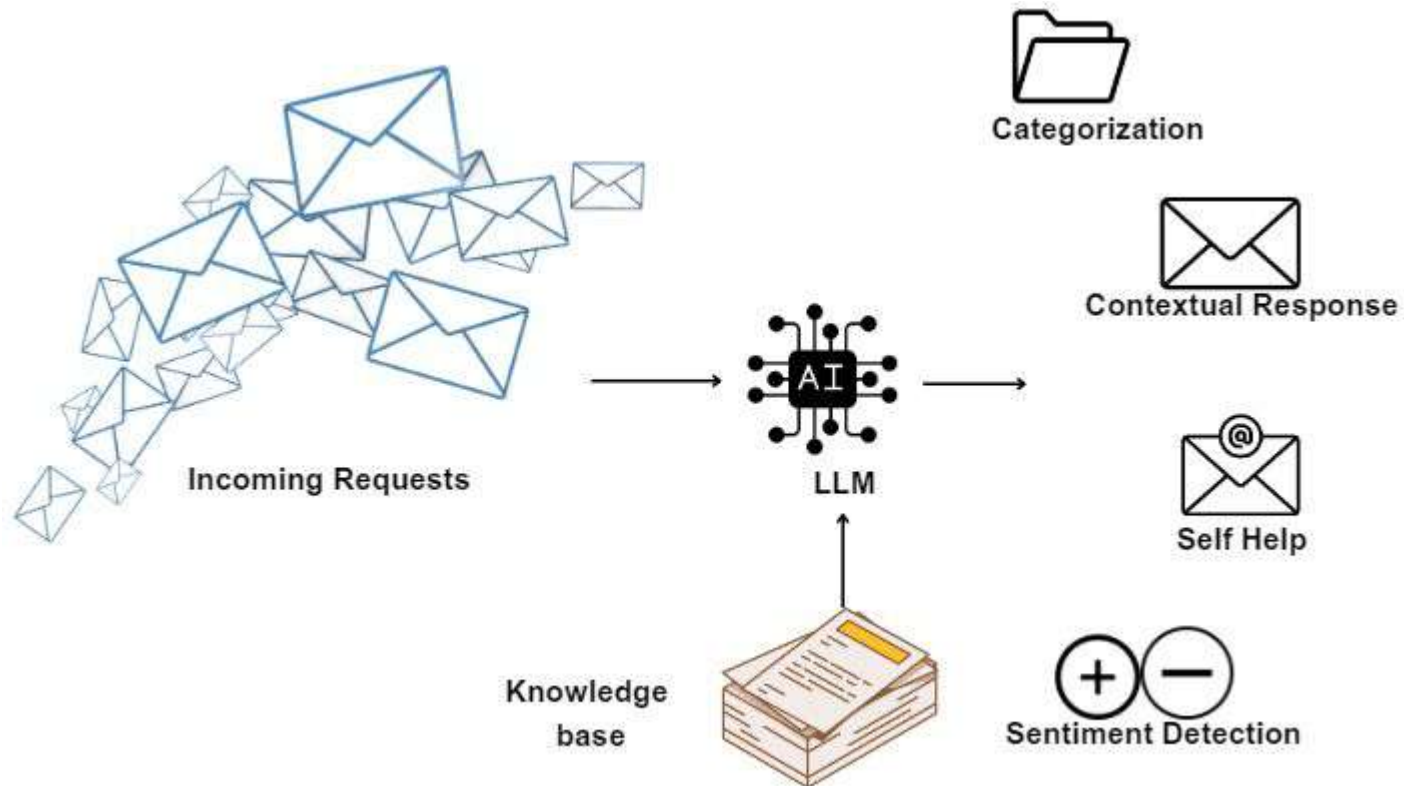


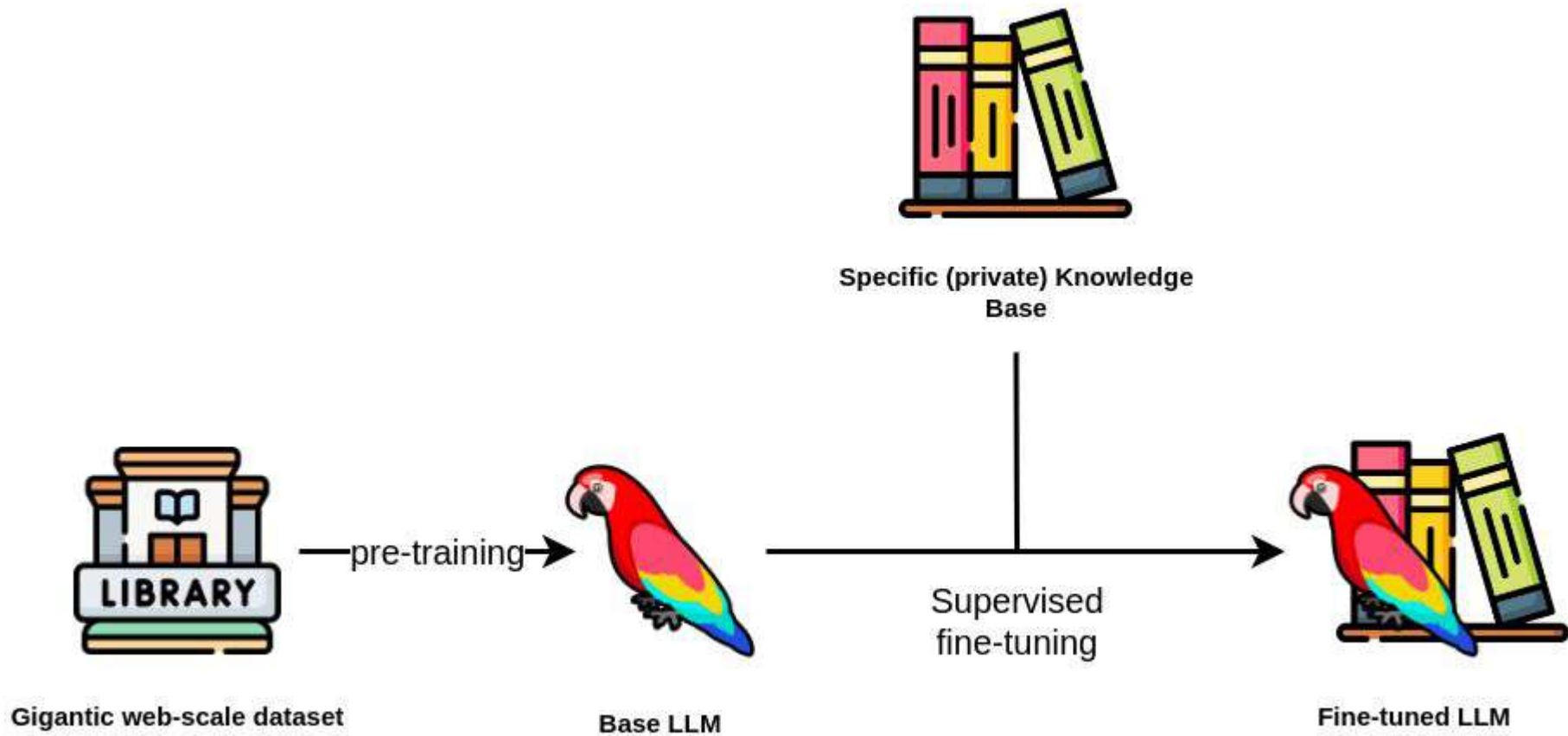


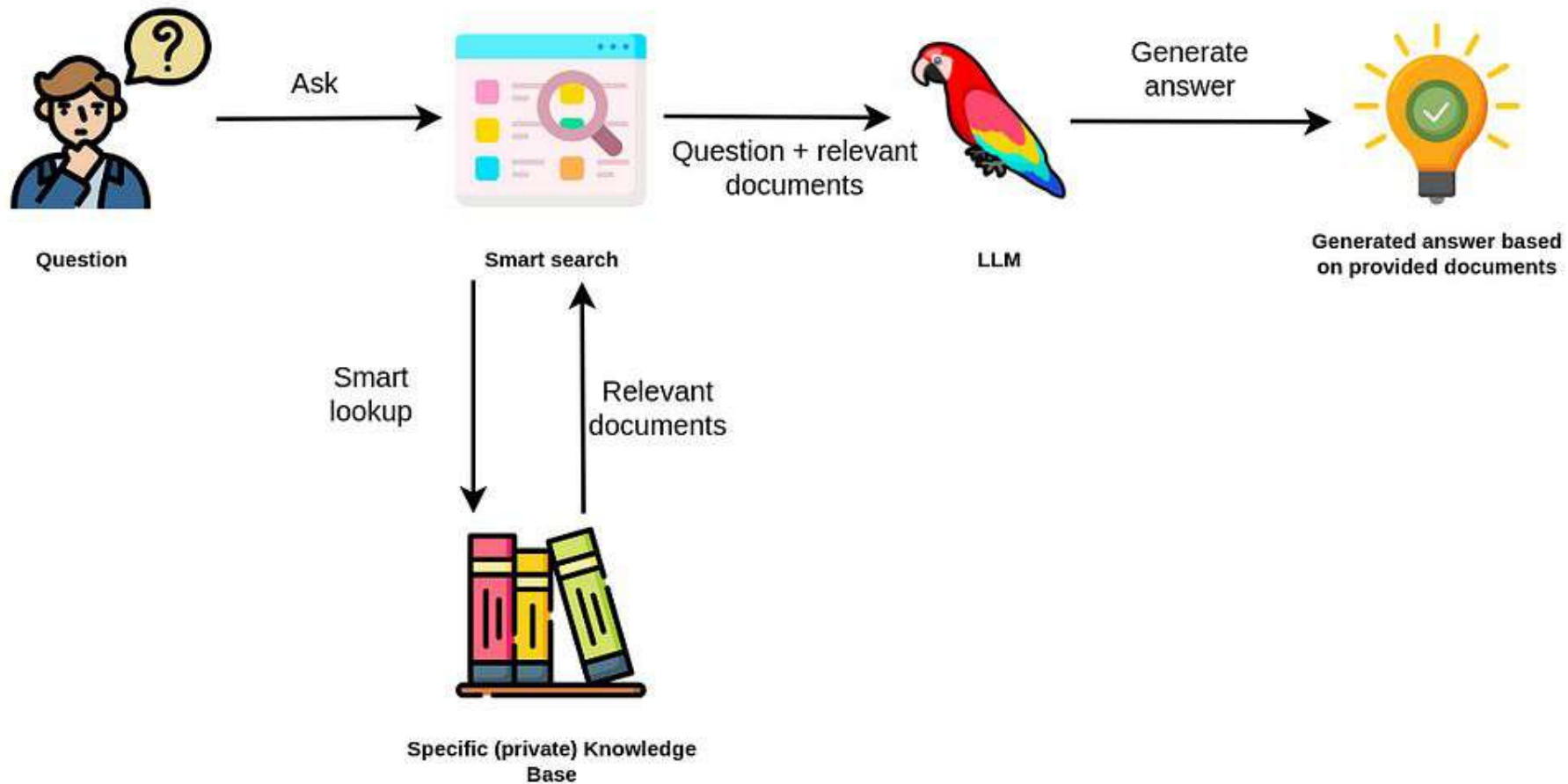


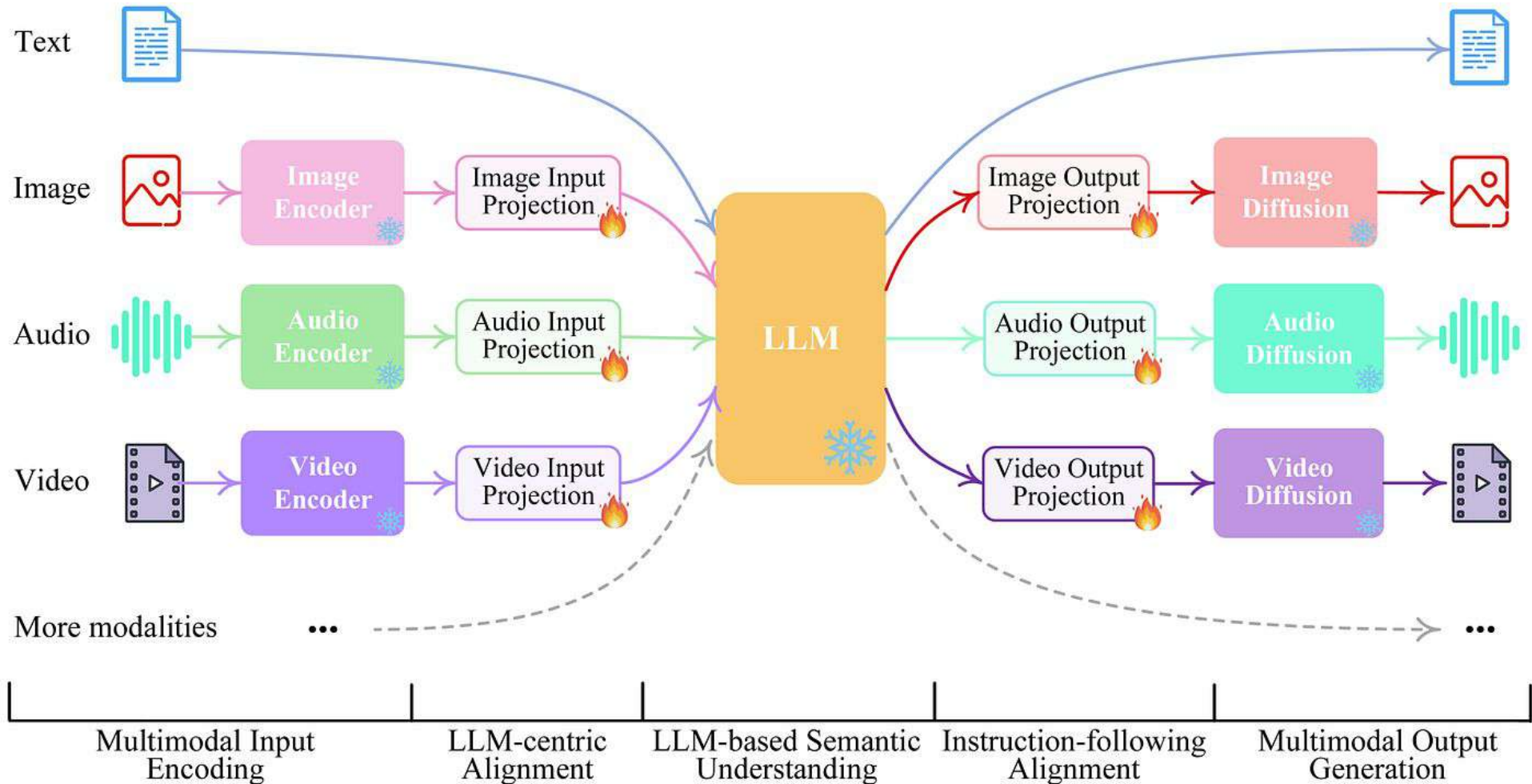
RAG Architecture Model





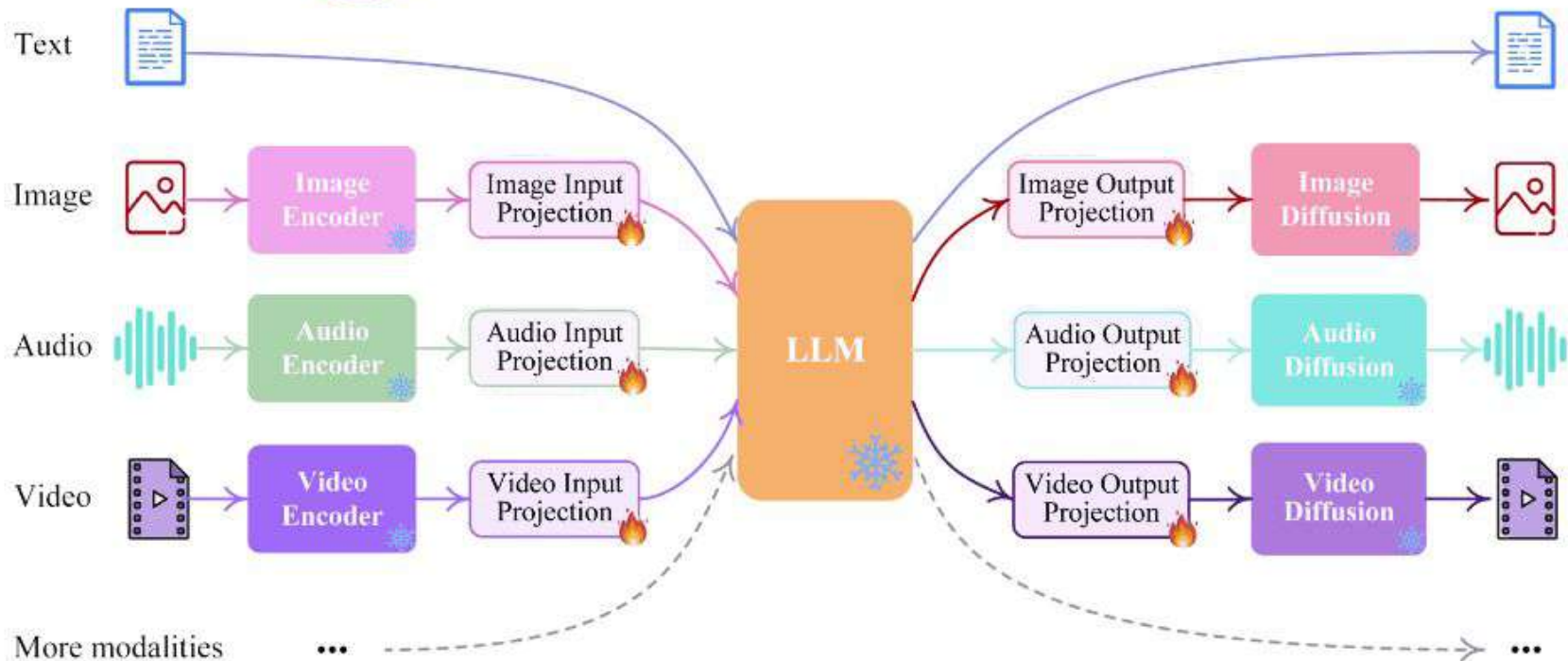


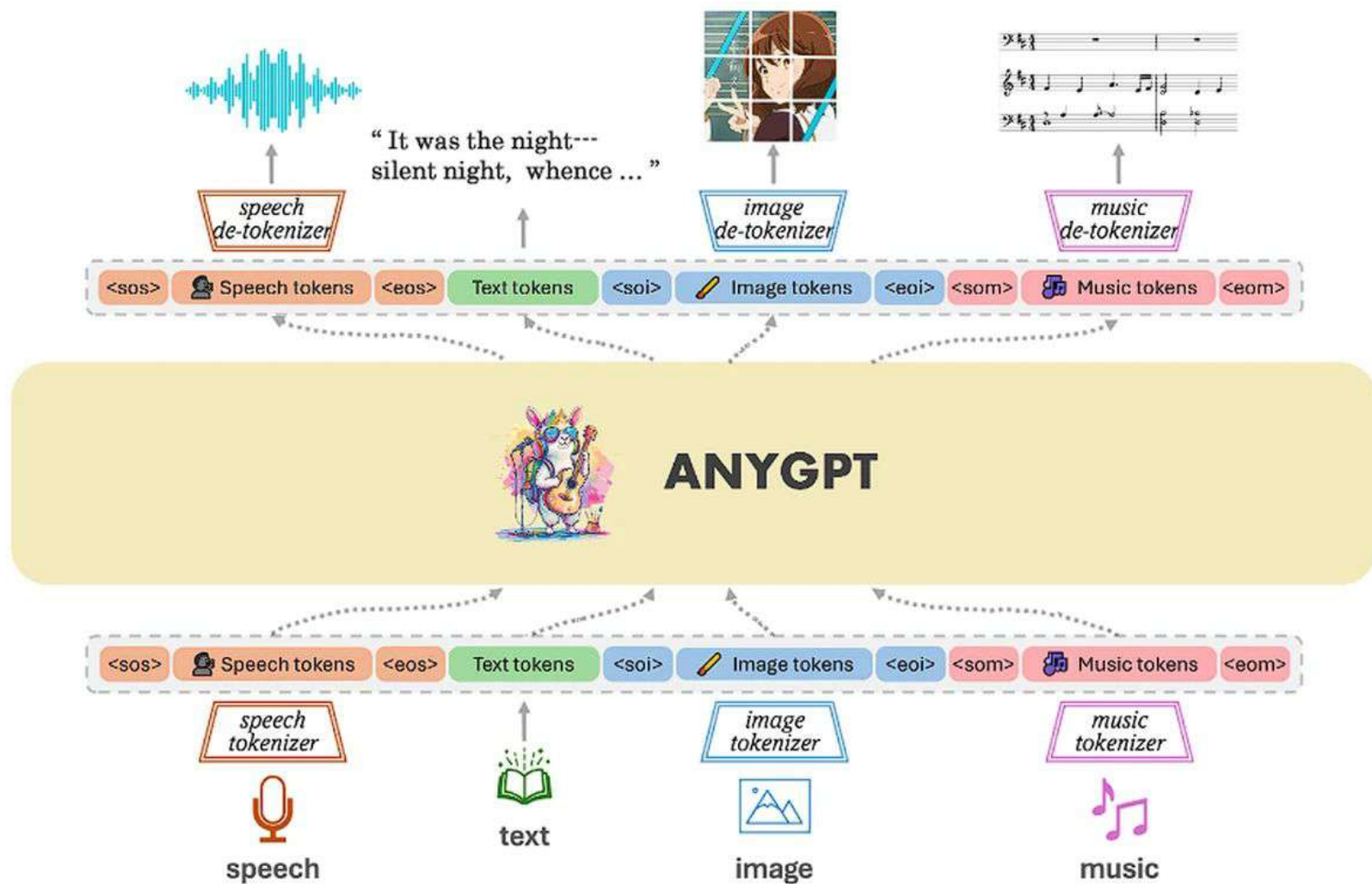




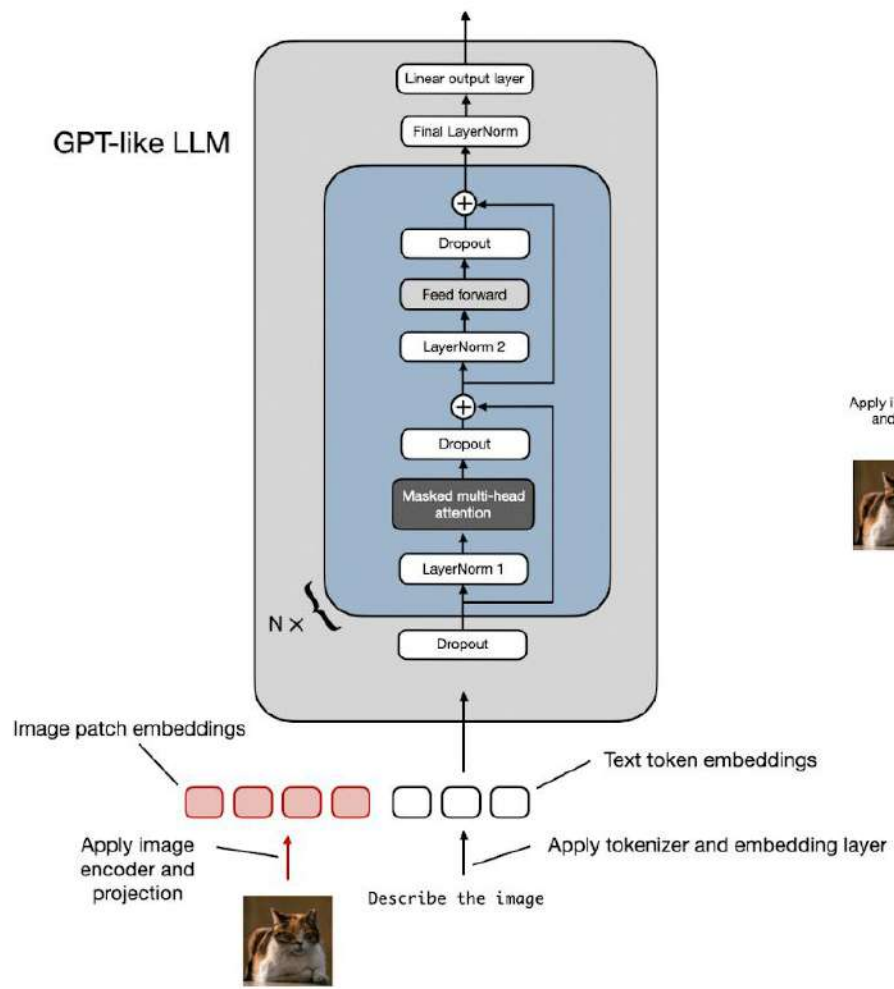


NExT-GPT

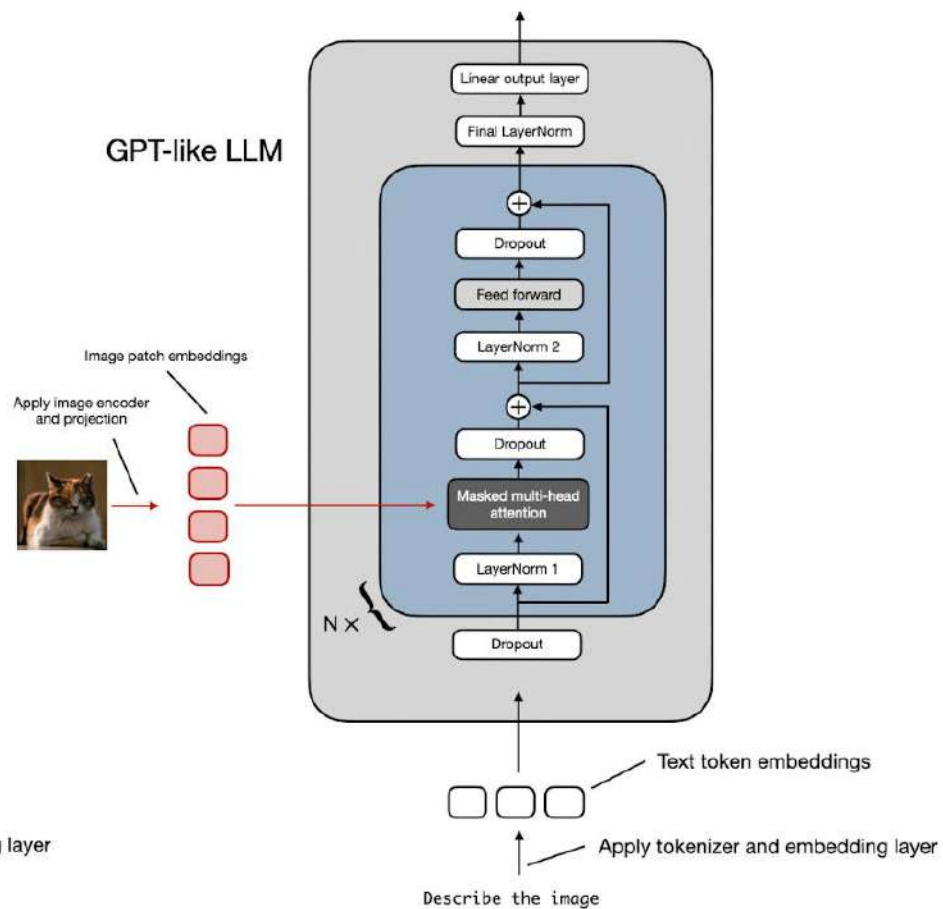


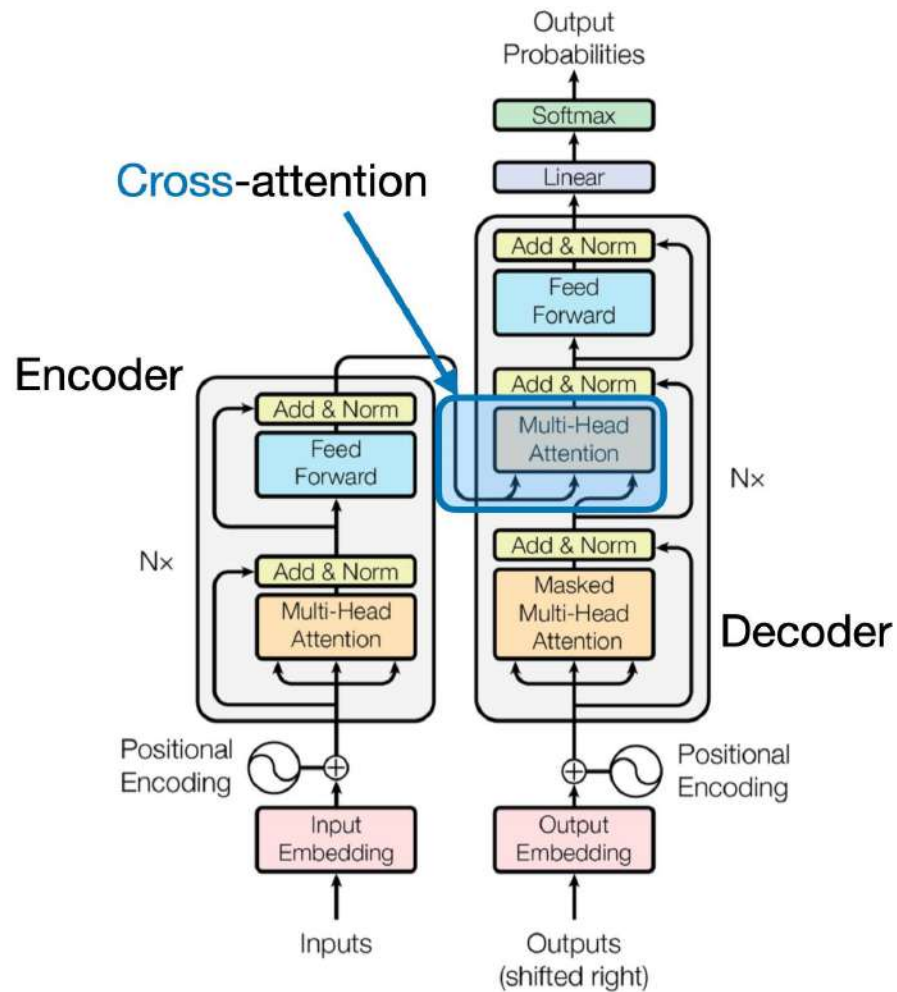


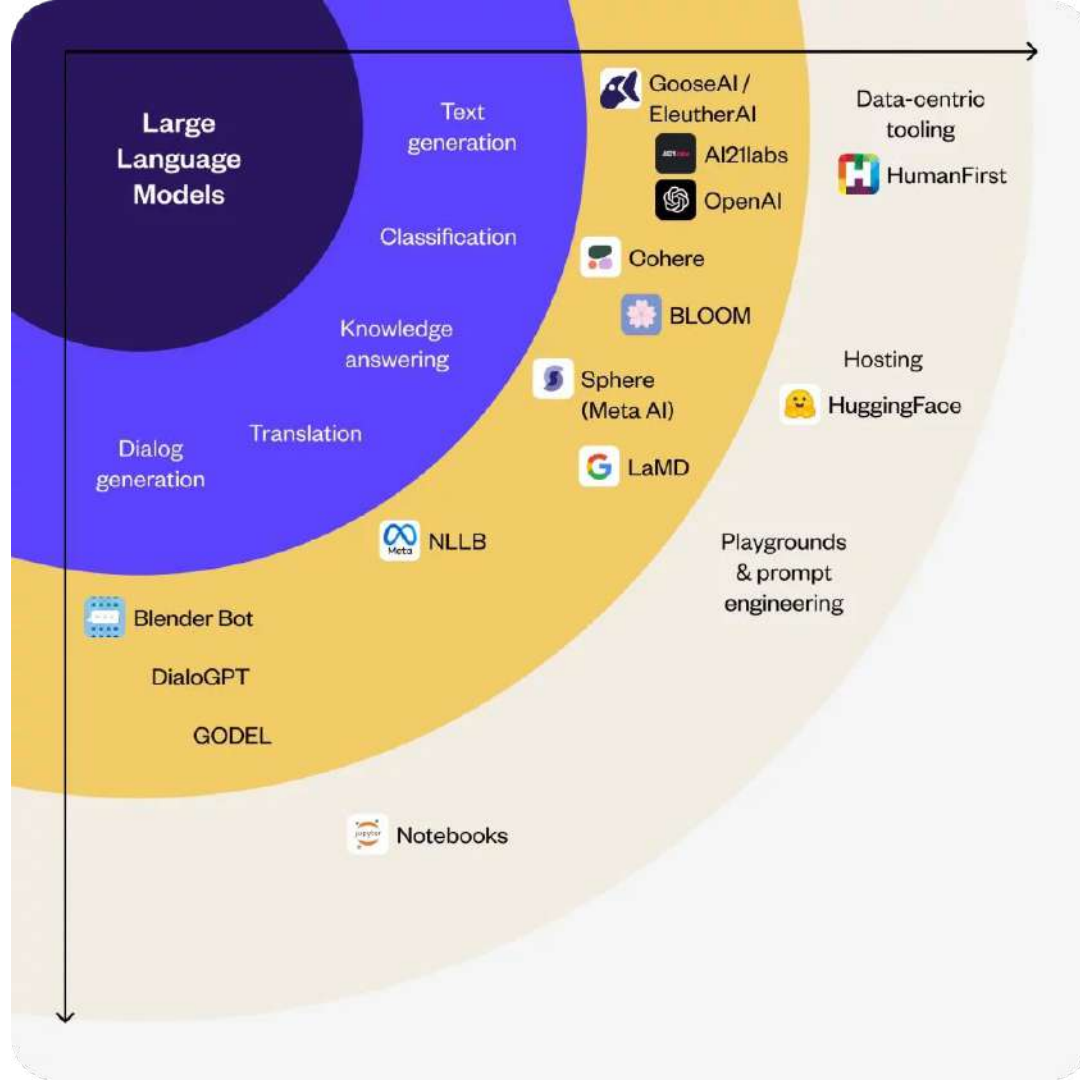
Method A: Unified Embedding Decoder Architecture



Method B: Cross-Modality Attention Architecture







Large Language Models

BCV

Application Layer

Copywriting



Coding



Dev Tools



Chat / Comms



BizOps



Infrastructure Layer

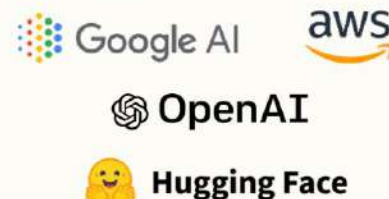
Model Creation



Hardware



Fine Tuning



Inference



Google Cloud Overview Solutions Products Pricing > Q

JUMP TO

Topics > AI Hallucinations

What are AI hallucinations?

AI hallucinations are incorrect or misleading results that [AI models](#) generate. These errors can be caused by a variety of factors, including insufficient training data, incorrect assumptions made by the model, or biases in the data used to train the model. AI hallucinations can be a problem for AI systems that are used to make important decisions, such as medical diagnoses or financial trading.

Con

When A.I. Chatbots Hallucinate

Share full article



By Karen Weise and Cade Metz

Karen Weise reported this story from Seattle and Cade Metz reported from San Francisco. Published May 1, 2023 Updated May 9, 2023

Opinion | Honestly, I love when AI hallucinates



By Josh Tyrangiel
Columnist

December 27, 2023 at 7:00 a.m. EST

NEXT GEN INVESTING

AI chatbots can 'hallucinate' and make things up—why it happens and how to spot it

Published Fri, Dec 22 2023 11:55 AM EST

IBM Products Solutions Consulting Support

Think Artificial intelligence Cloud Security Sustainability

What are AI hallucinations?

Discover watsons governance →

zapier

App picks Automation with Zapier Productivity Business growth Product & platform

What are AI hallucinations and how do you prevent them?

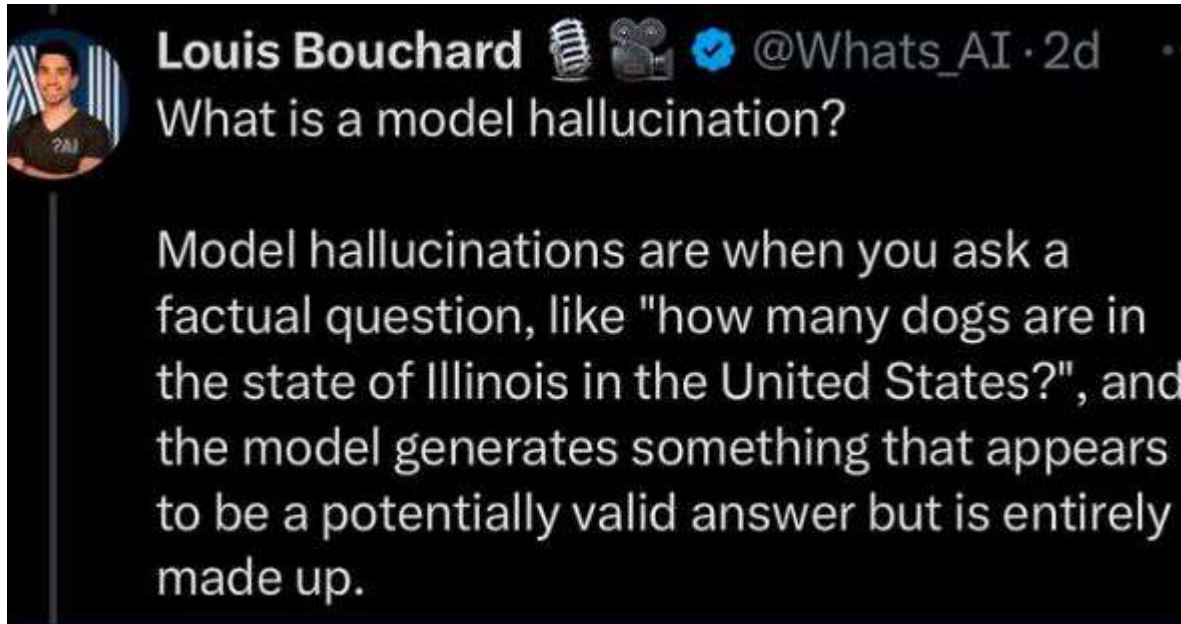
Here's how to encourage AI to stop hallucinating.

AP

de report US election Royal family sun

Chatbots sometimes make things up. Is AI's hallucination problem fixable?

AI Hallucinations





You

what is 49338/293



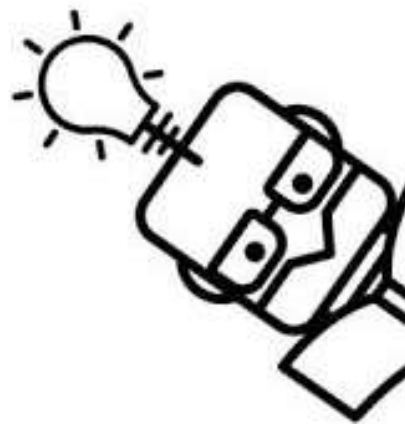
ChatGPT Classic

To find the result of $\frac{49338}{293}$, you perform the division:

$$\frac{49338}{293} \approx 168.441$$

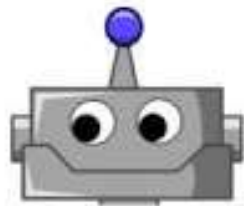
If you put this into any pocket calculator you'll find that it's (astonishingly) close, but wrong.

WHY LANGUAGE MODELS HALLUCINATE?



How is the weather today?

Weather is the state of the atmosphere at a particular place and time as regards heat, cloudiness, dryness, sunshine, wind, rain, etc.



Applications of Artificial Intelligence

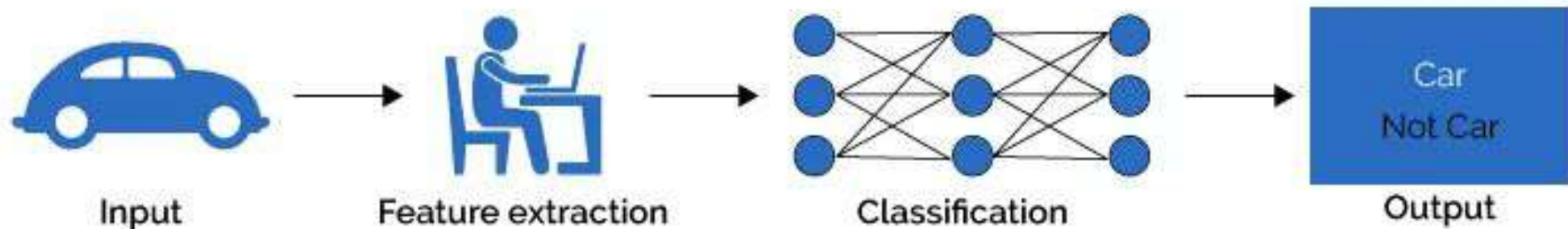
- Machine Translation such as Google Translate
- Self Driving Vehicles such as Google's Waymo
- AI Robots such as Sophia and Aibo
- Speech Recognition applications like Apple's Siri or OK Google



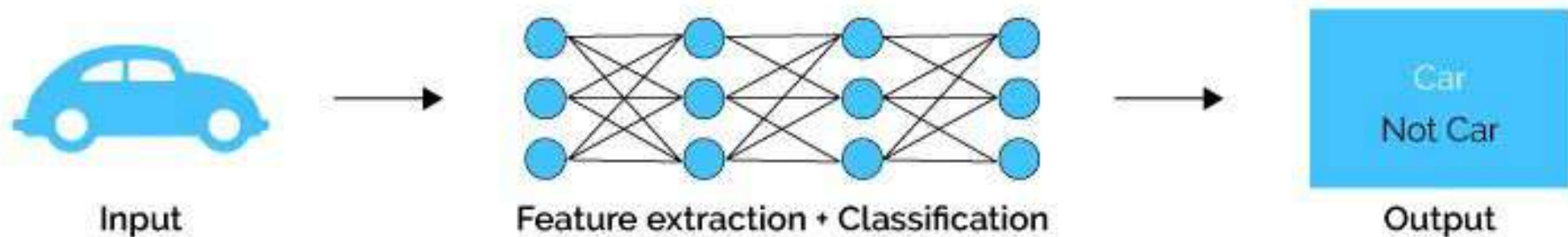
<https://www.youtube.com/watch?v=kJoAcEI2PXQ&t=14s>



Machine Learning



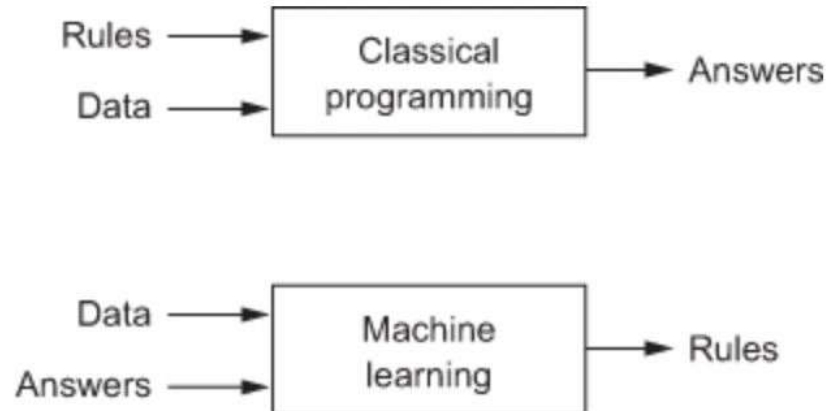
Deep Learning



What is Deep Learning?

Deep Learning is a subset of Machine Learning that uses mathematical functions to map the input to the output.

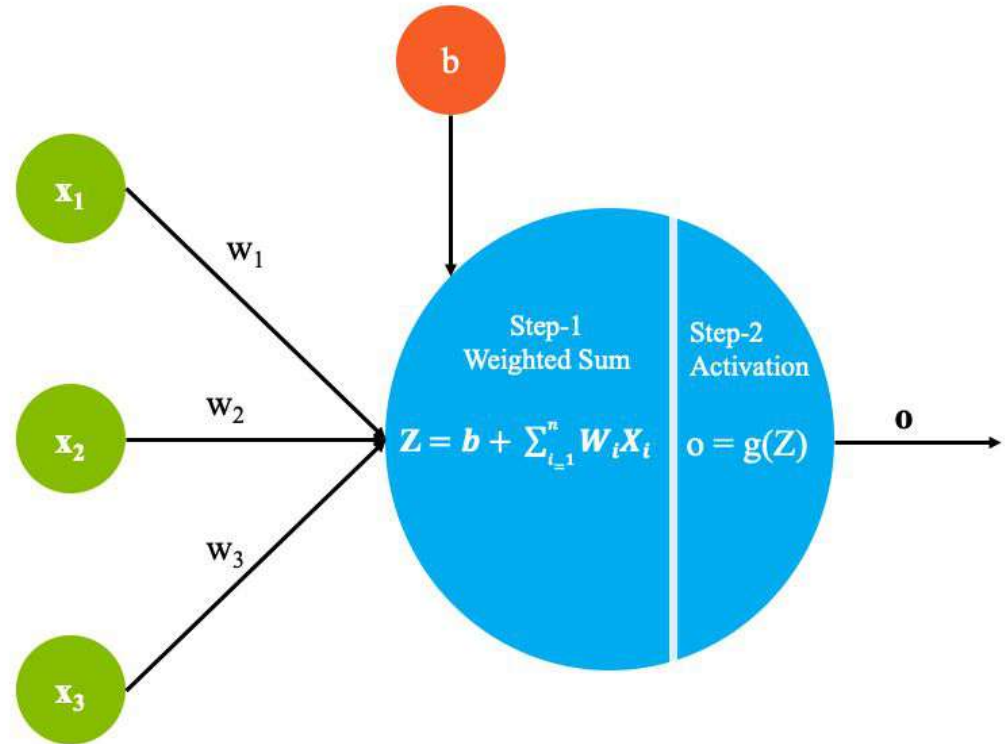
These functions can extract non-redundant information or patterns from the data, which enables them to form a relationship between the input and the output.



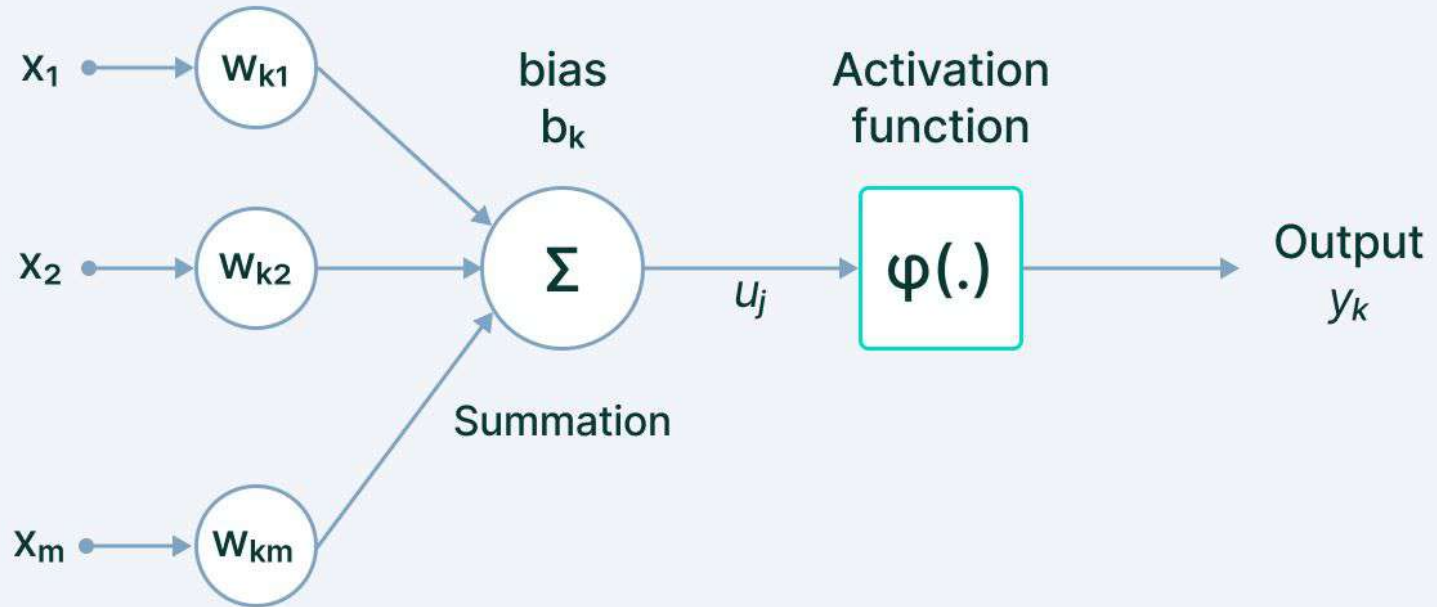
Artificial Neural Network (ANN)

This section provides an overview of the architecture behind deep learning, artificial neural networks (ANN), and discusses some of the key terminology.

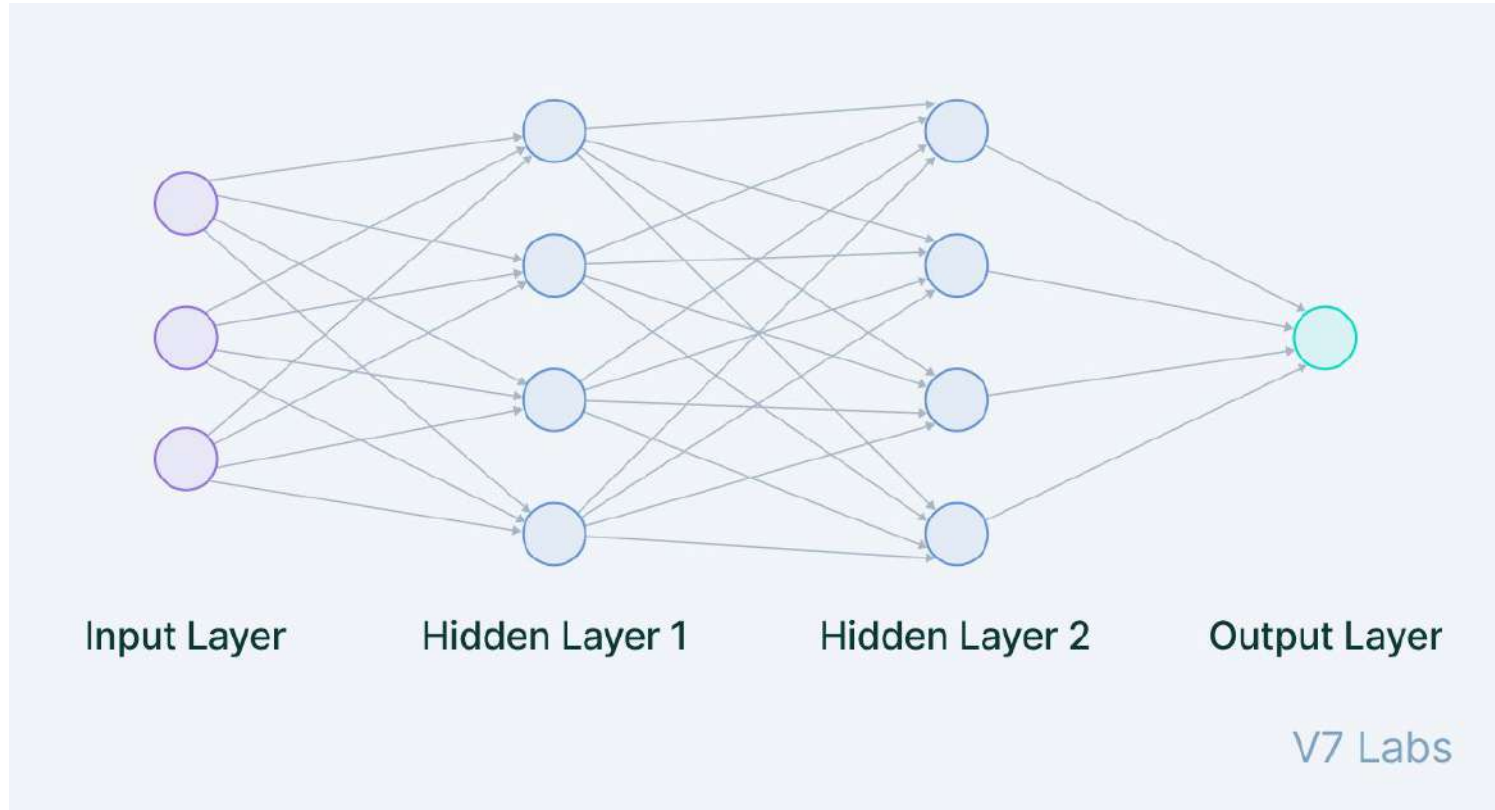
As shown in the following figure, each perceptron is made up of the following parts:



Neuron



How does Deep Learning work?



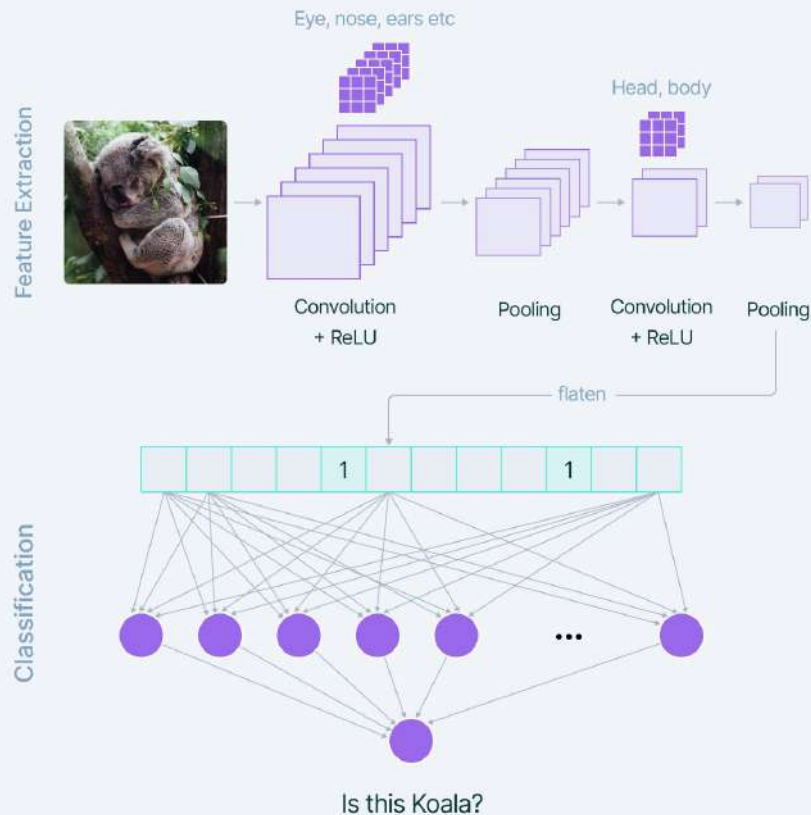
The Convolutional Neural Networks or CNNs

The Convolutional Neural Networks or CNNs are primarily used for tasks related to computer vision or image processing.

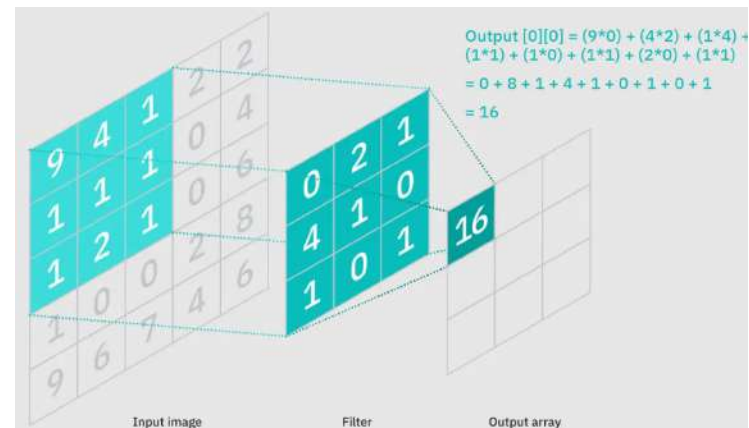
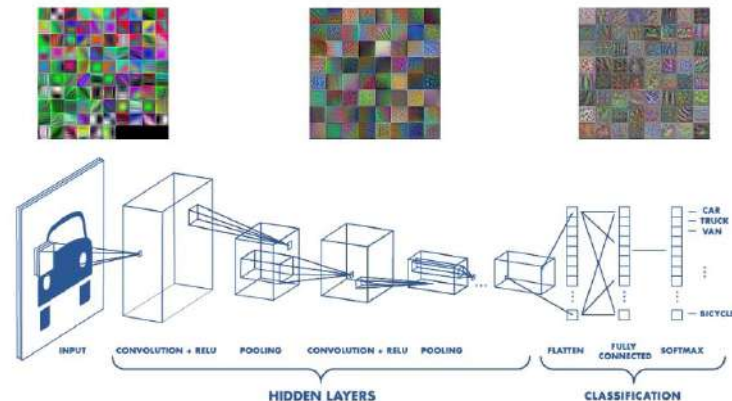
CNNs are extremely good in modeling spatial data such as 2D or 3D images and videos.

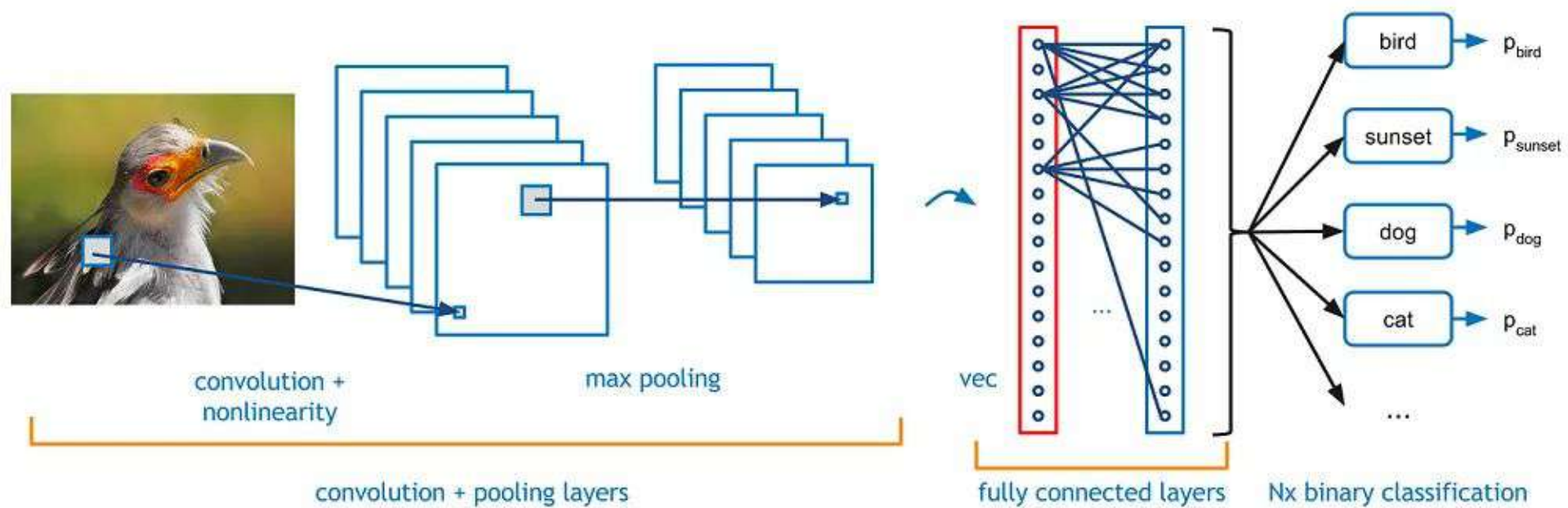
They can extract features and patterns within an image, enabling tasks such as image classification or object detection.

Convolutional Neural Networks

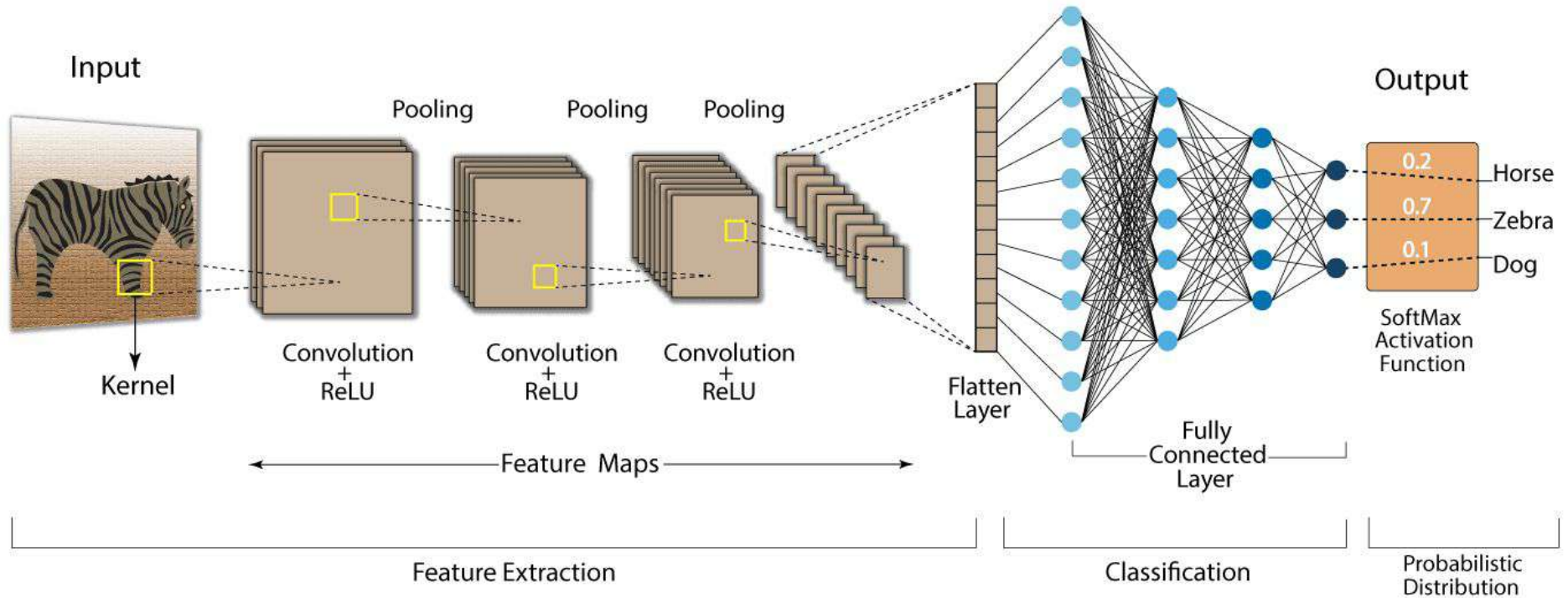


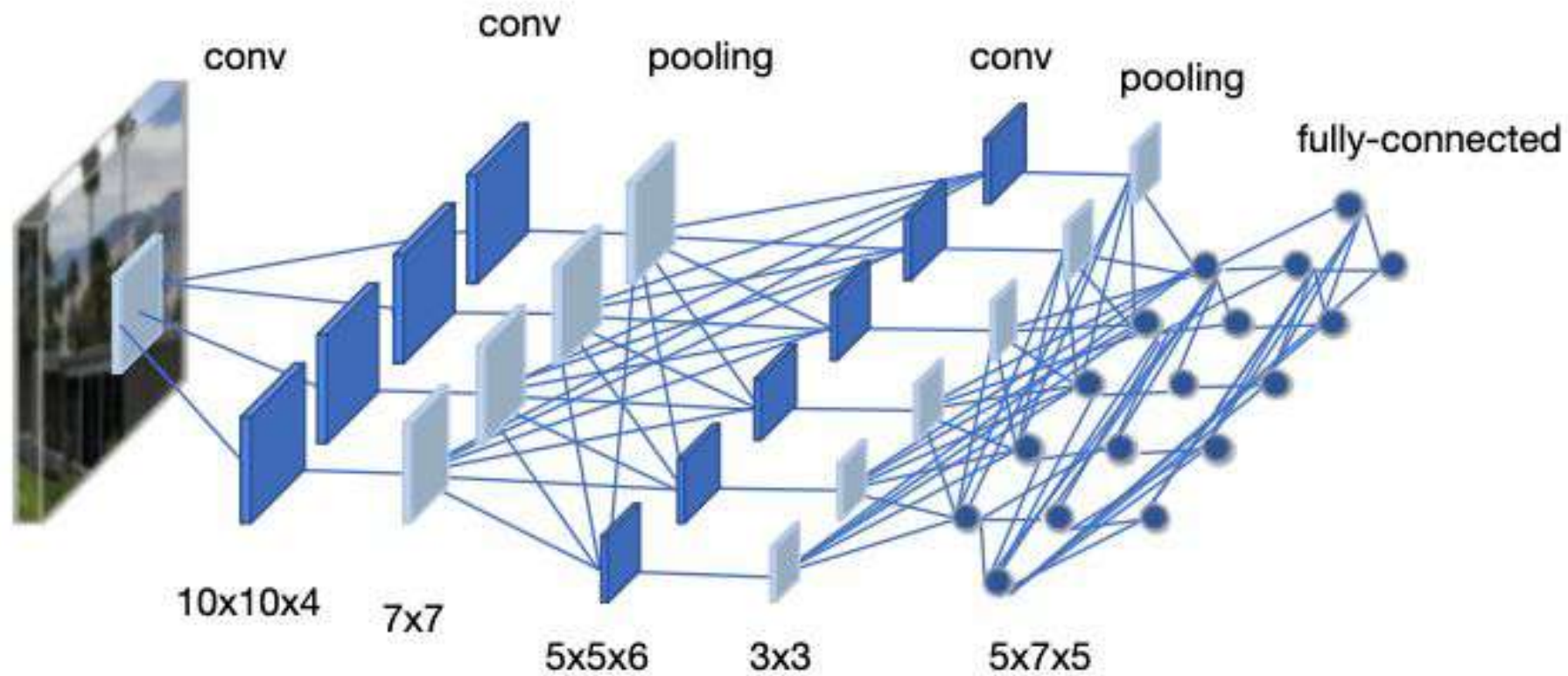
Convolutional Neural Network





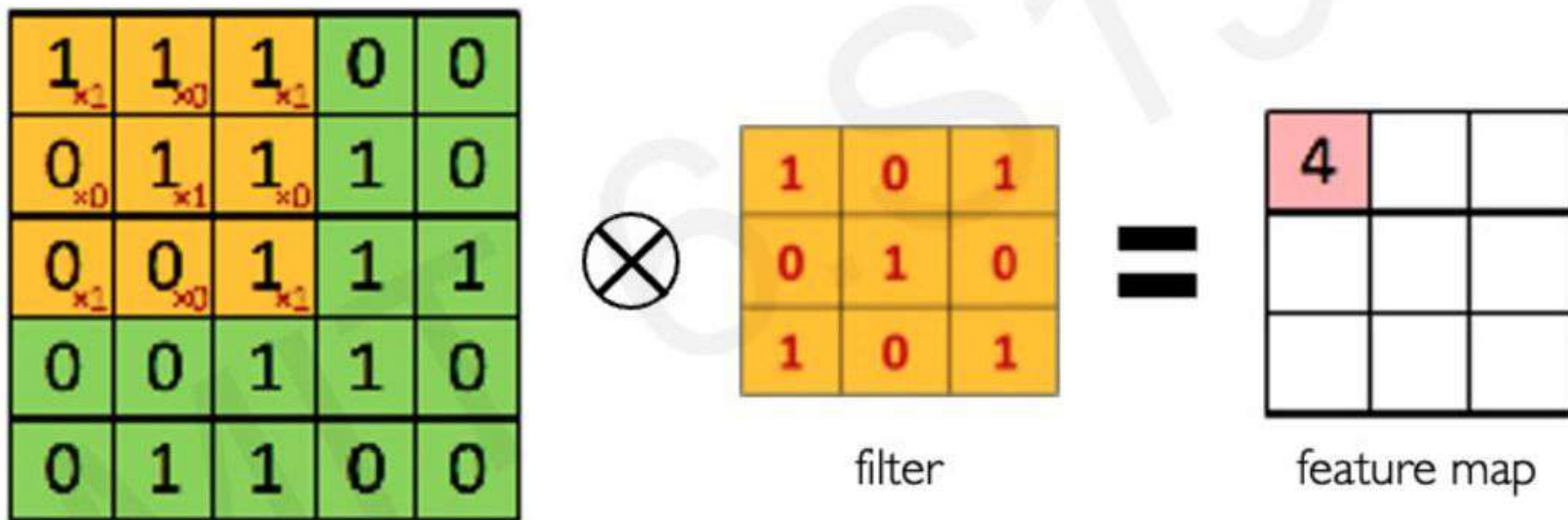
Convolution Neural Network (CNN)

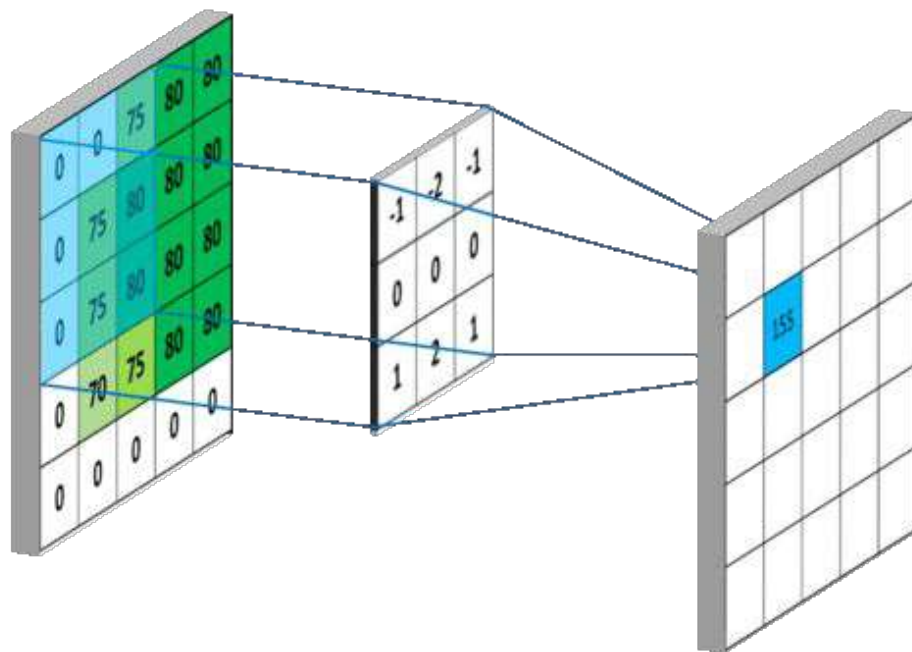




The Convolution Operation

We slide the 3x3 filter over the input image, element-wise multiply, and add the outputs:





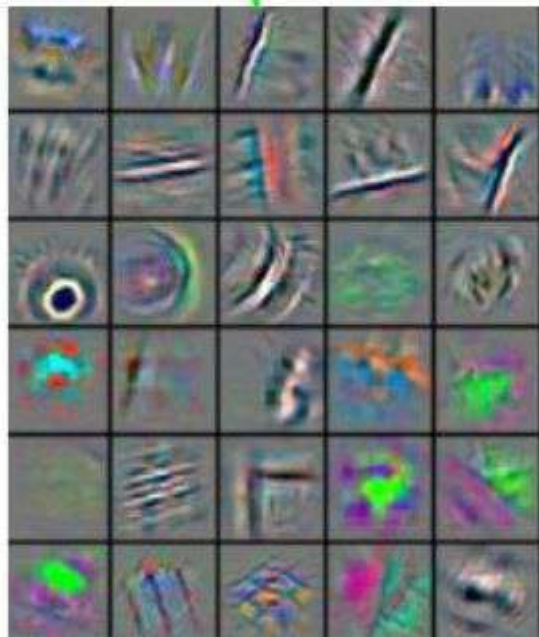


Low-Level
Feature

Mid-Level
Feature

High-Level
Feature

Trainable
Classifier

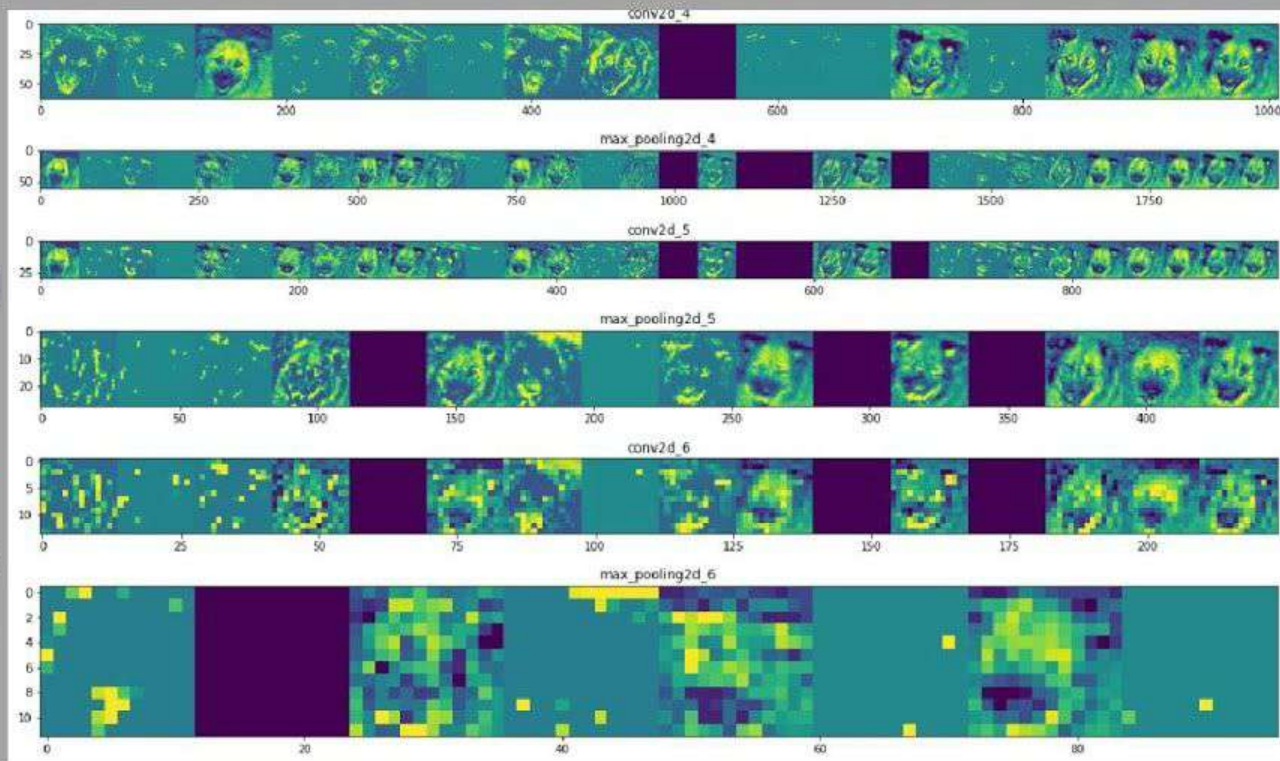


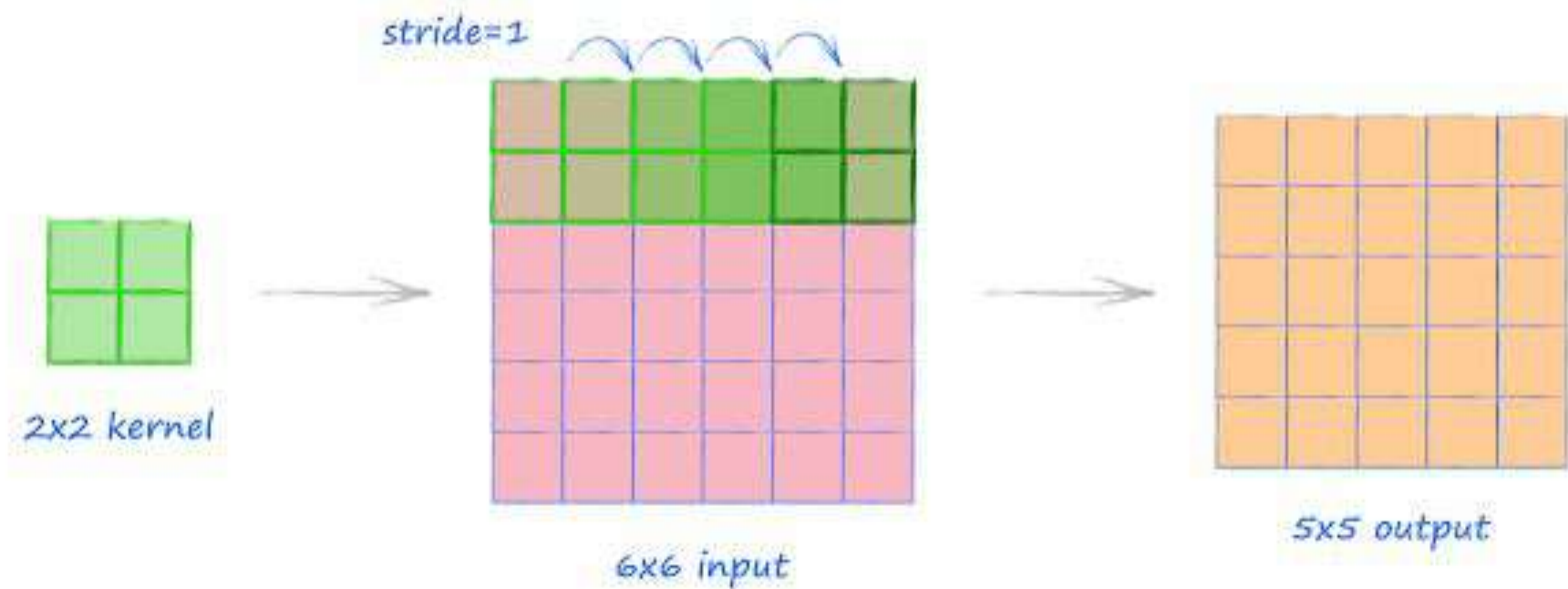


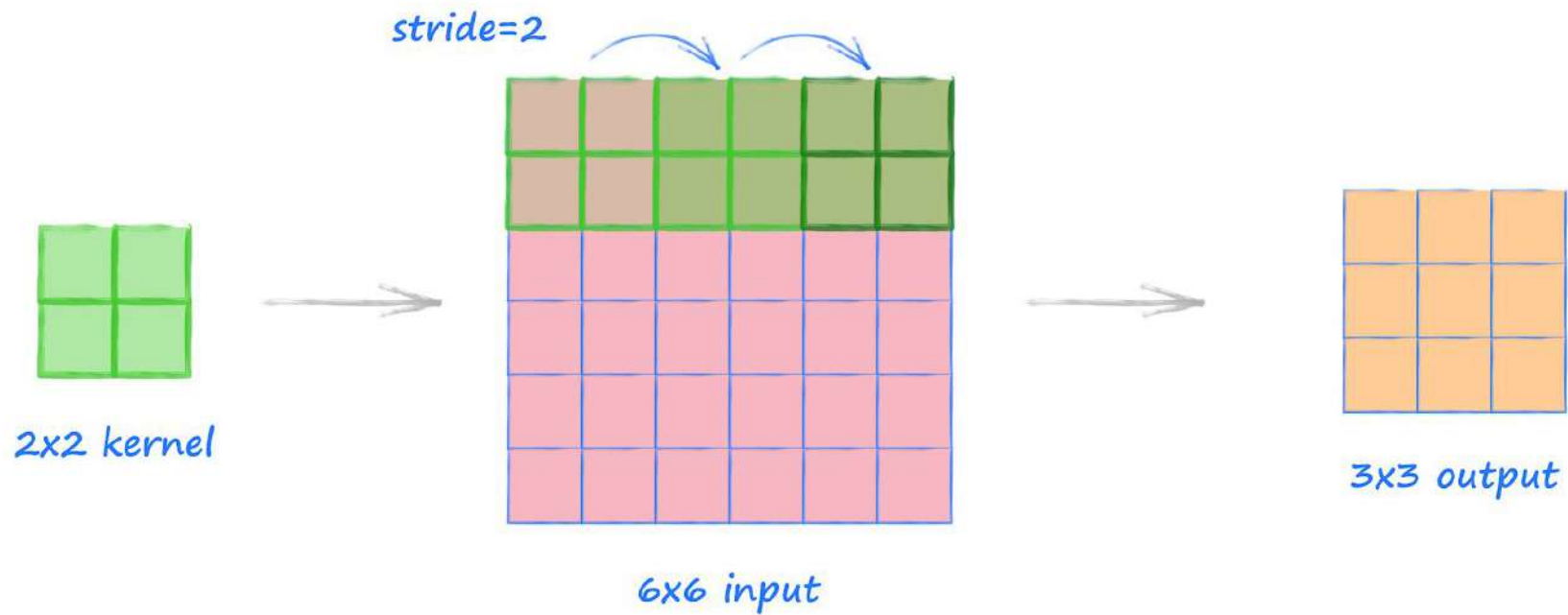
Visualize
Feature
Maps of CNN
Layers to
Understand

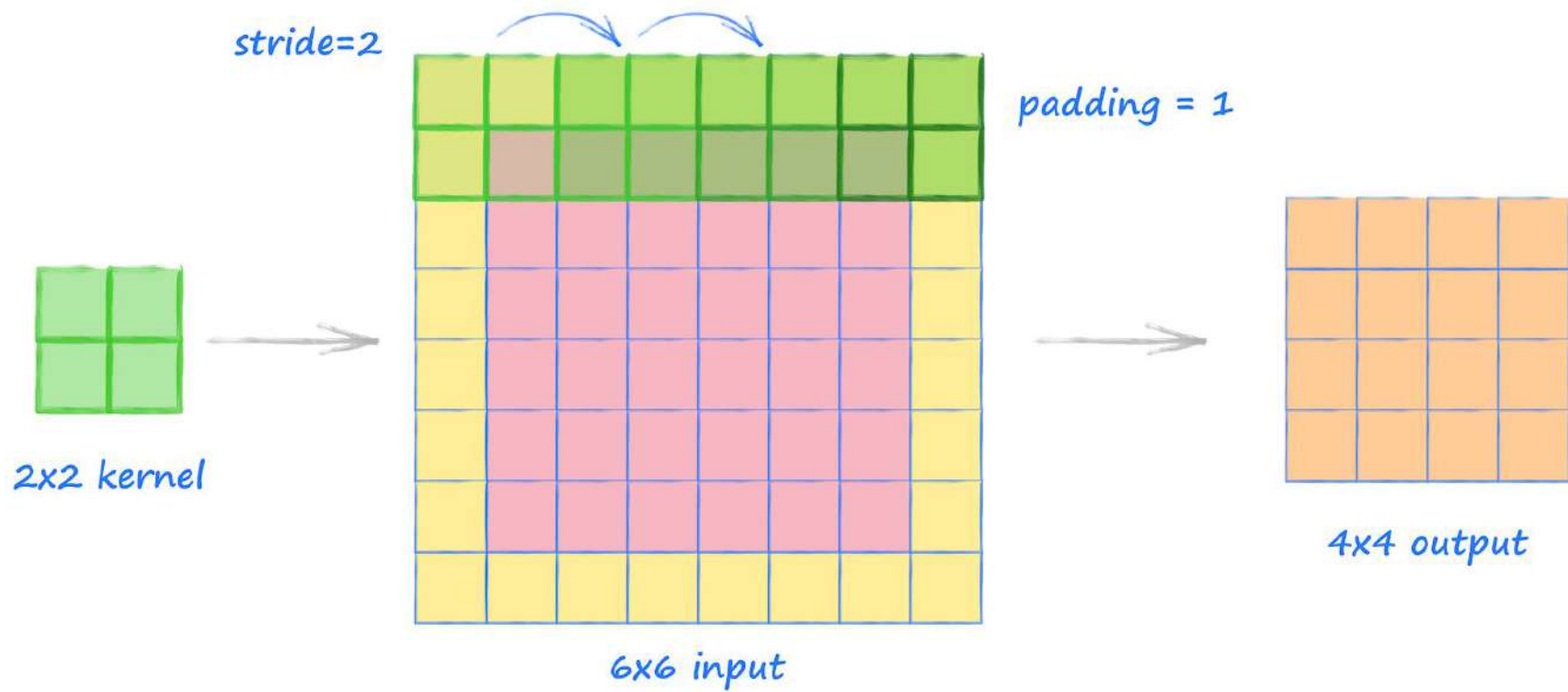


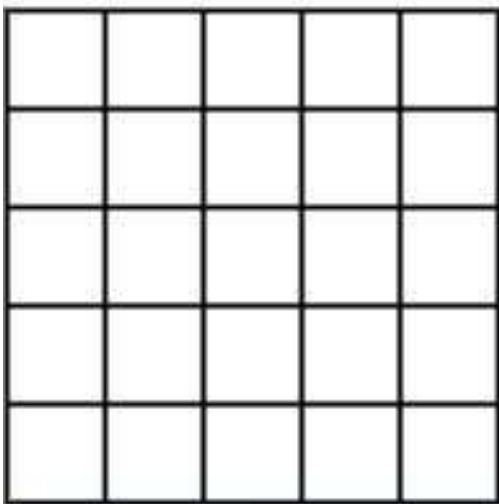
What Layers
are Learning



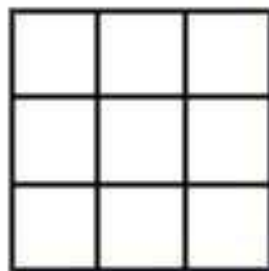






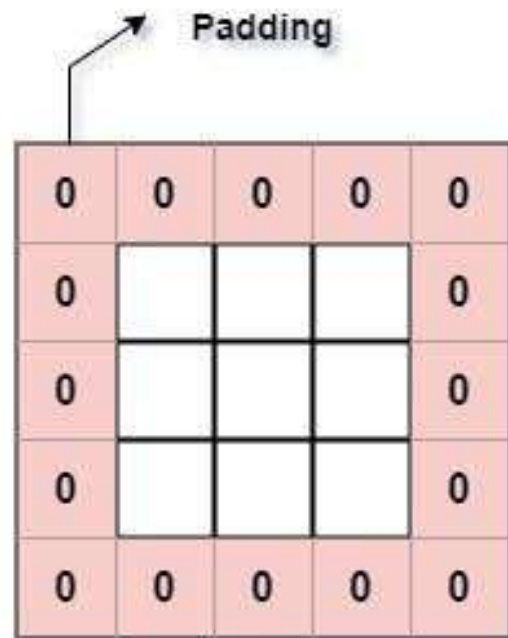


Original Image: 5 x 5



Filter: 3 x 3

Same Padding

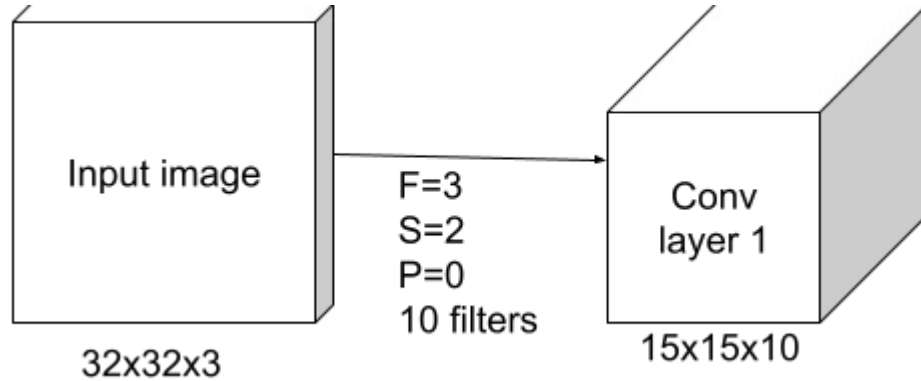


Feature Map: 5 x 5

Calculating Output Sizes

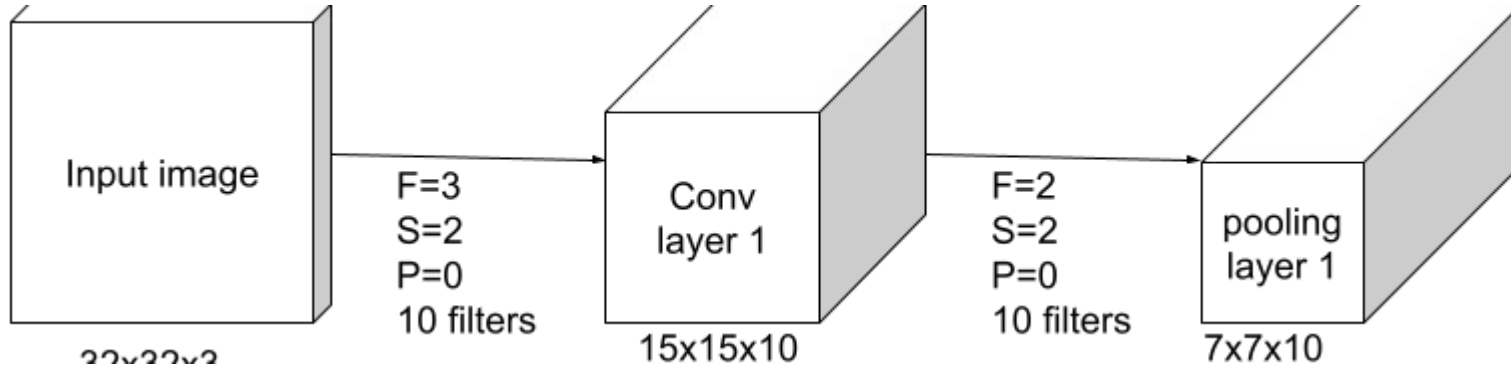
$$\text{output size} = \left\lfloor \frac{(\text{input size}) + 2 * \text{padding} - (\text{kernel size} - 1) - 1}{\text{stride}} \right\rfloor + 1$$

Calculate Output Size of Convolutional and Pooling layers



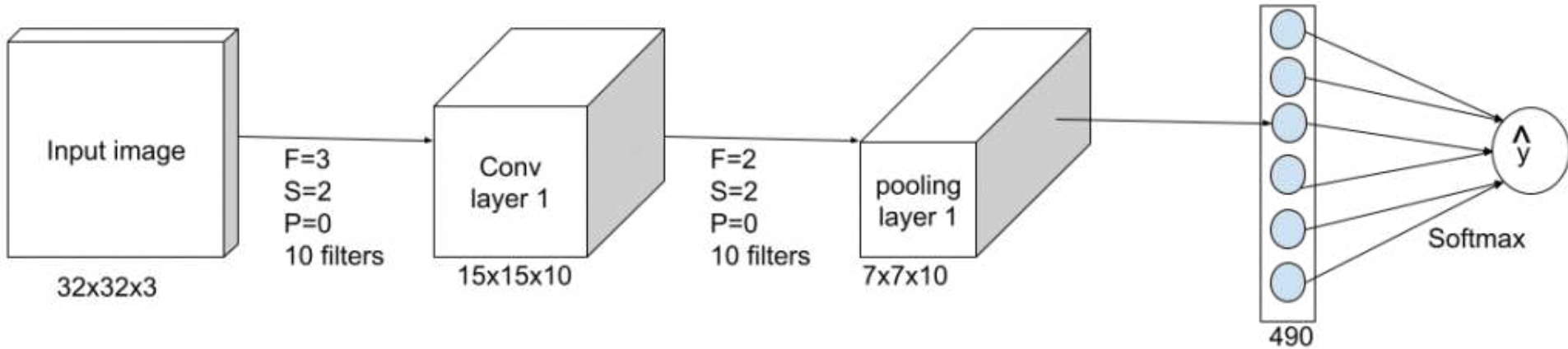
$$\text{Floor}\left(\frac{(W-F+2P)}{S} + 1\right)$$

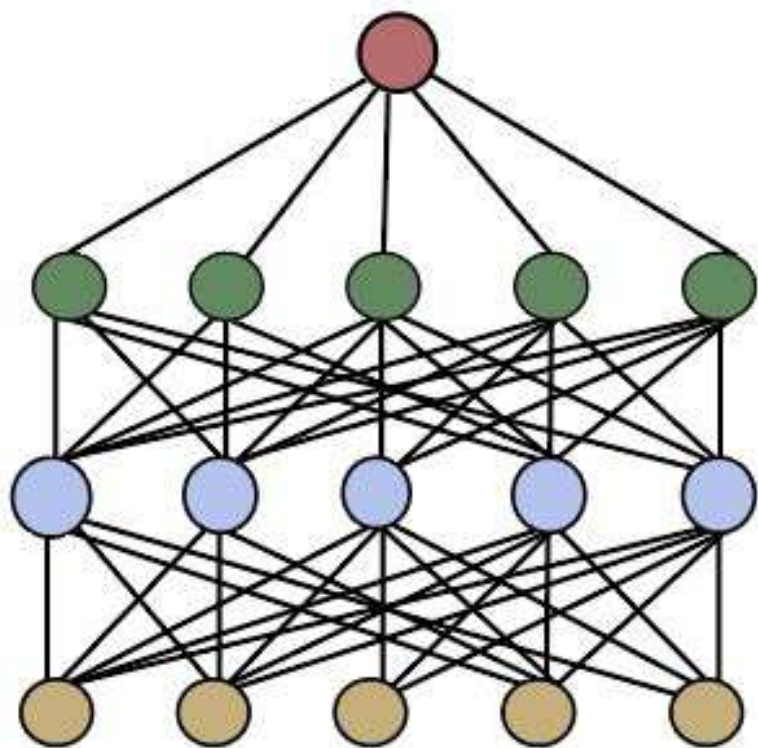
Calculate Output Size of Convolutional and Pooling layers



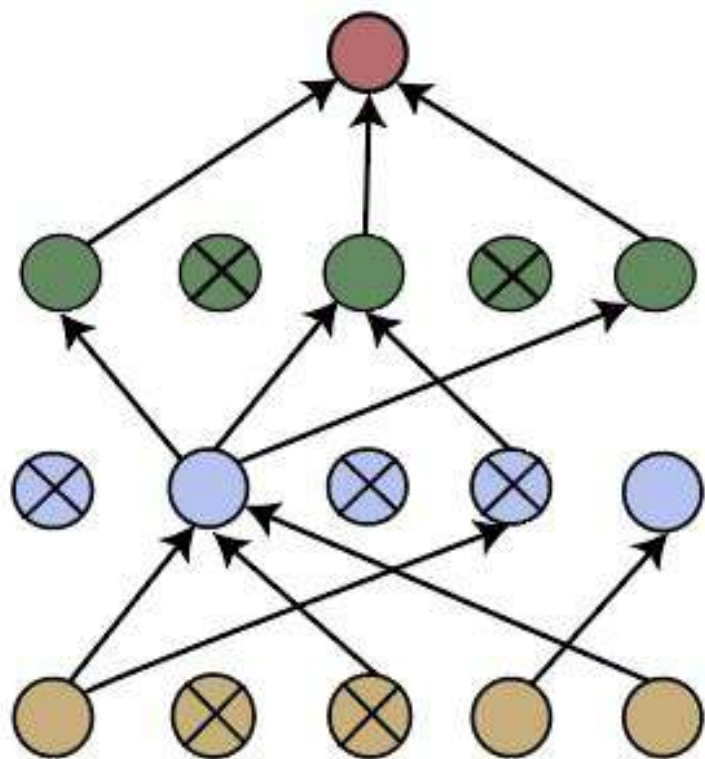
$$\text{Floor} \left(\frac{(W-F)}{S} + 1 \right)$$

Fully Connected layer





Standard Neural Network



After Applying Dropout

What is an Activation Function?

An Activation Function is a mathematical function used in Artificial Neural Networks (ANNs) to introduce non-linearity into the network. It determines whether a neuron should be activated (fired) or not based on the input. Activation functions help neural networks learn complex patterns and make decisions.

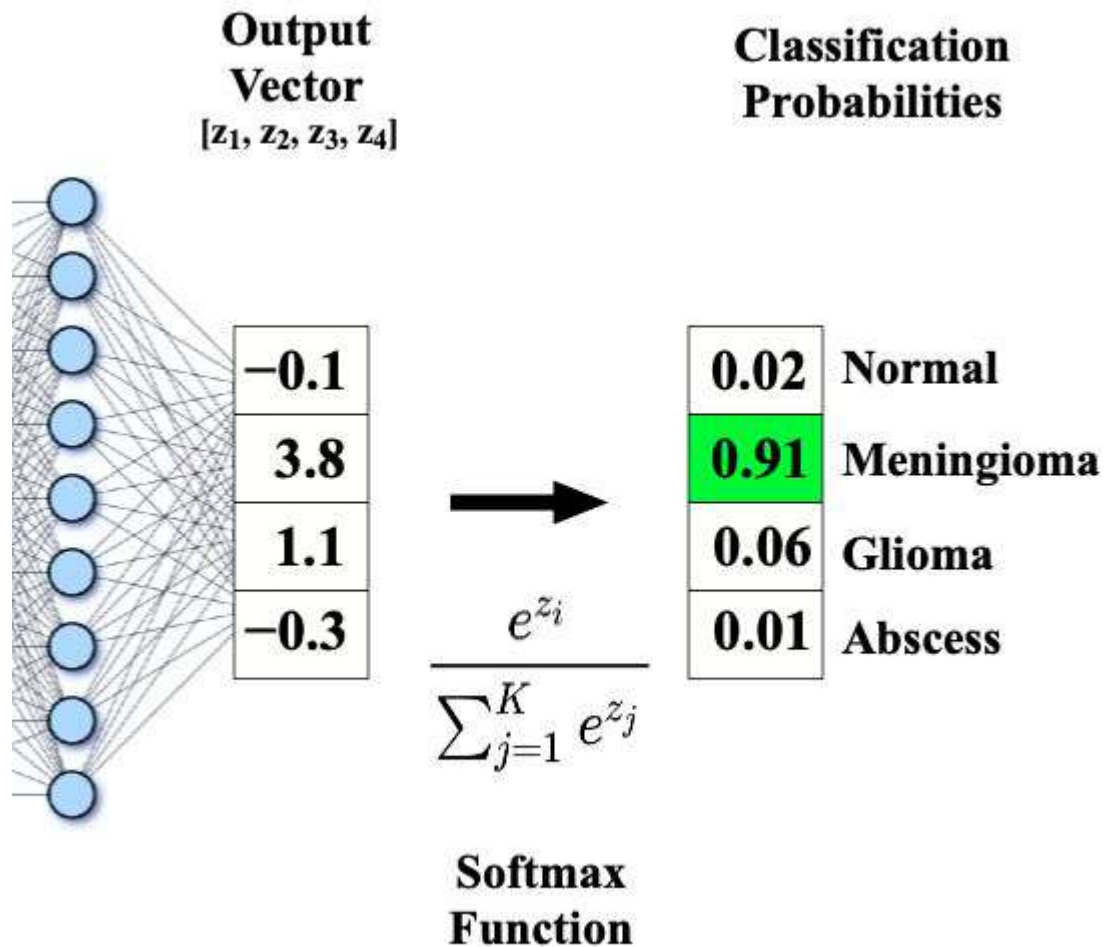
Why are Activation Functions Important?

- Non-linearity: Activation functions introduce non-linearities to the network, allowing it to learn complex relationships between input and output.
- Decision Making: They help neurons make decisions based on their inputs, effectively controlling the flow of information through the network.
- Enabling Backpropagation: Activation functions make it possible for networks to adjust weights during training by providing meaningful gradients during the backpropagation process.

Common Activation Functions (Sample)

Softmax

- Formula: $\text{Softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$
- Used for multi-class classification problems.
- Converts a vector of values into probabilities that sum to 1.



RELU

ReLU (Rectified Linear Unit)

- Formula: $\text{ReLU}(x) = \max(0, x)$
- Output Range: $[0, \infty)$
- Often used in hidden layers to improve training speed and avoid vanishing gradient problem.

Input

-249	-91	-37
250	-134	101
27	61	-153

ReLU

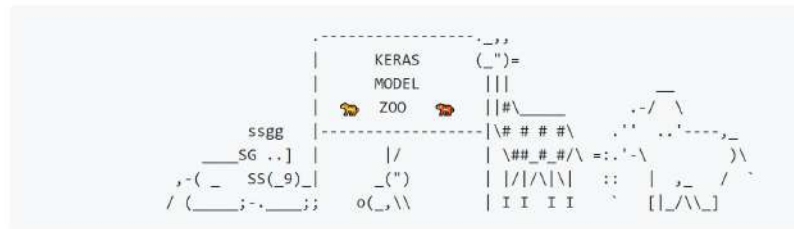
0	0	0
250	0	101
27	61	0

TensorFlow 2 Classification Model Zoo

https://github.com/tensorflow/models/blob/master/research/object_detection/g3doc/tf2_classification_zoo.md

The Keras Model Zoo

Ready to go, downloadable models for Keras



hosted with ❤️ from [Infinite Red](#)

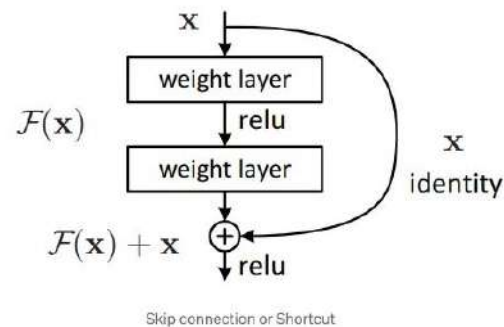
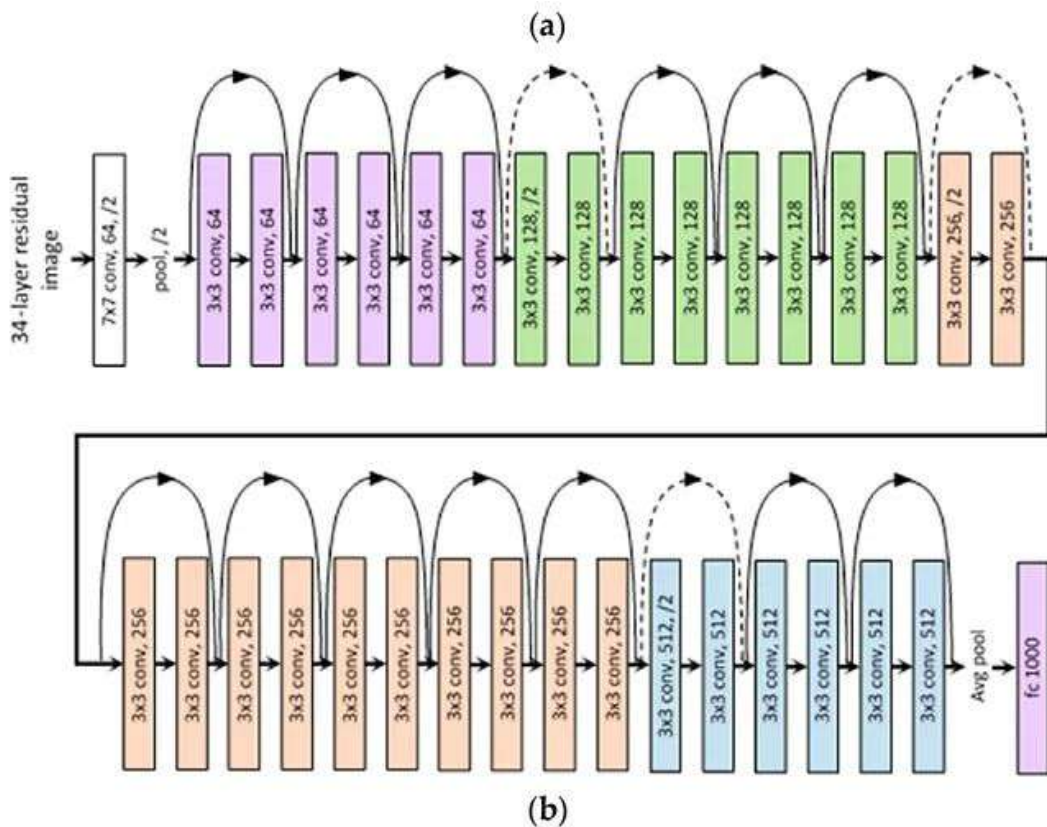
TensorFlow 2 Classification Model Zoo

TensorFlow 2.2 Python 3.6

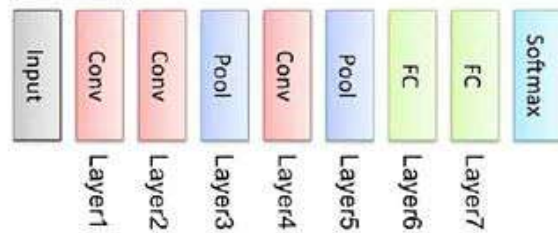
We provide a collection of classification models pre-trained on the [Imagenet](#). These can be used to initialize detection model parameters.

Model name
EfficientNet B0
EfficientNet B1
EfficientNet B2
EfficientNet B3
EfficientNet B4
EfficientNet B5
EfficientNet B6
EfficientNet B7
Resnet V1 50
Resnet V1 101
Resnet V1 152
Inception Resnet V2
MobileNet V1
MobileNet V2

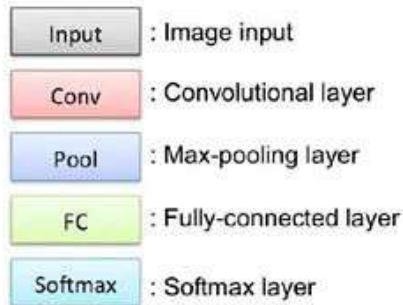
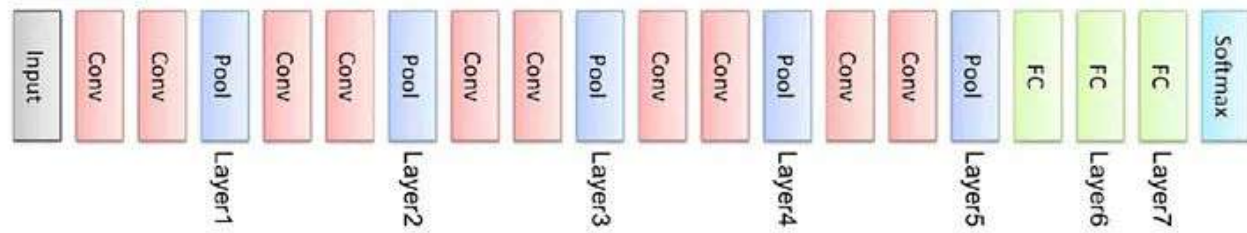
Resnet Architecture Explained



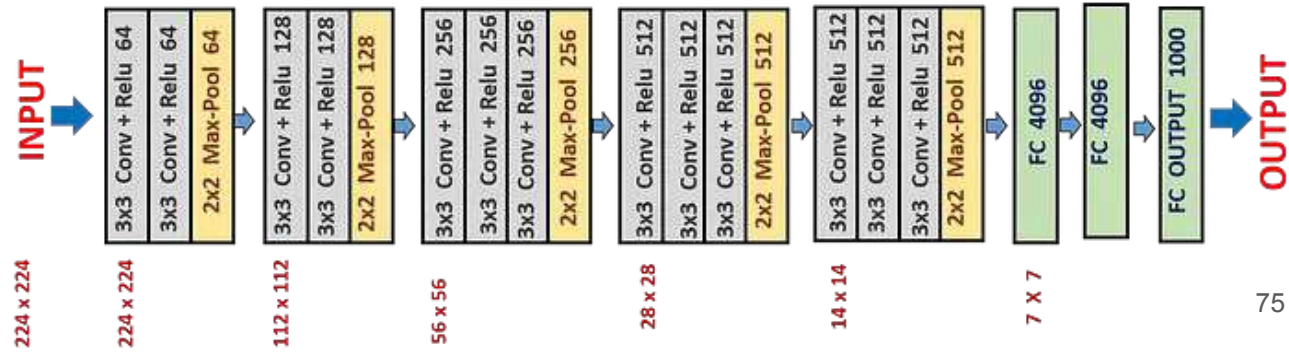
AlexNet

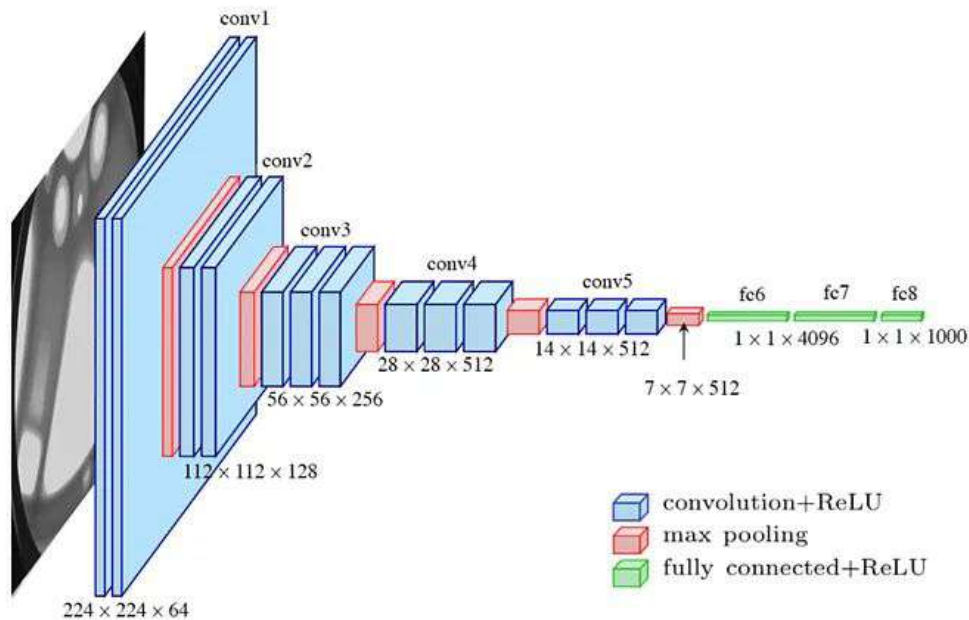


VGGNet



VGG-16



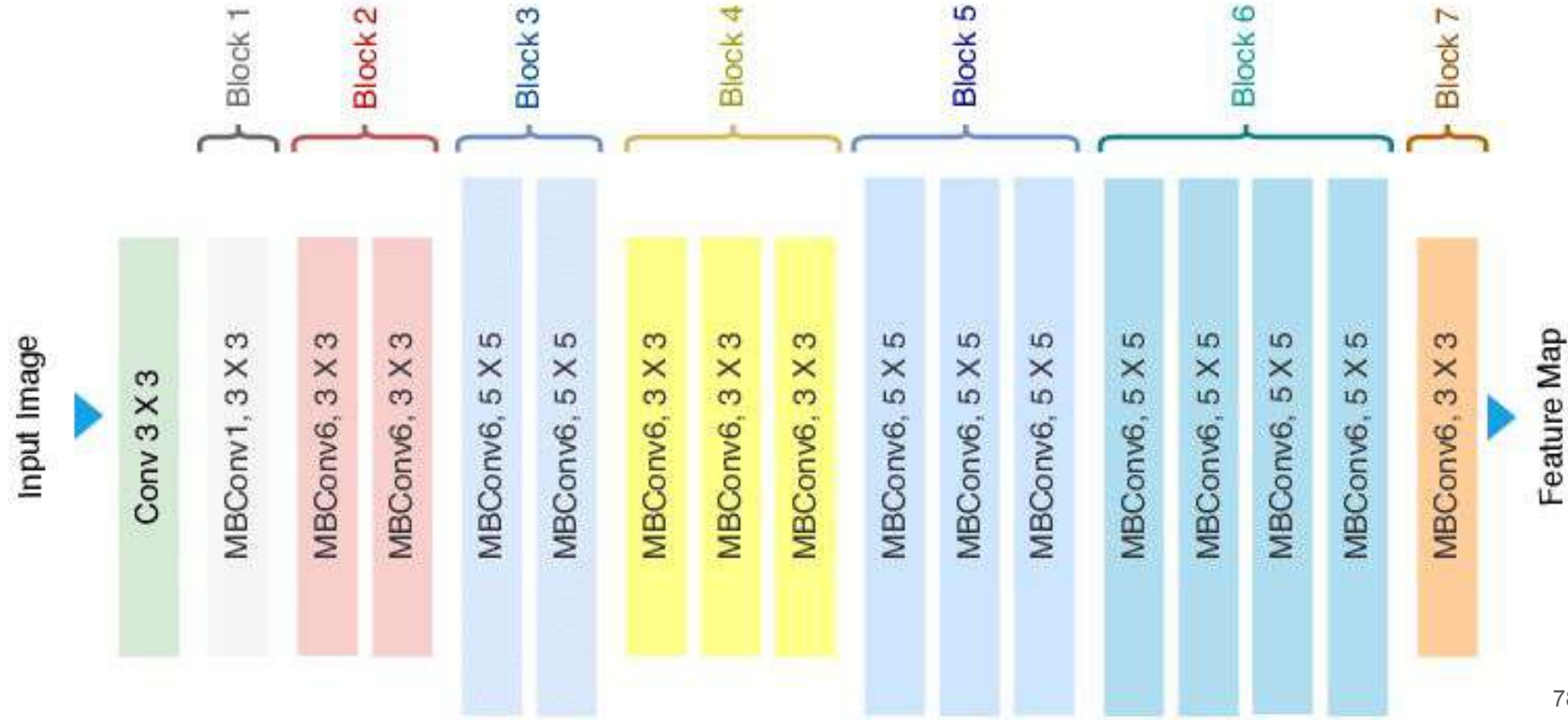


ConvNet Configuration					
A	A-LRN	B	C	D	E
11 weight layers	11 weight layers	13 weight layers	16 weight layers	16 weight layers	19 weight layers
input (224×224 RGB image)					
conv3-64	conv3-64 LRN	conv3-64 conv3-64	conv3-64 conv3-64	conv3-64 conv3-64	conv3-64 conv3-64
maxpool					
conv3-128	conv3-128	conv3-128 conv3-128	conv3-128 conv3-128	conv3-128 conv3-128	conv3-128 conv3-128
maxpool					
conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256 conv1-256	conv3-256 conv3-256 conv3-256	conv3-256 conv3-256 conv3-256 conv3-256
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 conv1-512	conv3-512 conv3-512 conv3-512	conv3-512 conv3-512 conv3-512 conv3-512
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 conv1-512	conv3-512 conv3-512 conv3-512	conv3-512 conv3-512 conv3-512 conv3-512
maxpool					
FC-4096					
FC-4096					
FC-1000					
soft-max					

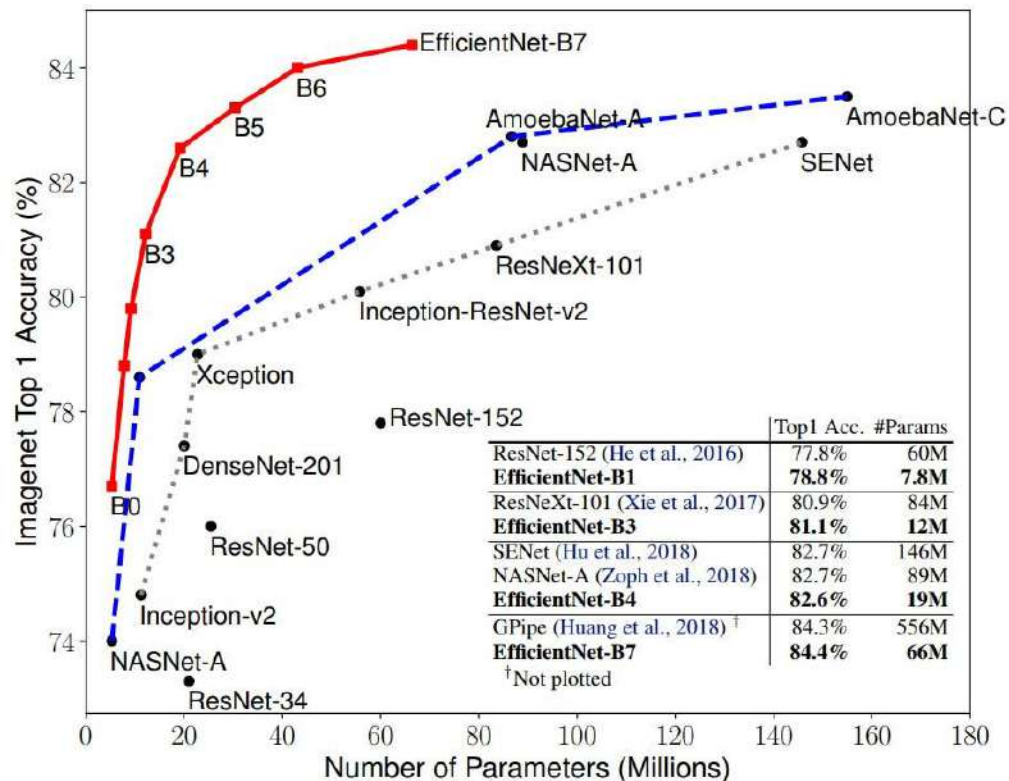
VGG-16 Summary

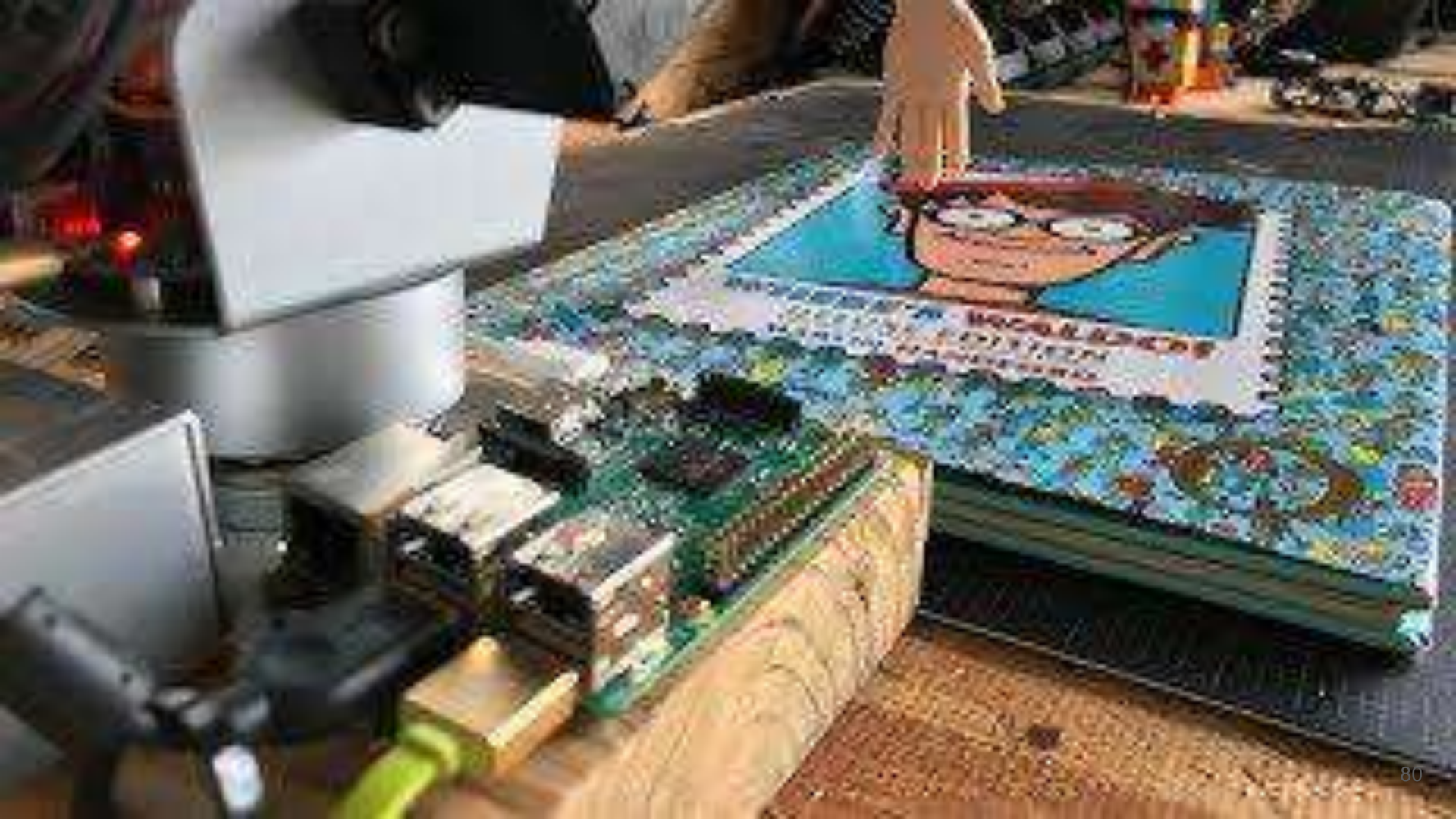
	Layer	Feature Map	Size	Kernel Size	Stride	Activation
Input	Image	1	224 x 224 x 3	-	-	-
1	2 X Convolution	64	224 x 224 x 64	3x3	1	relu
	Max Pooling	64	112 x 112 x 64	3x3	2	relu
3	2 X Convolution	128	112 x 112 x 128	3x3	1	relu
	Max Pooling	128	56 x 56 x 128	3x3	2	relu
5	2 X Convolution	256	56 x 56 x 256	3x3	1	relu
	Max Pooling	256	28 x 28 x 256	3x3	2	relu
7	3 X Convolution	512	28 x 28 x 512	3x3	1	relu
	Max Pooling	512	14 x 14 x 512	3x3	2	relu
10	3 X Convolution	512	14 x 14 x 512	3x3	1	relu
	Max Pooling	512	7 x 7 x 512	3x3	2	relu
13	FC	-	25088	-	-	relu
14	FC	-	4096	-	-	relu
15	FC	-	4096	-	-	relu
Output	FC	-	1000	-	-	Softmax

EfficientNet



EfficientNet Performance



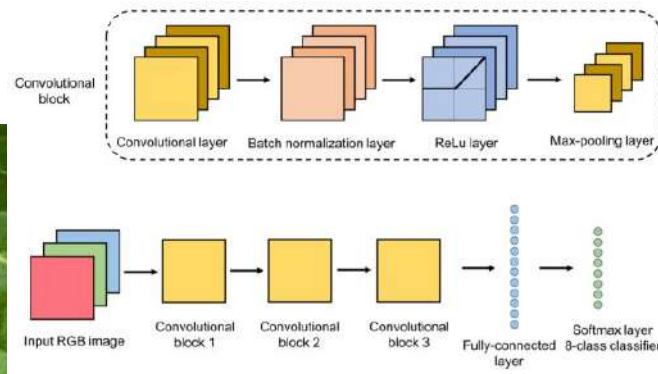




TensorFlow

Deep Learning Exercise:

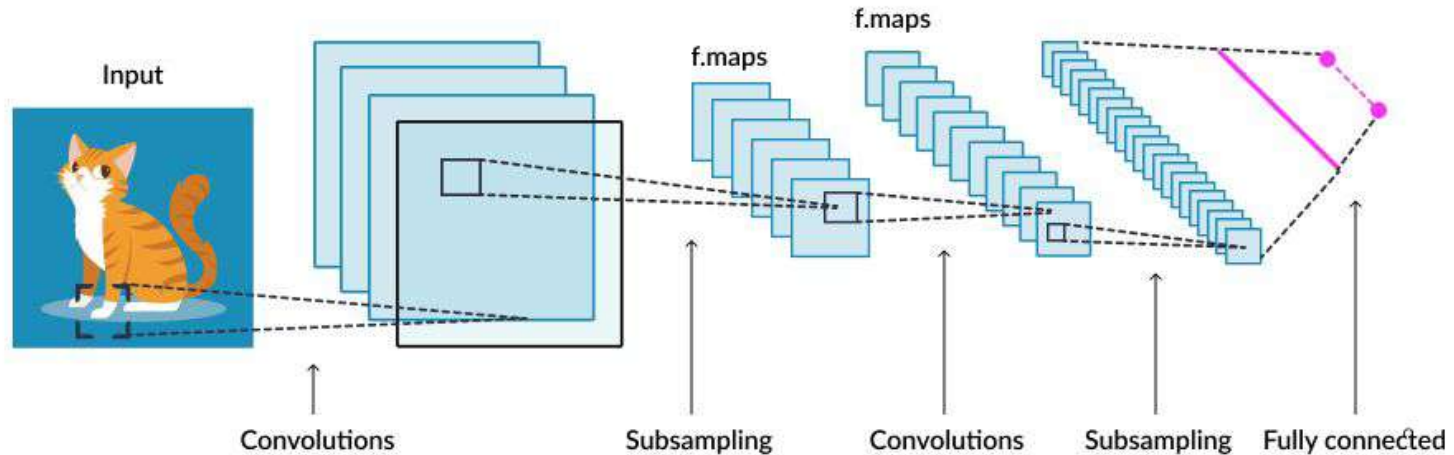
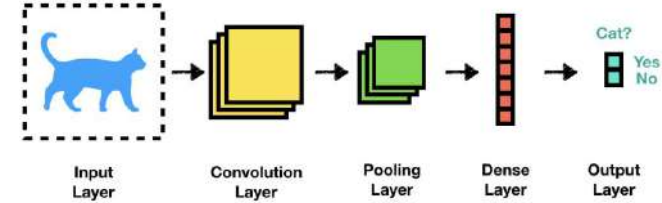
Mango 🥭 Leaf 🌿 🍂 Disease Dataset



Assignment

Build **Your Own Image** Classification in TensorFlow

This assignment will help you understand the fundamentals of image classification, TensorFlow, and neural network architectures.



Architecture of LeNet5

LeNet5 is a network made up of 7 layers. It consists of 3 convolution layers, two subsampling layers, and two fully connected layers.

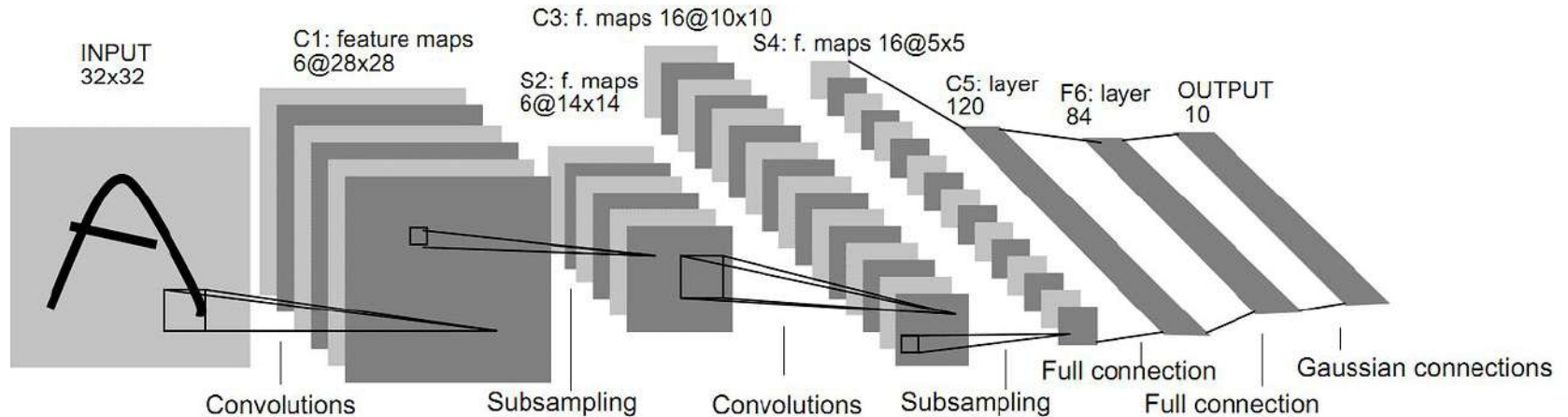
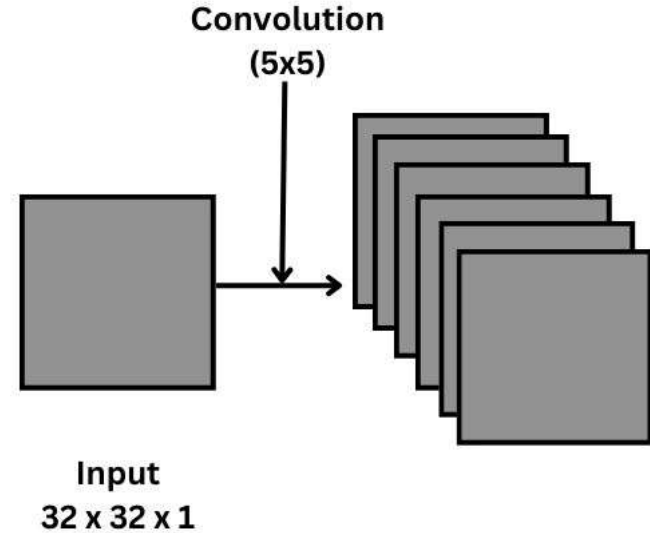


Fig. 2. Architecture of LeNet-5, a Convolutional Neural Network, here for digits recognition. Each plane is a feature map, i.e. a set of units whose weights are constrained to be identical.

LeNet5: First layer

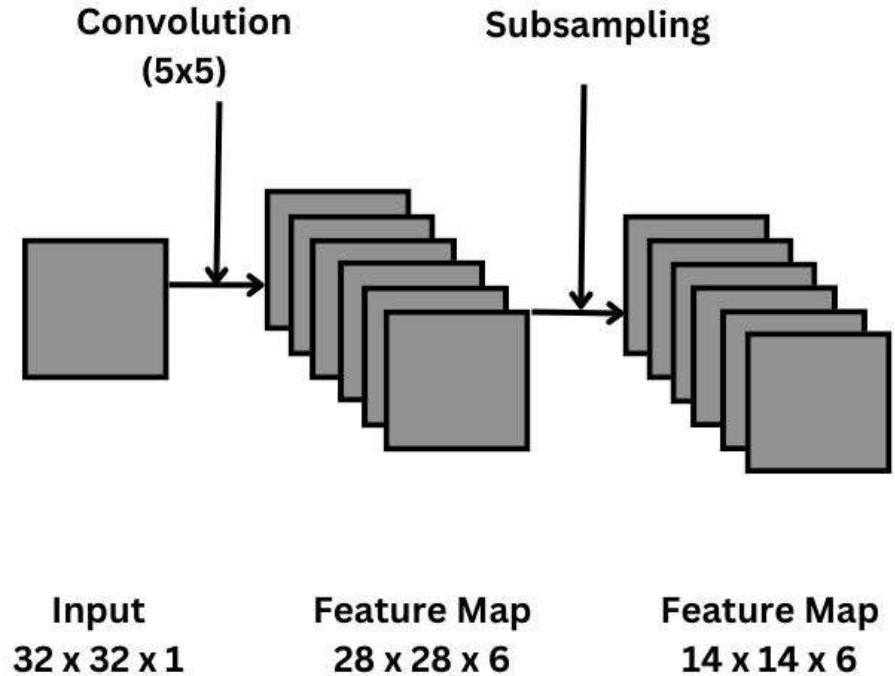
After this, the first convolution step takes place, and the input image is convoluted to the size of 28x28.



$$\begin{aligned}\text{Output Shape} &= ((32 - 5 + 1) \times (32 - 5 + 1) \times 6) \\ &= (28 \times 28 \times 6)\end{aligned}$$

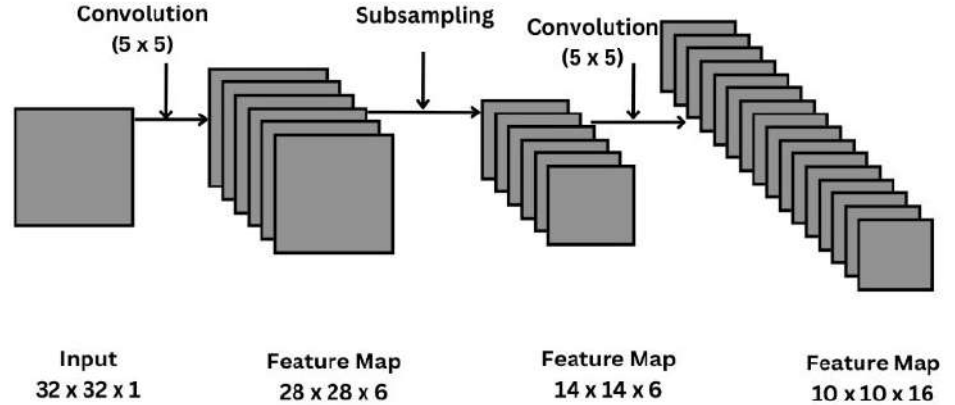
LeNet5: Second layer

Next is the subsampling layer, in which the size is reduced to half, i.e., 14x14.



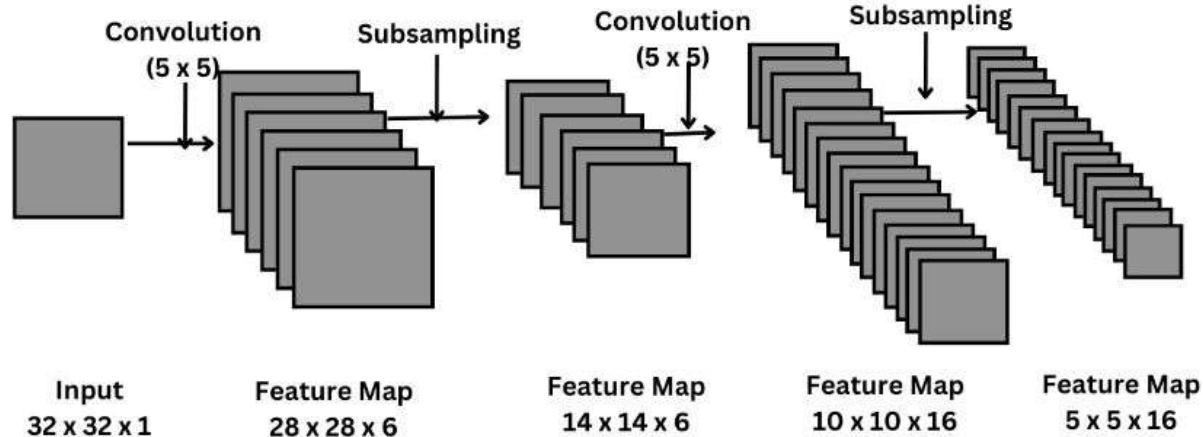
LeNet5: Third layer

In the third layer, convolution occurs again, but this time with 16 filters of 5x5 size. After this layer, the size of the input image is reduced to 10x10x16.



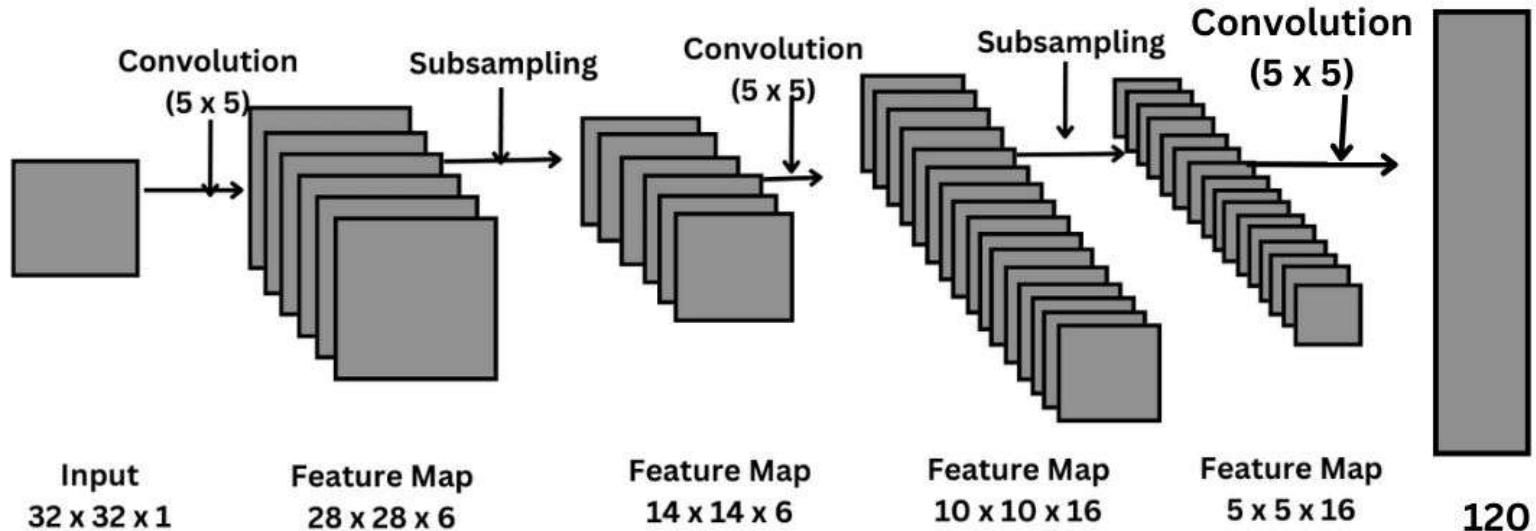
LeNet5: Fourth layer

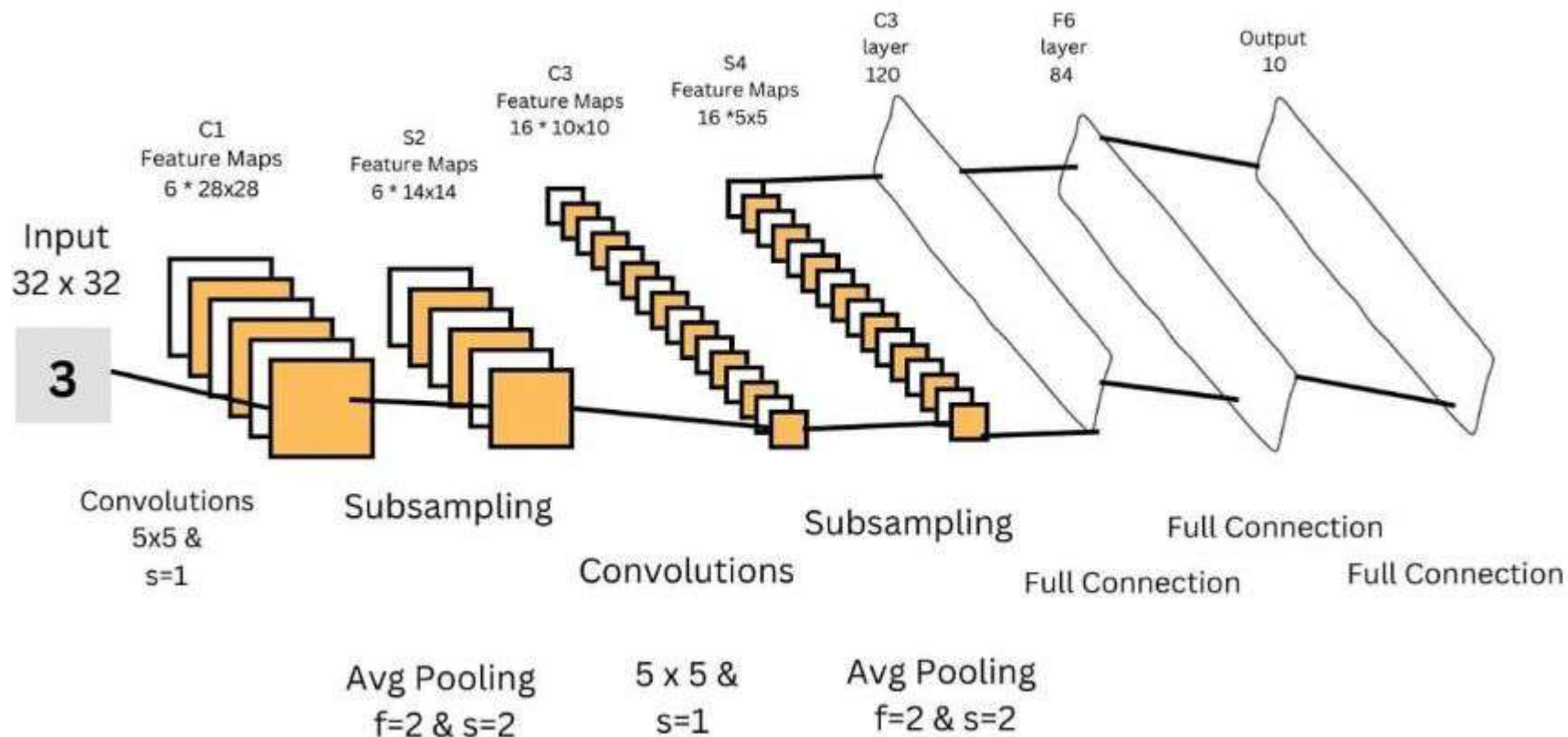
The subsampling takes place, and the image size in this step is reduced to $5 \times 5 \times 16$. In this layer, the input for the very last function diagram comes from all the remaining function diagrams.



LeNet5: Fifth layer

This is the final layer of the convolution-subsampling pair. There are 120 filters in this layer with a convolution size of 5×5 . After this layer, the feature map is $1 \times 1 \times 120$, and the flattened result is 120 values.

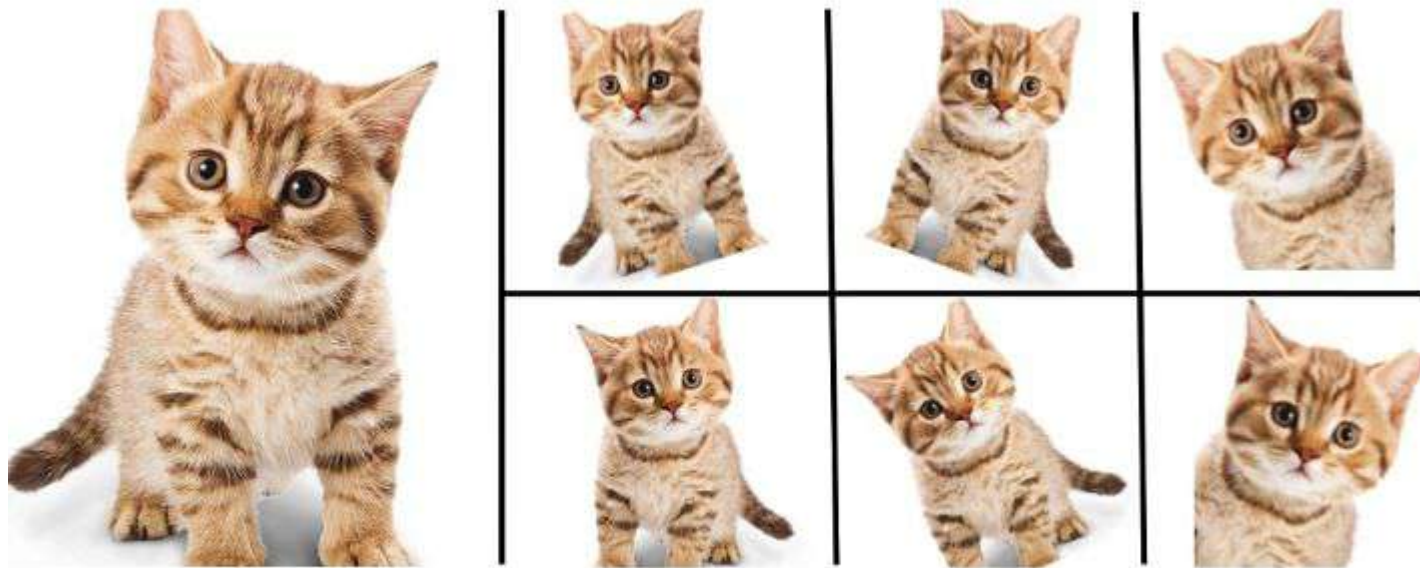




Summary of the architecture of LeNet5

Layer	Filter	Filter Size	Stride	Size of feature map	Activation Function
Input	-	-	-	32 x 32 x 1	-
Conv 1	6	5 x 5	1	28 x 28 x 6	tanh
Pooling 1	-	2 x 2	2	14 x 14 x 6	-
Conv 2	16	5 x 5	1	10 x 10 x 16	tanh
Pooling 2	-	2 x 2	2	5 x 5 x 16	-
Conv 3	120	5 x 5	1	120	tanh
Fully Connected 1	-	-	-	84	tanh
Fully Connected 2	-	-	-	10	Softmax

Data Augmentation



Enlarge your Dataset



Data Augmentation



Original Image



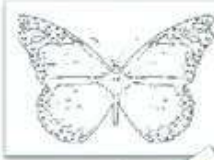
De-texturized



De-colored



Edge Enhanced



Salient Edge Map



Flip/Rotate

Horizontal

Vertically

+45 Rotation

-45 Rotation

Blur



Brighter

Noise added

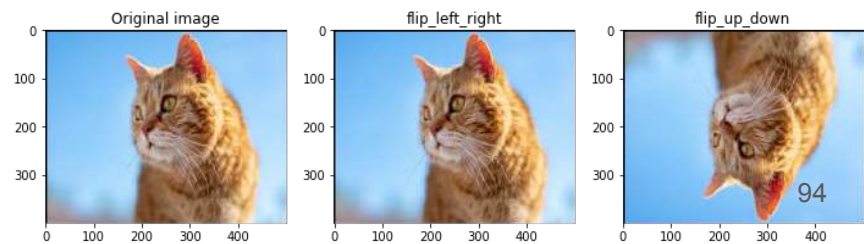
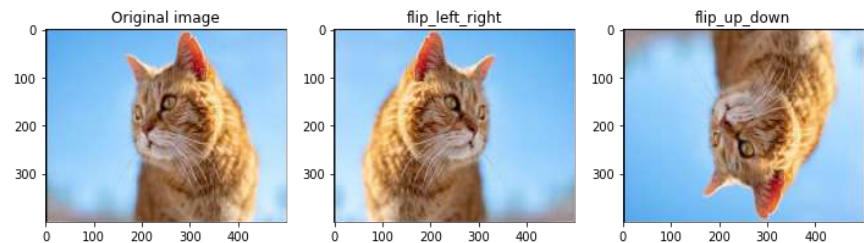
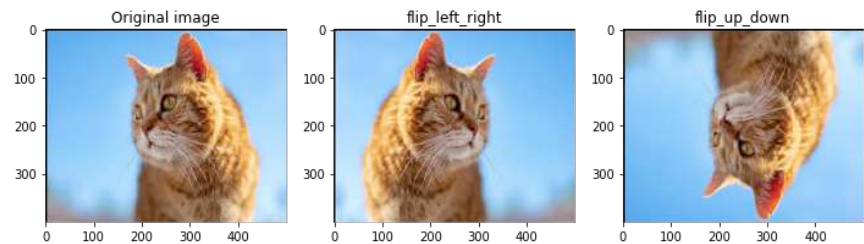
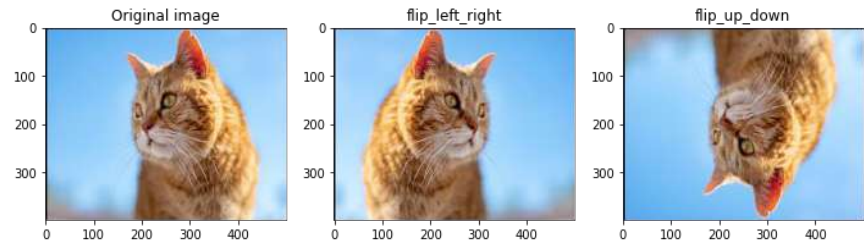
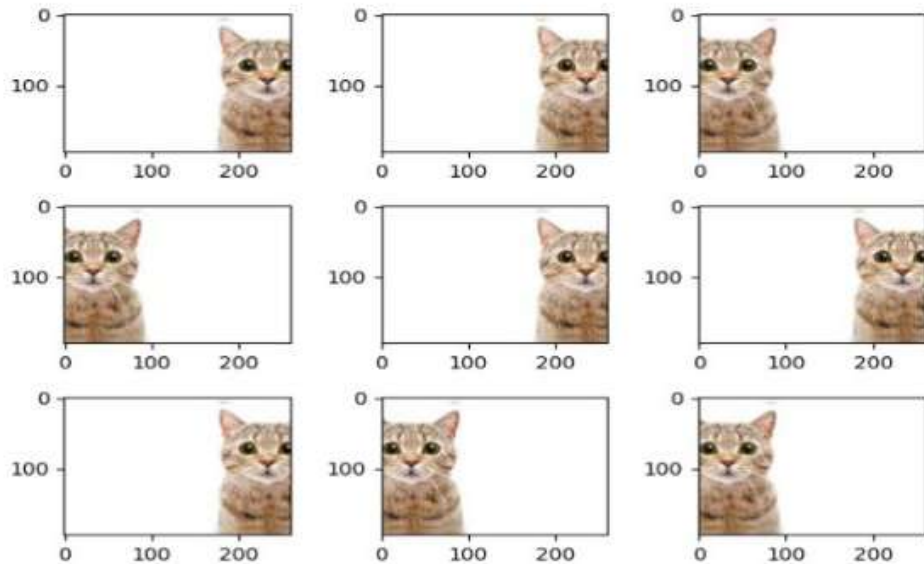
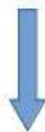
Darker

Grayscale

Crop



Augmented Images

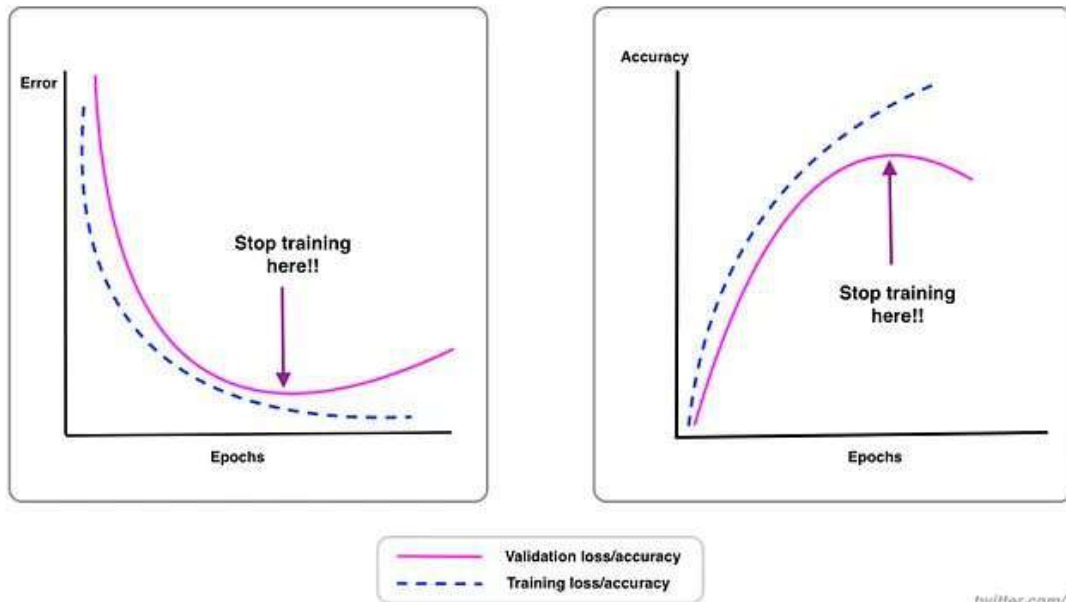


Early Stopping

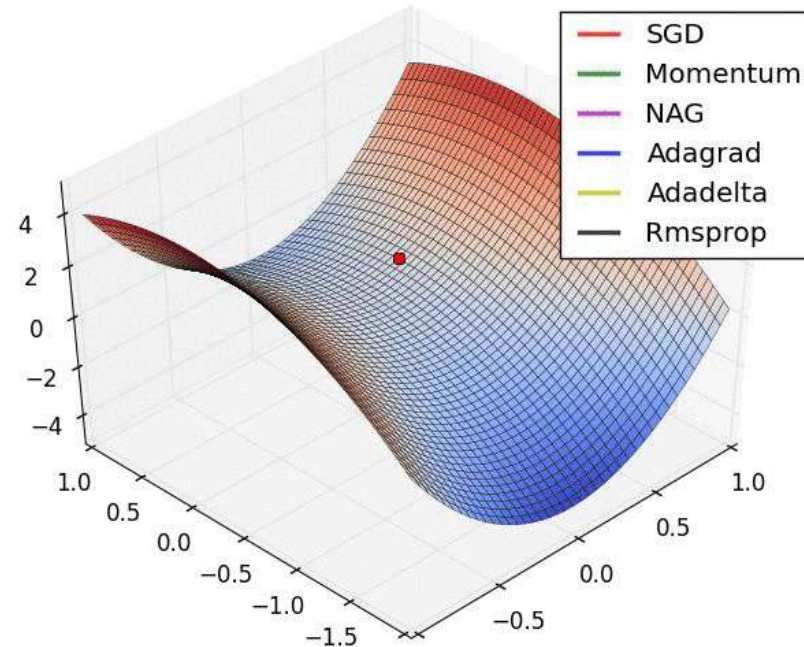
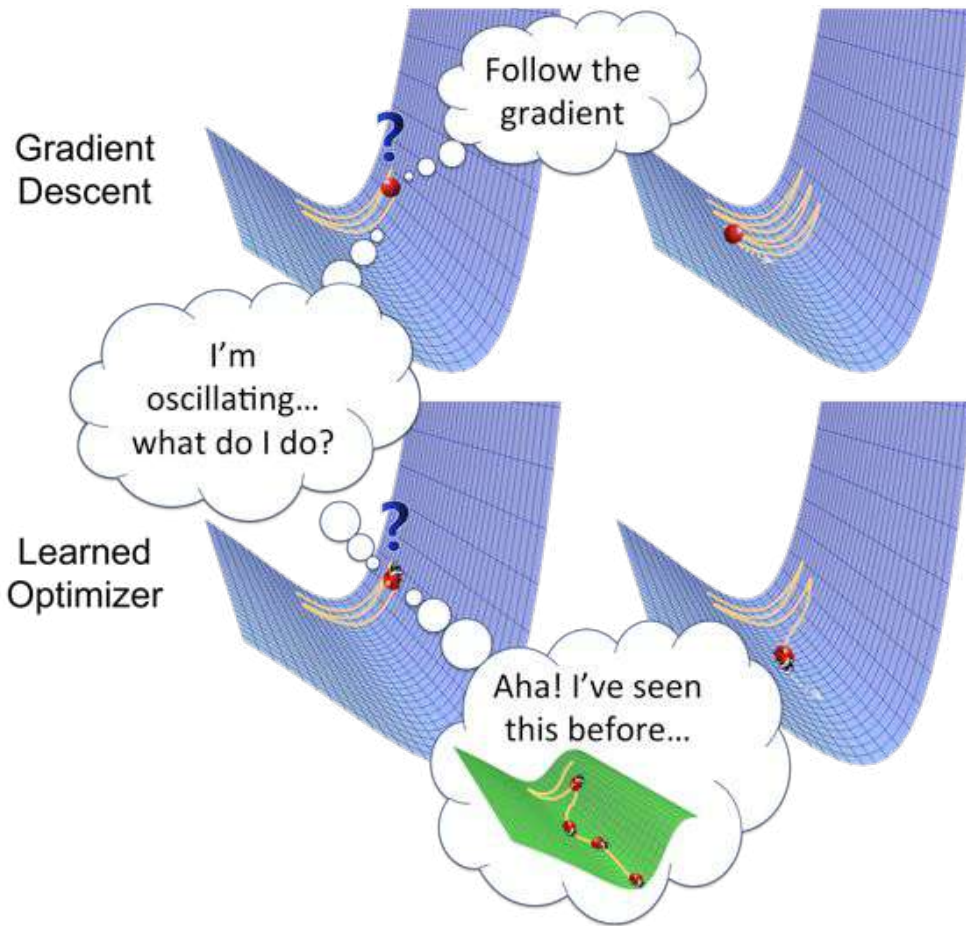
Early stopping is an optimization technique used to reduce overfitting without compromising on model accuracy.

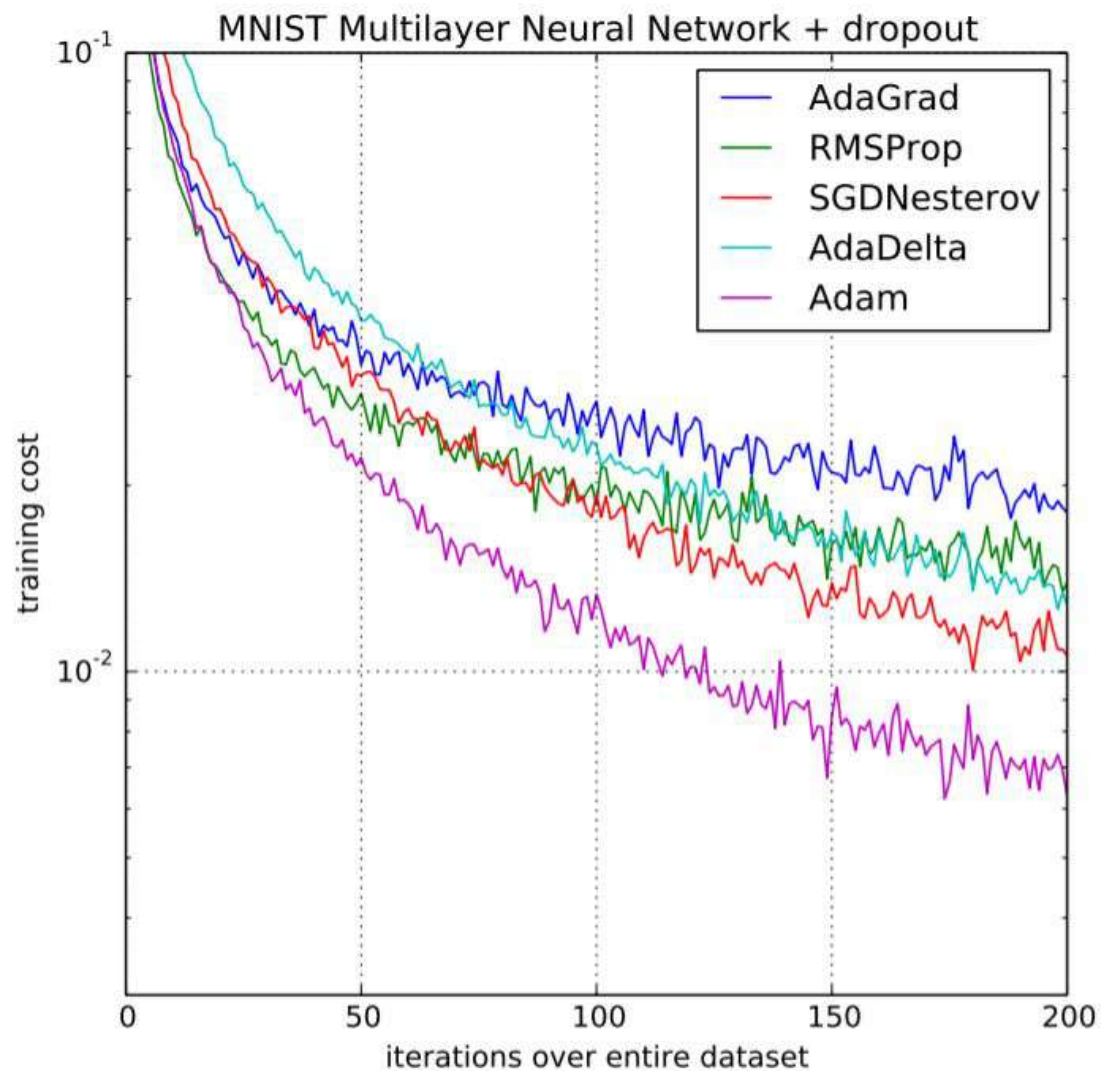
The main idea behind early stopping is to stop training before a model starts to overfit.

Early Stopping

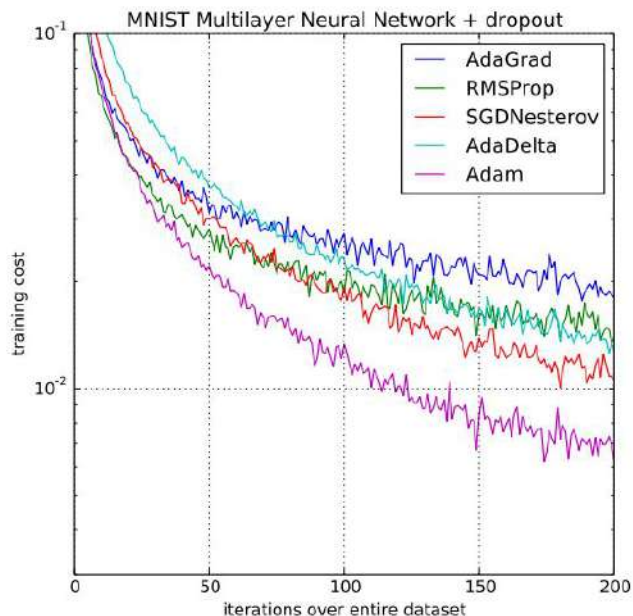


twitter.com/jeande_d





https://www.tensorflow.org/api_docs/python/tf/keras/optimizers



Classes

`class Adadelta` : Optimizer that implements the Adadelta algorithm.

`class Adafactor` : Optimizer that implements the Adafactor algorithm.

`class Adagrad` : Optimizer that implements the Adagrad algorithm.

`class Adam` : Optimizer that implements the Adam algorithm.

`class AdamW` : Optimizer that implements the AdamW algorithm.

`class Adamax` : Optimizer that implements the Adamax algorithm.

`class Ftrl` : Optimizer that implements the FTRL algorithm.

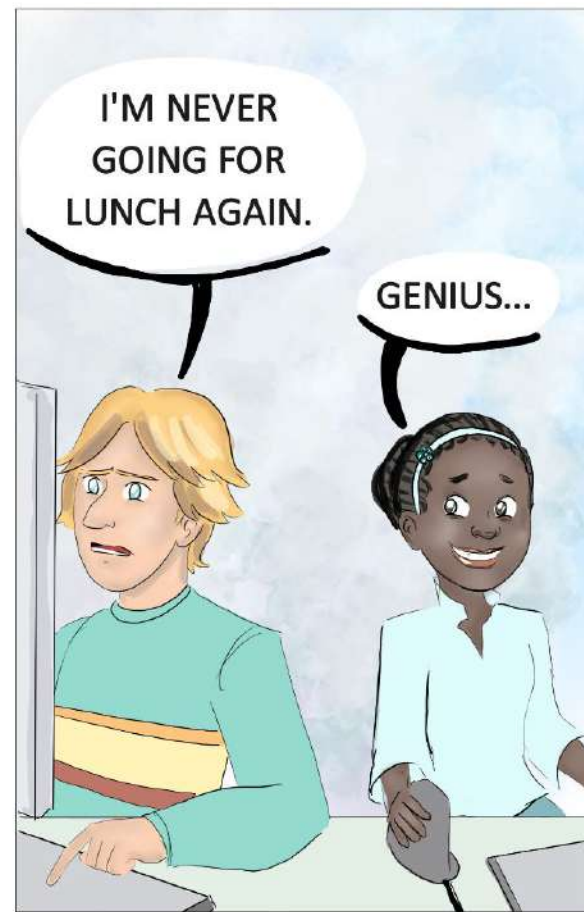
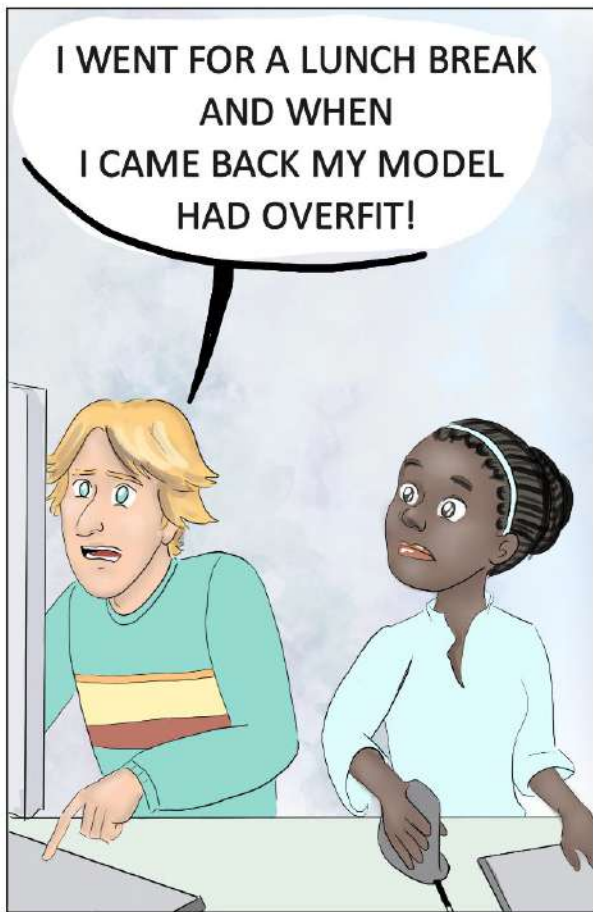
`class Lion` : Optimizer that implements the Lion algorithm.

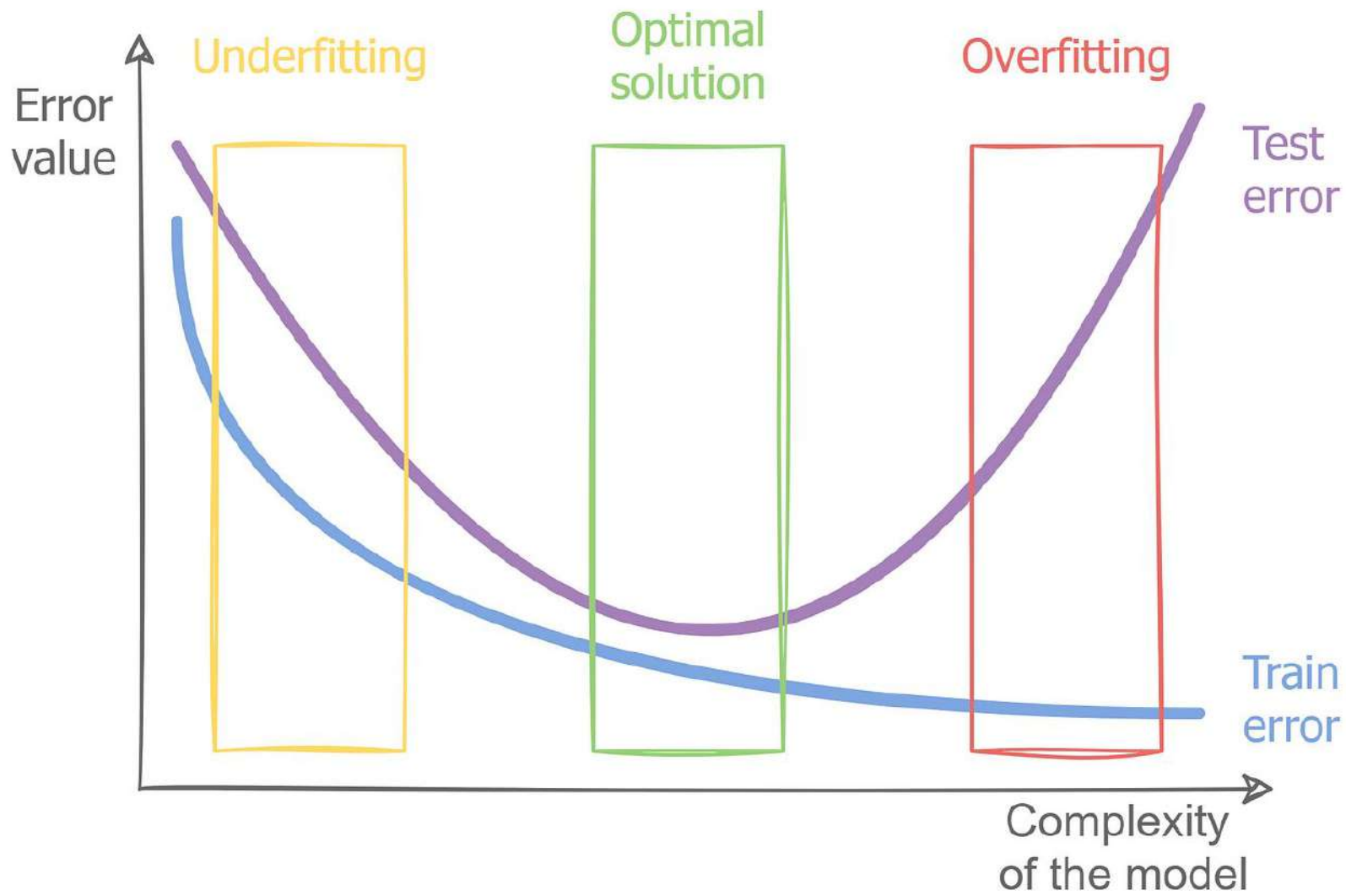
`class Nadam` : Optimizer that implements the Nadam algorithm.

`class Optimizer` : Abstract optimizer base class.

`class RMSprop` : Optimizer that implements the RMSprop algorithm.

`class SGD` : Gradient descent (with momentum) optimizer.





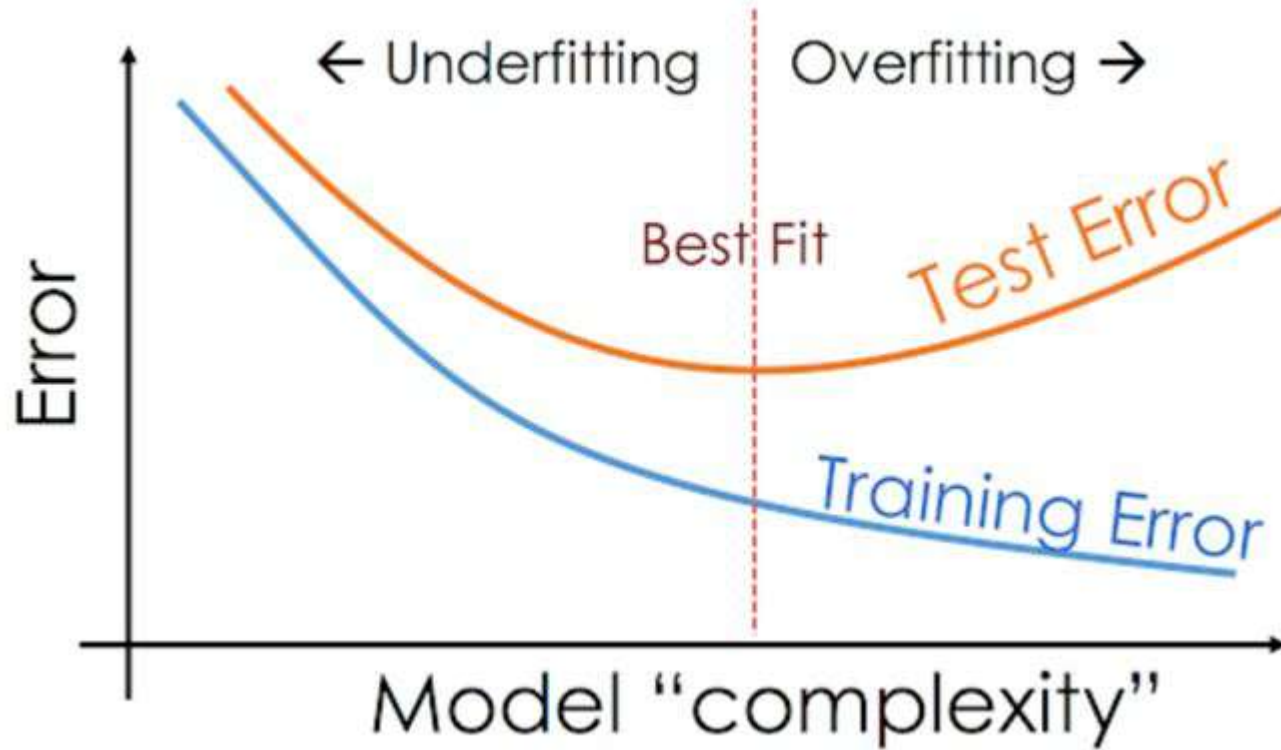
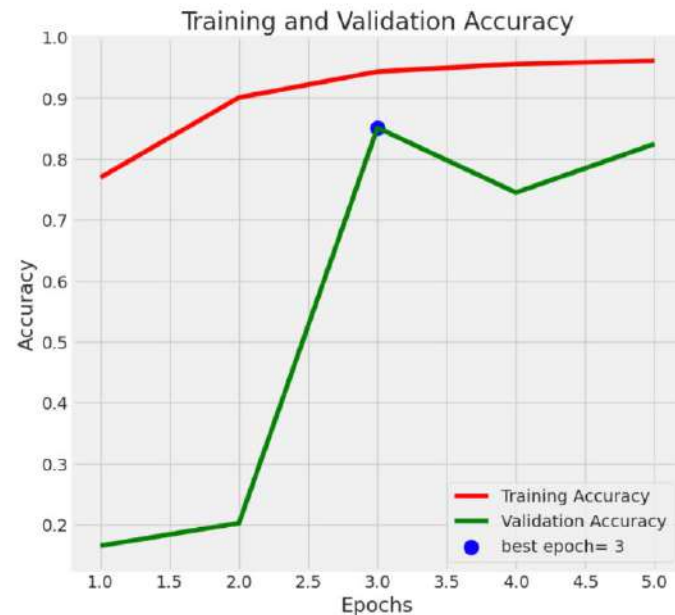


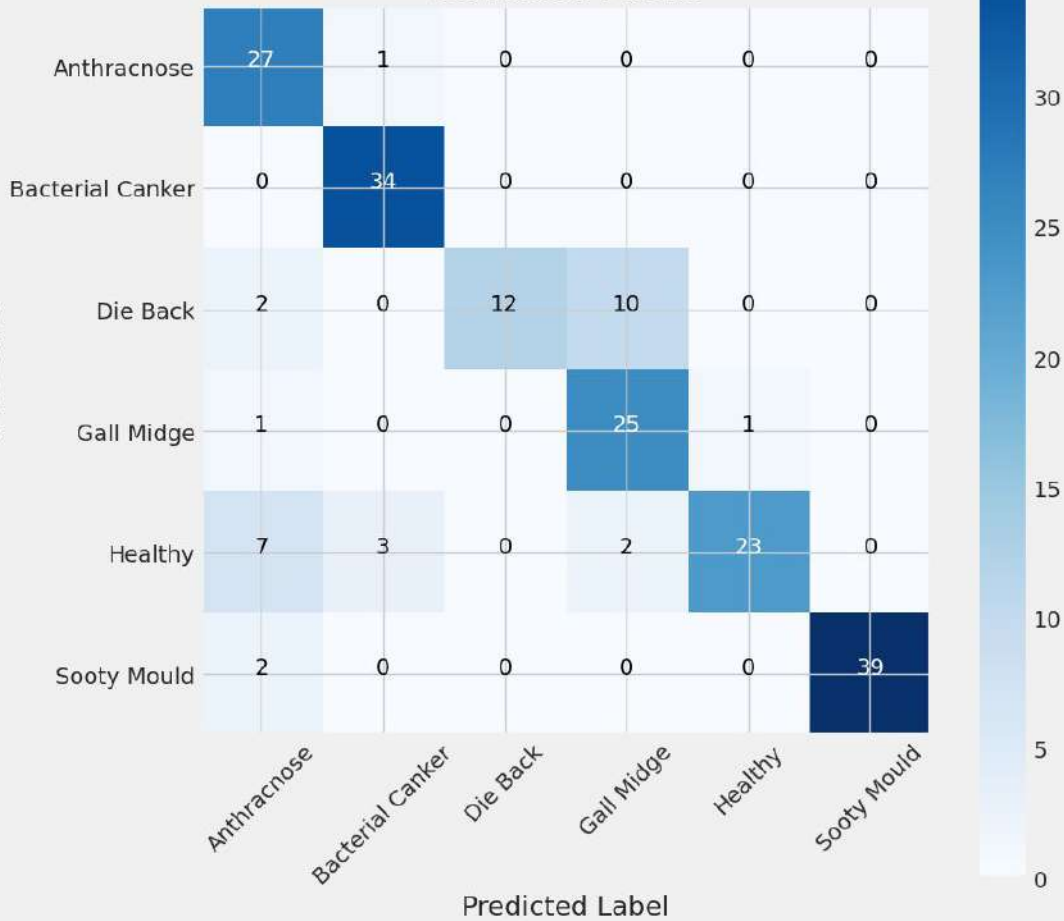
Table of contents

- Split dataframe into train, valid, and test
- Create image data generator
- Define data generators with augmentation
 - Show sample from train data
- Model Structure
 - Generic Model Creation
 - CNN Model Type1: EfficientNetB0
 - CNN Model Type2: ResNet50
 - CNN Model Type3: (Custom Deep Learning Network) LeNet-5 with Average Pooling
 - CNN Model Type4: (Custom Deep Learning Network) LeNet-5 with Max Pooling
 - CNN Model Type5: (Custom Deep Learning Network) LeNet-5 with ReLU Activation Function
 - CNN Model Type6: (Custom Deep Learning Network) VGG16 (Oxford U. Model)
- Homework
 - Train the model while implementing EarlyStopping.
 - Display model performance**
- Evaluate model
- Get Predictions
 - Confusion Matrics and Classification Report

```
plt.tight_layout()
plt.show()
```

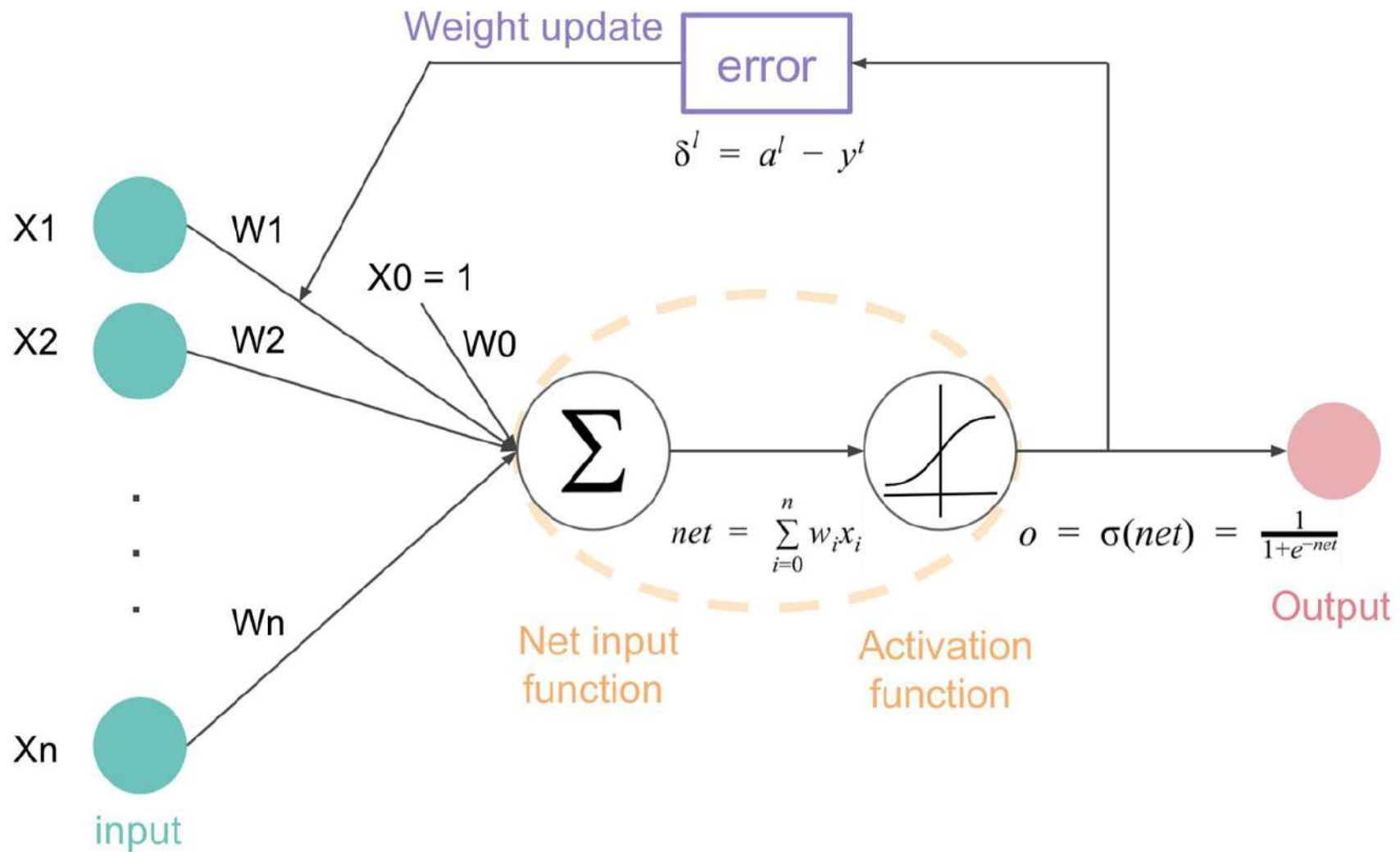


Confusion Matrix



```
[40] # Classification report
print(classification_report(test_gen.classes, y_pred, target_names= classes))
```

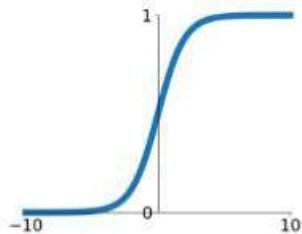
	precision	recall	f1-score	support
Anthracnose	0.69	0.96	0.81	28
Bacterial Canker	0.89	1.00	0.94	34
Die Back	1.00	0.50	0.67	24
Gall Midge	0.68	0.93	0.78	27
Healthy	0.96	0.66	0.78	35
Sooty Mould	1.00	0.95	0.97	41
accuracy			0.85	189
macro avg	0.87	0.83	0.83	189
weighted avg	0.88	0.85	0.84	189



Activation Functions

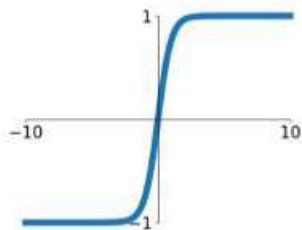
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



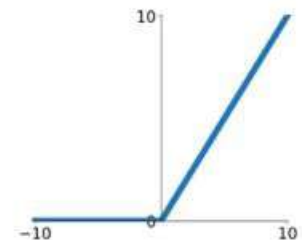
tanh

$$\tanh(x)$$



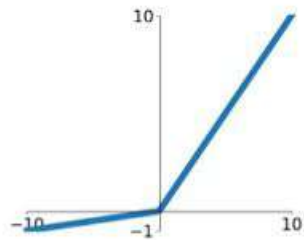
ReLU

$$\max(0, x)$$



Leaky ReLU

$$\max(0.1x, x)$$

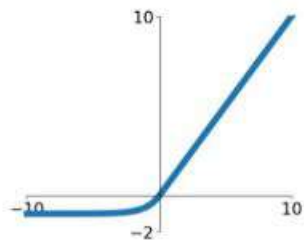


Maxout

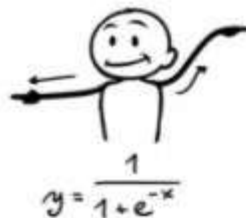
$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

ELU

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$



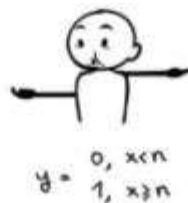
Sigmoid



Tanh



Step Function



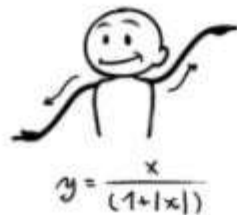
Softplus



ReLU



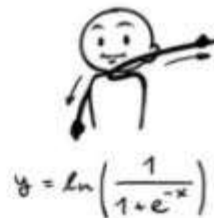
Softsign



ELU



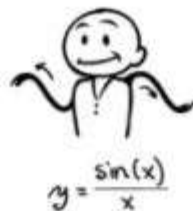
Log of Sigmoid



Swish



Sinc



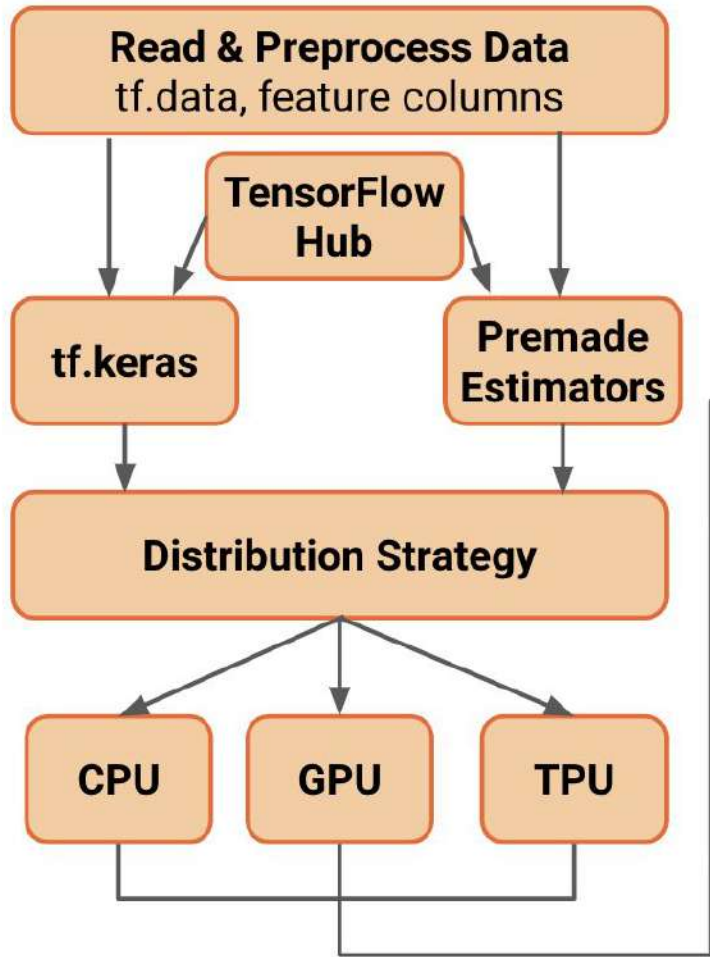
Leaky ReLU



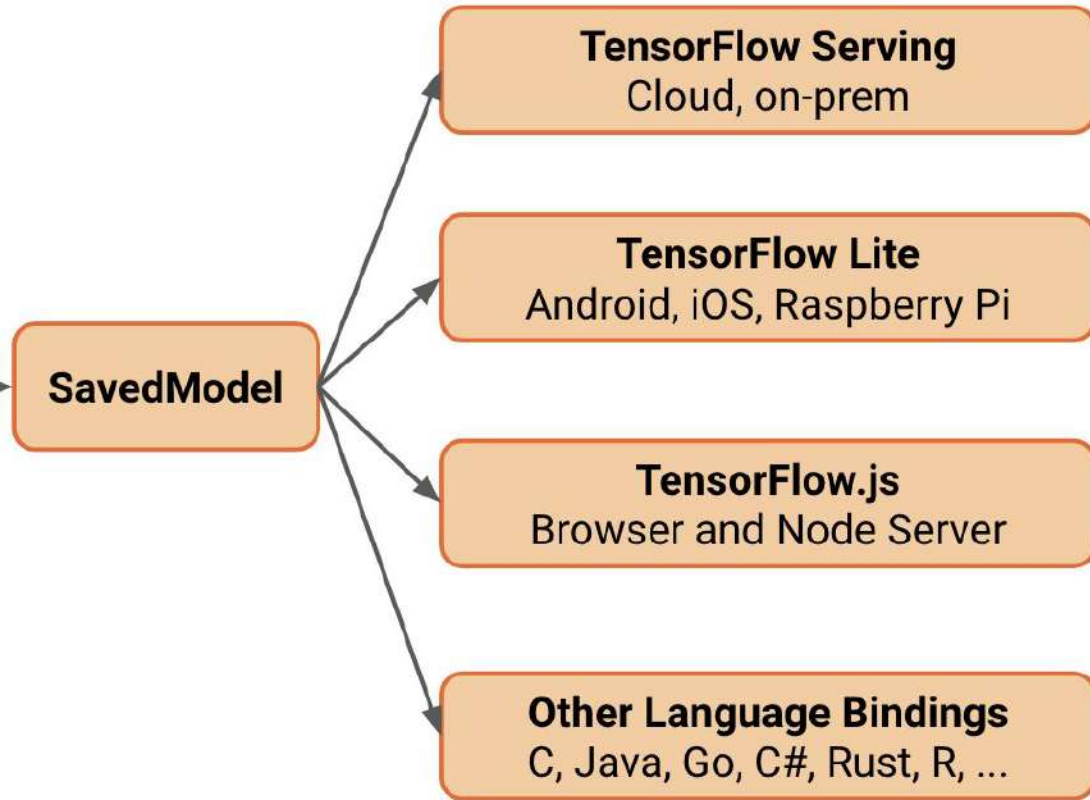
Mish



TRAINING



DEPLOYMENT



✓ 0s [32] `from tensorflow.keras.callbacks import EarlyStopping`

✓ 2m  `from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint`

```
batch_size = 16 # set batch size for training
epochs = 5 # number of all epochs in training

early_stopping = EarlyStopping(monitor='val_loss', patience=3, restore_best_weights=True)

# Add ModelCheckpoint callback to save the model weights
model_checkpoint = ModelCheckpoint('cnn_mango_leaf_best_model.h5', save_best_only=True)

history = model.fit(x=train_gen, epochs=epochs, verbose=1, validation_data=valid_gen,
                    validation_steps=None, shuffle=False,
                    callbacks=[early_stopping, model_checkpoint])
```



Epoch 1/5

36/36 [=====] - 53s 497ms/step - loss: 8.7925 - accuracy: 0.7695 - val_loss: 36.5454

Epoch 2/5

36/36 [=====] - 11s 309ms/step - loss: 7.6616 - accuracy: 0.9007 - val_loss: 10.4449

Epoch 3/5

36/36 [=====] - 11s 310ms/step - loss: 6.8699 - accuracy: 0.9433 - val_loss: 7.0636

Epoch 4/5

36/36 [=====] - 12s 335ms/step - loss: 6.3583 - accuracy: 0.9557 - val_loss: 6.9716

Epoch 5/5

36/36 [=====] - 12s 324ms/step - loss: 5.8728 - accuracy: 0.9610 - val_loss: 6.3440

Files



- {x}
- ..
 - ▶ MangoLeafBD_dataset_small_v2
 - ▶ __MACOSX
 - ▶ drive
 - ▶ sample_data
 - Mango Diseases-class_dict.csv
 - cnn_mango_leaf_best_model.h5
 - resnet50-Mango Diseases-84.66.h5
 - resnet50-Mango Diseases-weights...

+ Code + Text

▼ Save model

```

✓ [41] model_name = model.input_names[0][: -6]
      2s subject = 'Mango Diseases'
      acc = test_score[1] * 100
      save_path = ''

      # Save model
      save_id = str(f'{model_name}-{subject}-{ "%.2f" %round(acc, 2)}.h5')
      model_save_loc = os.path.join(save_path, save_id)
      model.save(model_save_loc)
      print(f'model was saved as {model_save_loc}')

      # Save weights
      weight_save_id = str(f'{model_name}-{subject}-weights.h5')
      weights_save_loc = os.path.join(save_path, weight_save_id)
      model.save_weights(weights_save_loc)
      print(f'weights were saved as {weights_save_loc}')

```

model was saved as resnet50-Mango Diseases-84.66.h5
weights were saved as resnet50-Mango Diseases-weights.h5

✓
0s

```
[43] from tensorflow.keras.models import load_model  
      from tensorflow.keras.preprocessing import image  
      import numpy as np  
      import matplotlib.pyplot as plt  
      import ipywidgets as widgets  
      from IPython.display import display  
      import os, io
```

✓
3s

```
[44] # Load your pre-trained model  
      model = load_model('/content/cnn_mango_leaf_best_model.h5')
```


✓
0s

```
[56] # Get value counts
```

```
value_counts = df['labels'].value_counts()
```

```
# Create class indices dictionary
```

```
class_indices = {idx: label for idx, label  
                  in enumerate(value_counts.index)}
```

```
class_indices
```

```
{0: 'Die Back',  
 1: 'Bacterial Canker',  
 2: 'Sooty Mould',  
 3: 'Healthy',  
 4: 'Gall Midge',  
 5: 'Anthracnose'}
```

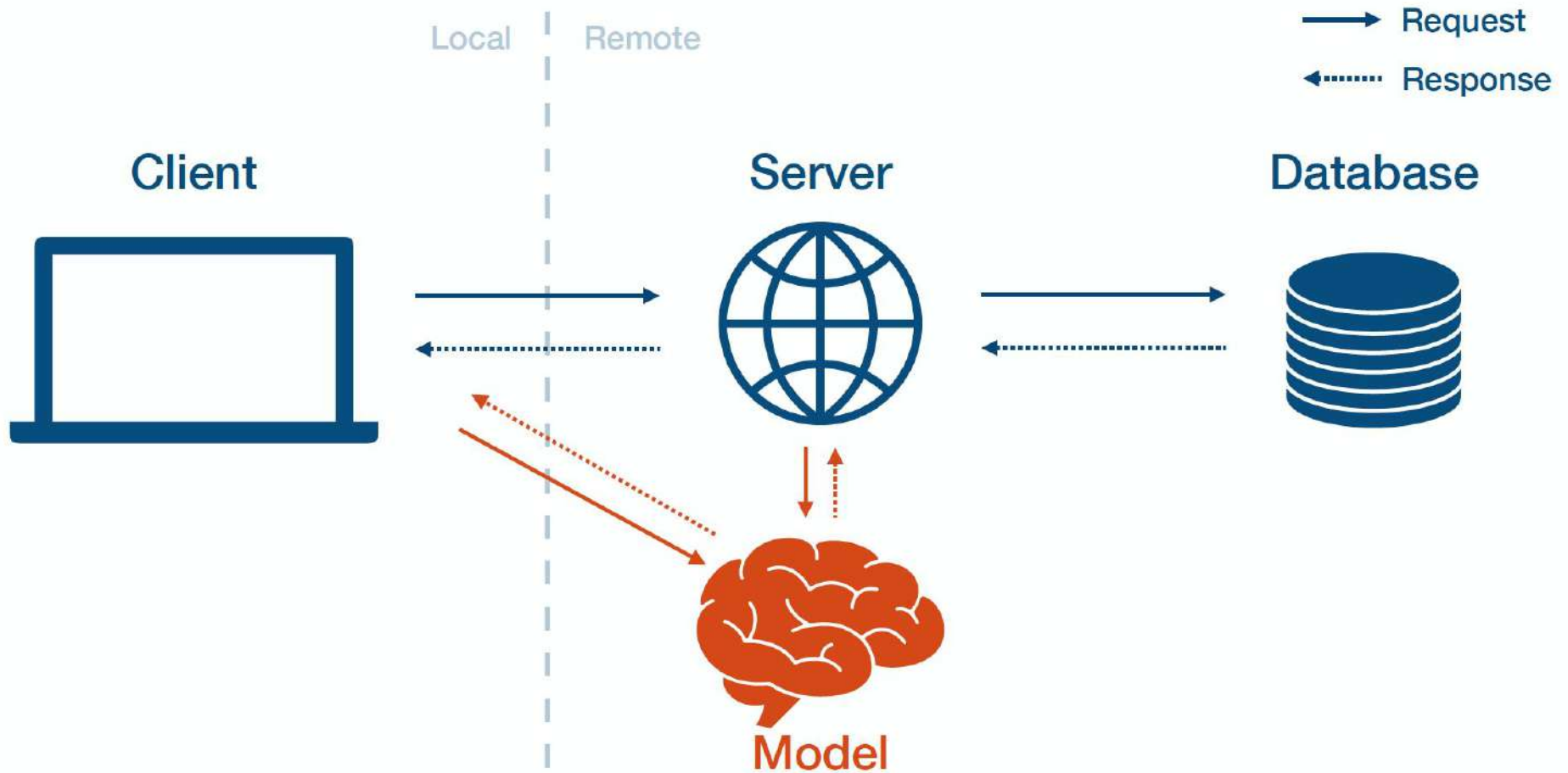
Upload (3)

1/1 [=====] - ETA: 0s

1/1 [=====] - 0s 22ms/step

Prediction: Sooty Mould, Confidence: 0.3687029778957367

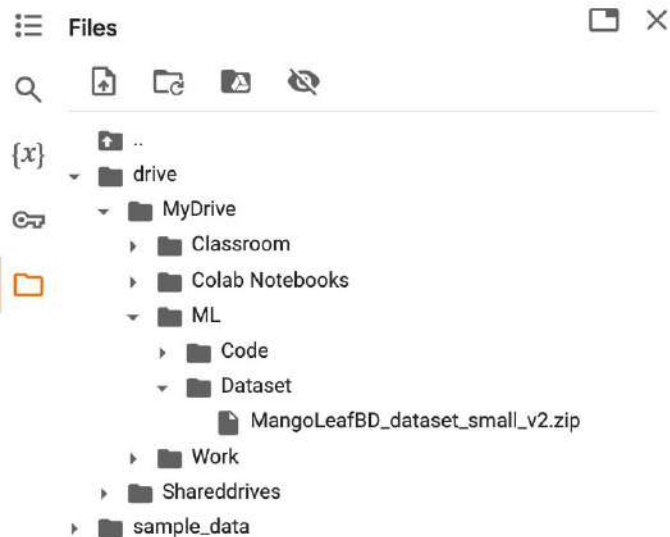




How to Import Files from Google Drive to Colab



 mango_leaf_train_and_save_weight_2024v3_part2.ipynb ☆
File Edit View Insert Runtime Tools Help [All changes saved](#)



+ Code + Text

Download Sample Dataset (From Google Drive)

```
✓ [4] from google.colab import drive  
21s drive.mount('/content/drive')
```

Mounted at /content/drive

```
[ ] # e.g., sample path: /content/drive/MyDrive/ML/Dataset/MangoLeafBD_dataset_small_v2.zip
```

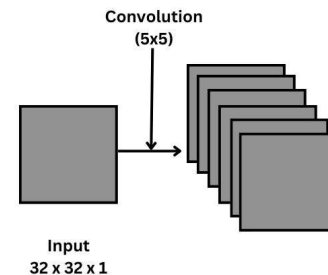
Don't forget to uncomment this line before training the model to unzip the dataset, which format.

```
[ ] # !unzip MangoLeafBD_dataset_small_v2.zip  
!unzip /content/drive/MyDrive/ML/Dataset/MangoLeafBD_dataset_small_v2.zip
```

Assignment

Build Your Own Deep Learning Architecture

Design and implement your own deep-learning architecture for a given task or dataset.



$$\begin{aligned}\text{Output Shape} &= ((32 - 5 + 1) \times (32 - 5 + 1) \times 6) \\ &= (28 \times 28 \times 6)\end{aligned}$$

